

Review

Enantioselective dialkylzinc-mediated alkynylation, arylation and alkenylation of carbonyl groups



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ABSTRACT

Enantiopure diarylmethanols, propargylic and allylic alcohols are important building blocks for the synthesis of a plethora of natural products and pharmaceutically relevant compounds. Among several methods employed for their synthesis, the enantioselective addition of diorganozinc compounds to carbonyl compounds in the presence of appropriate ligands becomes increasingly significant. In this comprehensive review, methods based on the use of readily available diethyl- and dimethylzinc for the formation of more complex compounds are compared and discussed. Available data on the mechanisms and transition states of reactions are presented.

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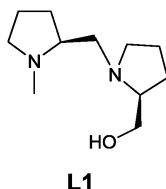
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1. Introduction

Enantioselective alkynylation, alkenylation and arylation reactions are now a well established tool in the arsenal of a contemporary organic chemist. At first glance, excellent results obtained so far indicate that synthetic chemistry cannot add much to this. However, many of these results were reported for model compounds, e.g. phenylacetylene and, though they add to our knowledge about this chemistry, they are of little value for the practical applications in the synthesis. This review focuses on the dialkylzinc-mediated, enantioselective addition to aldehydes and ketones leading to allylic alcohols, propargylic alcohols and diaryl-methanols since the first publications on each topic appeared in print. Several reviews on these topics were published [1–14], many of which being personal accounts. Numerous ligands were applied for both alkynylation and arylation or alkenylation; therefore, a review dealing with all these topics seems to be justified. The main goal of this review is to summarise the results obtained in this area, indicating existing and new challenges, and problems which still remain to be solved. The author attempted to write this review as comprehensively as possible. However, in order to make it more compact and easier to read, many schemes presenting detailed information about substrates, yields and enantioselectivities were placed in supplementary materials. These schemes are numbered separately; however, the numbering of ligands and compounds is consecutive both in the review and supplementary materials in order to facilitate the simultaneous reading of both parts.

The addition of organometallic reagents to carbonyl groups is one of the most common methods for the formation of carbon–carbon bonds. Performing the organometallic additions in a chiral environment leads to the chiral secondary and tertiary alcohols, which are of great synthetic importance. The most popular and most readily available organometallic reagents are Grignard reagents. Unfortunately, their addition to aldehydes and ketones is difficult to control due to their high reactivity. Recent reports show new prospects for such catalytic, enantioselective reactions [15–22], but so far more efficient methods have been based on organozinc compounds. The addition of dialkylzincs to carbonyl compounds is a very slow process in the absence of the catalyst. The situation changes in the presence of an appropriate ligand, usually β -amino alcohol. The first efficient diethylzinc addition to benzaldehyde giving 1-phenylpropanol in the presence of L-proline-derived β -amino alcohol **L1** was reported by Mukaiyama et al. [23].



This reaction proceeded with a good chemical yield (78%), but no asymmetric induction was observed. Several years later Oguni and Omi reported the synthesis of 1-phenylpropanol catalysed by (S)-leucinol with 49% ee [24]. The breakthrough was made by Noyori, who used DAIB (3-*exo*-dimethylaminoisoborneol) **L2** as a very efficient ligand for the addition of diethylzinc to benzaldehyde. For the first time 1-phenylpropanol was obtained in 99% ee and 98% yield [25]. Since then, hundreds of reports have been published, and the addition of dialkylzinc to carbonyl compounds has been reviewed thoroughly [26–28,5,29,30,8].

It has been shown that after coordinating to an amino alcohol ligand, the linear structure of dialkylzinc is converted to approximately tetrahedral. The carbon–zinc bond becomes longer, which results in a lower bond energy and a higher nucleophilicity of

the alkyl group [31,32]. The mechanism of the addition and the stereochemical model proposed by Noyori [26] and others [33–35] (Scheme 1) is still widely accepted. It is important to stress that 2 equiv. of dialkylzinc are required for this reaction, since compound **I** does not react with aldehyde; the addition of the second dialkylzinc molecule forming **II** is necessary for an alkyl transfer to the carbonyl group. The enantioselectivity of the reaction depends on the presence or absence of steric repulsions between zinc's alkyl group and the aryl moiety of an aldehyde (**III** or **IV**, respectively).

In 1989 Yoshioka, Ohno and Kobayashi showed the effectiveness of bis(sulphonamides) as ligands in the dialkylzinc addition to aldehydes in the presence of titanium tetrakisopropoxide [36]. They used a series of diaminocyclohexane-derived sulphonamides and the best results were obtained for triflic acid derivative **L3** (Scheme 2).

The originally postulated mechanism of the reaction of (bis)sulphonamides in the presence of $\text{Ti}(\text{OiPr})_4$ suggests an exchange reaction leading to complex **VII** [36]. Walsh proved that **L3** does not react with even a 5-fold excess of $\text{Ti}(\text{OiPr})_4$ [37,38], and the active complex **VII** is formed through (bis)sulphonamide zinc species, as proposed earlier by Denmark [39].

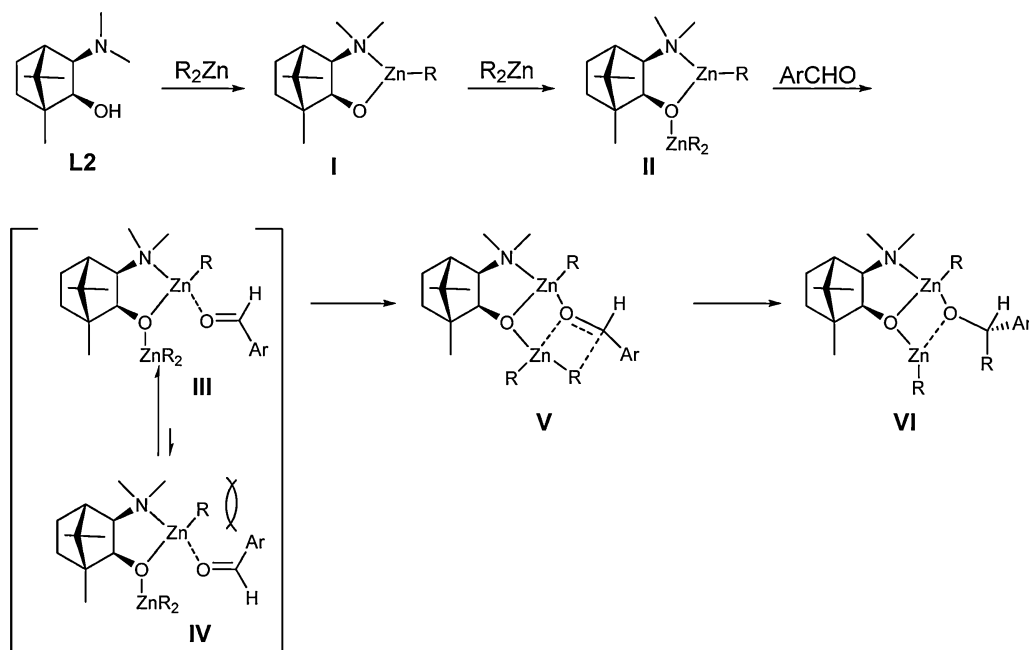
The pool of ligands used for reactions performed in the presence of titanium tetrakisopropoxide was soon extended to chiral *N*-sulphonamido alcohols [40], as well as TADDOLs [41], BINOLs [42–44] and their derivatives [45,46]. All these classes of compounds found broad use as ligands, and as Professor Seebach described it [41], the dialkylzinc addition “[became] a sport in organic synthesis, generating a number of unusual achievements.”

The diol- and sulphonamide-catalysed additions of dialkylzinc to aldehydes in the absence of titanium tetrakisopropoxide proceed with low enantioselectivities and rates; therefore, the mechanisms of titanium-promoted reactions were investigated thoroughly by research groups of Seebach [46], Walsh [47–50], Yus [51] and Gau [52], giving deep insight into the reaction's pathway. Seebach and Walsh concentrated on TADDOLs and BINOLs, respectively, while Yus and Gau studied the addition in the presence of β -hydroxy sulphonamides. The mechanism of titanium-promoted reactions, which emerged from very carefully designed studies of Walsh and Gau, is very similar for diols (such as TADDOLs and BINOLs) and β -hydroxy sulphonamides; its slightly simplified version is shown in Scheme 3.

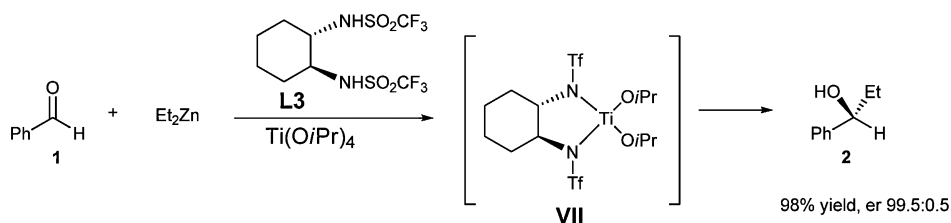
The dimeric complex **VIII** is created initially but it has no catalytic activity and, in the presence of the excess of titanium tetrakisopropoxide, is transformed into binuclear L^*Ti_2 complex **IX**. The role of the dialkylzinc reagent is to transfer the alkyl group to titanium and to form complex **X**. Dialkylzinc is incapable of delivering the alkyl group to **IX**, however it reacts with $\text{Ti}(\text{OiPr})_4$, giving alkyltitanium compound $\text{R}^*\text{Ti}(\text{OiPr})_3$, which transfers R^* to **IX**, yielding complex **X**. The more Lewis acidic titanium atom coordinates aldehyde, while the other one delivers the alkyl group to the carbonyl moiety. The addition proceeds with simultaneous transfer of the aryl aldehyde oxygen to adjacent titanium atom. The excess of $\text{Ti}(\text{OiPr})_4$ now reacts with **XII**, thus forming alkoxide of the newly formed chiral alcohol and regenerating complex **VIII** or **IX**. The alkoxide is incapable of interacting with chiral complexes; therefore, the possibility of the background autoinduction reaction altering the enantioselectivity of the process is minimised [49].

The addition reactions preformed in the absence or presence of a Lewis acid, most often $\text{Ti}(\text{OiPr})_4$, are usually catalysed by different groups of ligands and have a different mechanisms. Therefore, these reactions are discussed in separate chapters of this review.

Due to their high functional group tolerance, diorganozincs are excellent reagents for organic synthesis. To make them useful, easy access to diorganozinc other than the simplest dimethyl- and diethylzinc that are commercially available is required. While there are several methods for the synthesis of such compounds, the



Scheme 1. The mechanism of the addition and the stereochemical model of the dialkylzinc addition.



Scheme 2. (Bis)sulphonamide-catalysed diethylzinc addition in the presence of $\text{Ti}(\text{O}i\text{Pr})_4$.

purification of products is difficult and requires specialist equipment, as well as having to dealing with dangerous pyrophoric diorganozincs. The use of non-purified solutions of zinc reagents has a detrimental effect on the selectivity of the asymmetric synthesis, because lithium or magnesium salts, which are usual by-products of the synthesis, are also Lewis acidic and substantially diminish the enantioselectivity of the reaction [53].

Optically active allylic and propargylic alcohols as well as diaryl-methanols constitute a large class of very important precursors for the synthesis of biologically active compounds. Divinylzinc and diarylzinc reagents are readily obtained *via* mixing organolithium reagents with zinc chlorides. However, as mentioned above, the salts, being by-products of the synthesis, can greatly influence the organozinc addition reactions, resulting in a decreased enantioselectivity. Recent years have seen the arrival of new methodologies based on the use of dialkylzinc compounds, usually dimethyl- and diethylzinc, as precursors of alkynyl, aryl or alkenyl zinc compounds containing non-transferable, or more precisely, less-transferable methyl (ethyl) group. Such an approach allows for access to various organozinc reagents which are unavailable or otherwise very difficult to obtain.

2. Alkynylation

Optically active propargylic alcohols are versatile building blocks for asymmetric synthesis. Several examples of total syntheses based on the chiral propargylic alcohols are presented later in this review.

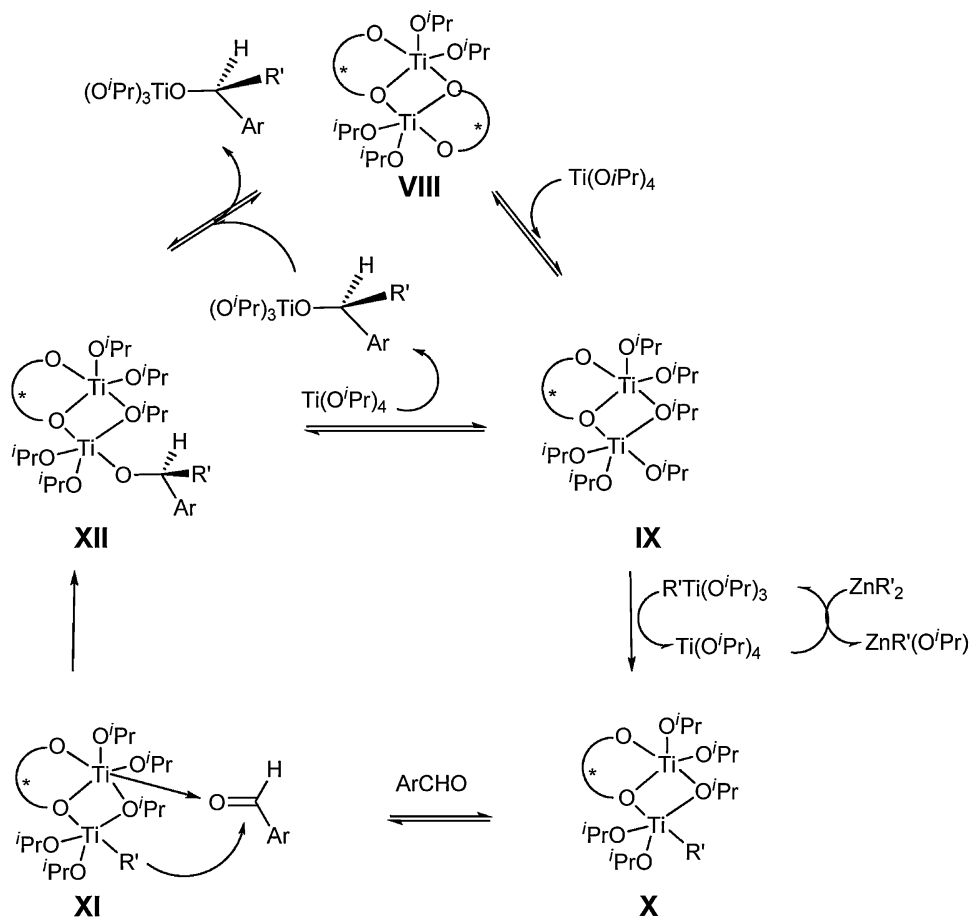
The first enantioselective, organometallic addition of alkyne to aldehyde was reported by Mukaiyama [54]. The addition of lithium acetylide derived from trimethylsilylacetylene to benzaldehyde in the presence of ligand **L1** at -123°C led to propargylic alcohol in 92% ee.

Low chemoselectivity of lithium acetylides prompted researchers to seek another approach. It soon became obvious that zinc acetylides were an excellent choice due to the very high tolerance of various functional groups. Two efficient methodologies have been developed. The formation of zinc acetylide could be achieved by the reaction with $\text{Zn}(\text{OTf})_2$, which was originated by Carreira [55–60], or by the reaction with commercially available diethyl- or dimethylzinc. The latter method is discussed in detail in this review.

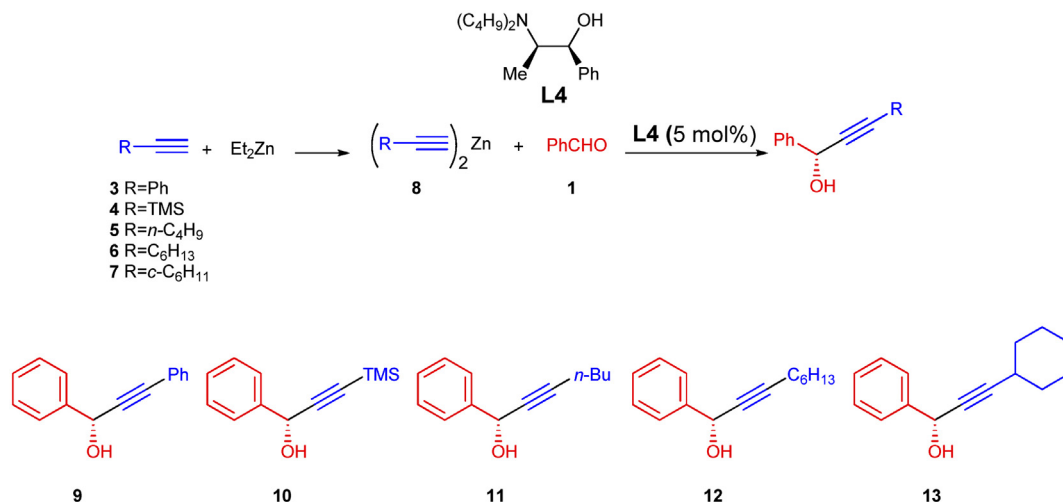
2.1. Alkynylation in the absence of Lewis acid additives

2.1.1. Addition to aldehydes

The first enantioselective addition of a zinc-based alkyne nucleophile to aldehyde was reported by Niwa and Soai [61]. Two methods were used for the preparation of alkynylzinc reagent. First, alkynes **3–7** were heated to 70°C with 2 equiv. of diethylzinc to yield (bis)alkynylzincs **8**, subsequently reacted with benzaldehyde **1** in the presence of 5 mol% of amino alcohol **L4**. The reaction furnished propargylic alcohols **9–13** in very good yields up to 99% (Scheme 4), but with a low enantioselectivity (<35% ee). In the second method, equimolar amounts of dialkylzinc and alkyne were reacted to give alkylalkynylzinc. The addition to aldehydes in the presence of **L4** resulted in a slightly better enantioselectivity (39–40% ee).

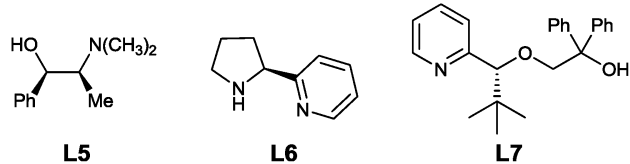


Scheme 3. The mechanism of the titanium-promoted addition.



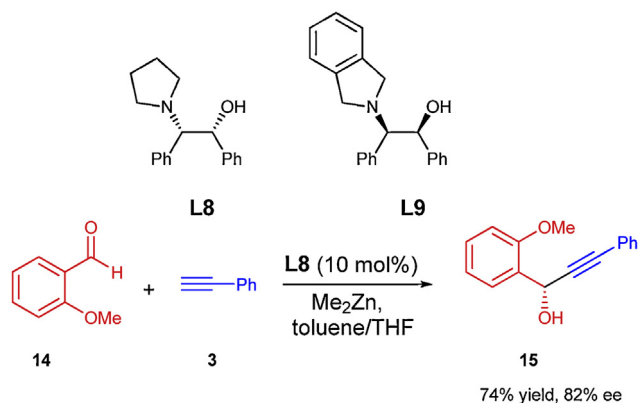
Scheme 4. Enantioselective alkynylation with bisalkynylzincs.

The following decade brought no substantial progress to this area. *N*-Methylephedrine **L5** [62] as well as ligands **L6** [63] and **L7** [64] gave addition products with moderate ee.



In 1999 Li and co-workers reported the use of amino alcohols **L8** and **L9** (Scheme 5). The addition of phenylacetylene to benzaldehyde and a group of 2-substituted aromatic aldehydes was investigated. Better results were observed for 2-substituted benzaldehydes; the best ones were 2-fluoro and 2-methoxy benzaldehydes [65]. The reaction was strongly accelerated by a ligand [66].

The catalysed reaction was complete after 3 h at 0 °C while, in the absence of the ligand, only 30% conversion was observed after



Scheme 5. Dimethylzinc-mediated alkynylation catalysed by amino alcohols **L8** and **L9**.

17 h at 0 °C. ^{13}C NMR studies revealed a lack of reaction between dialkylzinc and acetylene in the absence of an amino alcohol ligand [66].

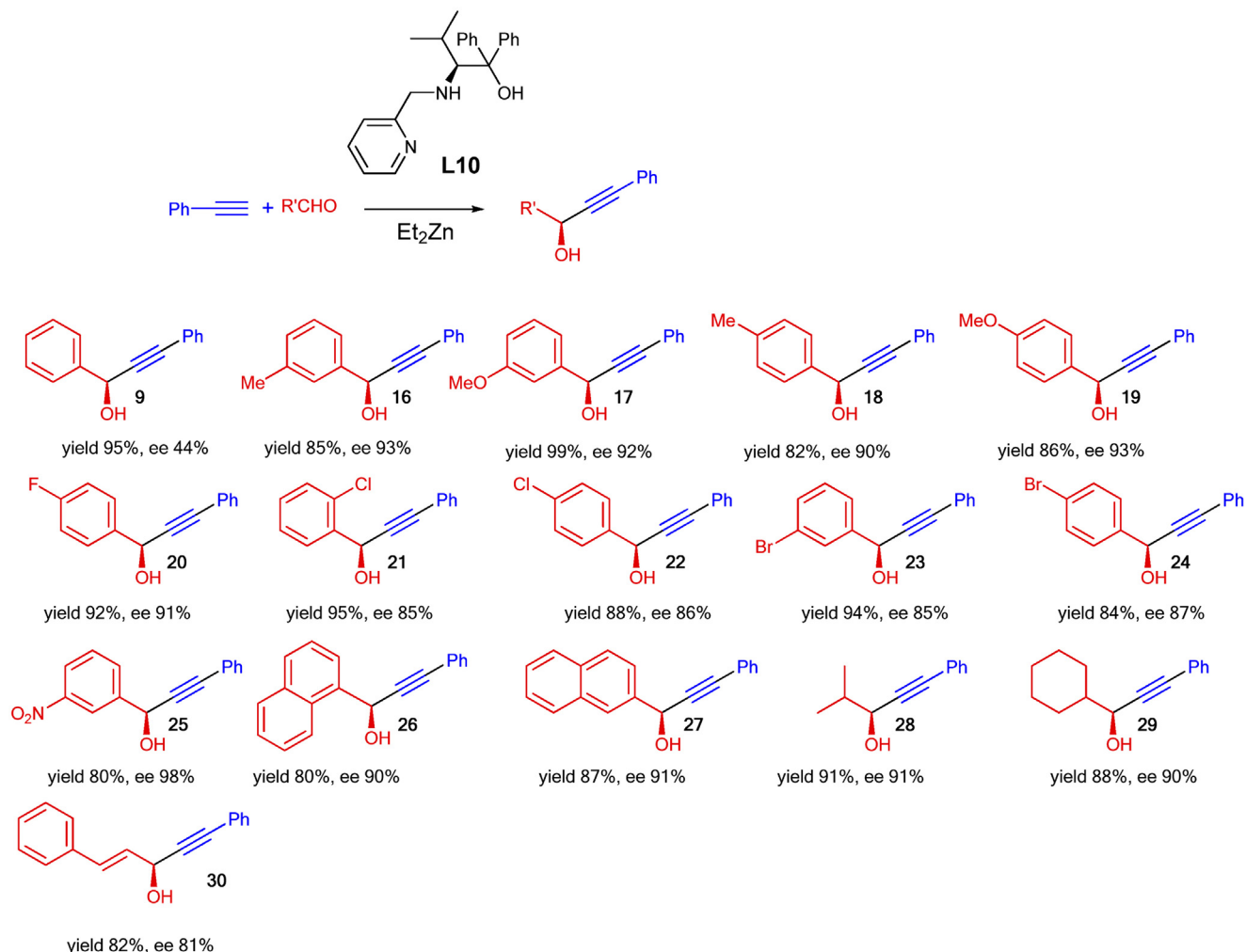
This experiment has been very important, because many reports (see below) claim synthesis of alkylalkynylzinc compounds simply by mixing an alkyne with a dialkylzinc.

Another interesting observation was the suppression of the methylation of aldehyde when the reaction was carried out in a 2.75:1 toluene:THF mixture.

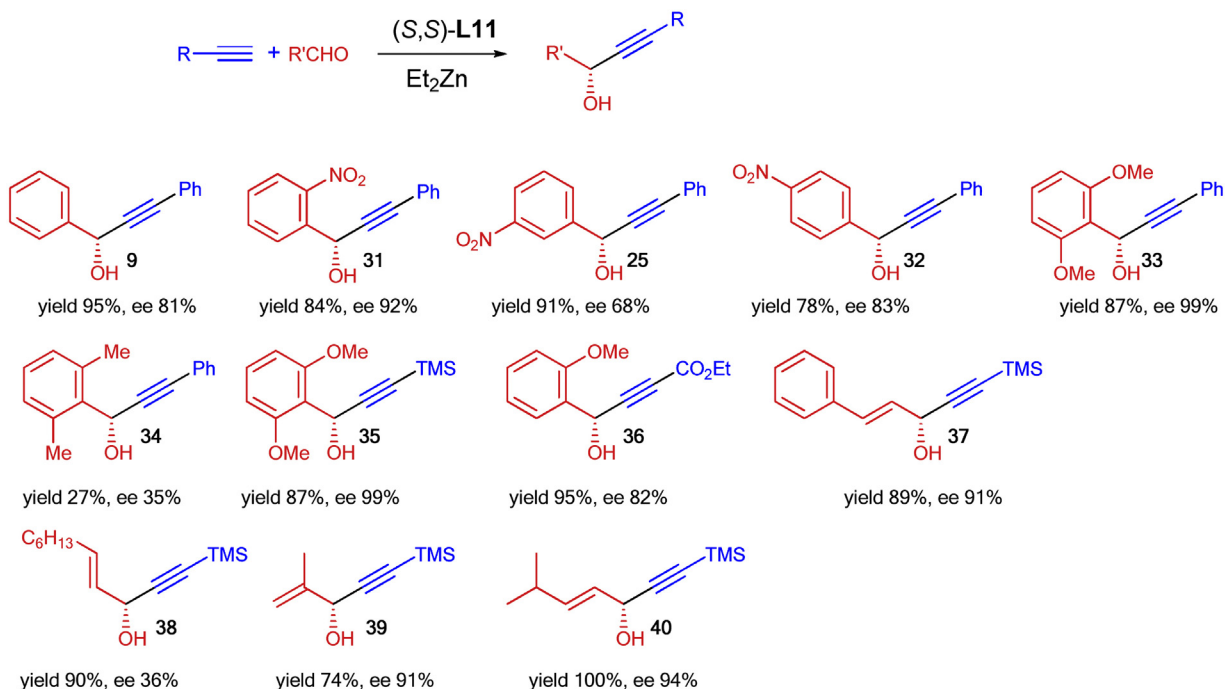
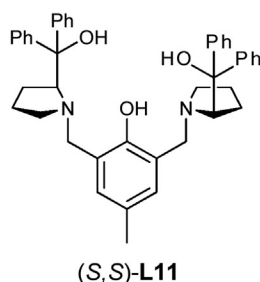
Wang et al. prepared the bifunctional ligand **L10** derived from L-valine [67]. The presence of pyridyl moiety and isopropyl side chain had a beneficial effect on the enantioselectivity. The replacement of the pyridyl ring with a phenyl one or the isopropyl side-chain with benzyl or isobutyl had a deteriorating effect on the enantiomeric excess. The level of ee was highly dependent on the solvent used, reaching its maximum (94%) at a 3:1 THF–hexane mixture. The results using ligand **L10** for the addition to aldehydes other than benzaldehyde were very good (81–93% ee) and only slightly dependent on the nature of the aldehyde (Scheme 6).

The influence of the pyridyl moiety on the enantiodiscrimination was explained by the transition state model **XIII** depicted in Fig. 1. This model was built on the basis of the model used earlier for the rationalisation of the enantioselectivity of the diethylzinc addition in the presence of ligand **L10** [68].

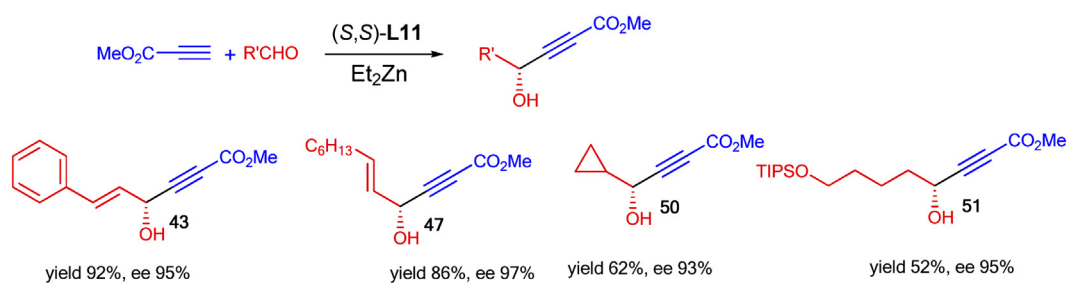
Trost examined his L-proline-derived ProPhenol ligand **L11**, previously used for several enantioselective reactions as aldol [69–73], nitroaldol [74], Mannich [75,76], and desymmetrization reactions [77], in the addition of various acetylene derivatives to aromatic, aliphatic and α,β -unsaturated aldehydes (Schemes 7 and 8) with a good to excellent enantioselectivity [78]. However, the best results were achieved for the addition to



Scheme 6. Asymmetric addition of phenylacetylene to aldehydes promoted by 20 mol% of ligand **L10**.



Scheme 7. Alkynylation of aldehydes catalysed by 10 mol% of ligand (S,S)-**L11**.



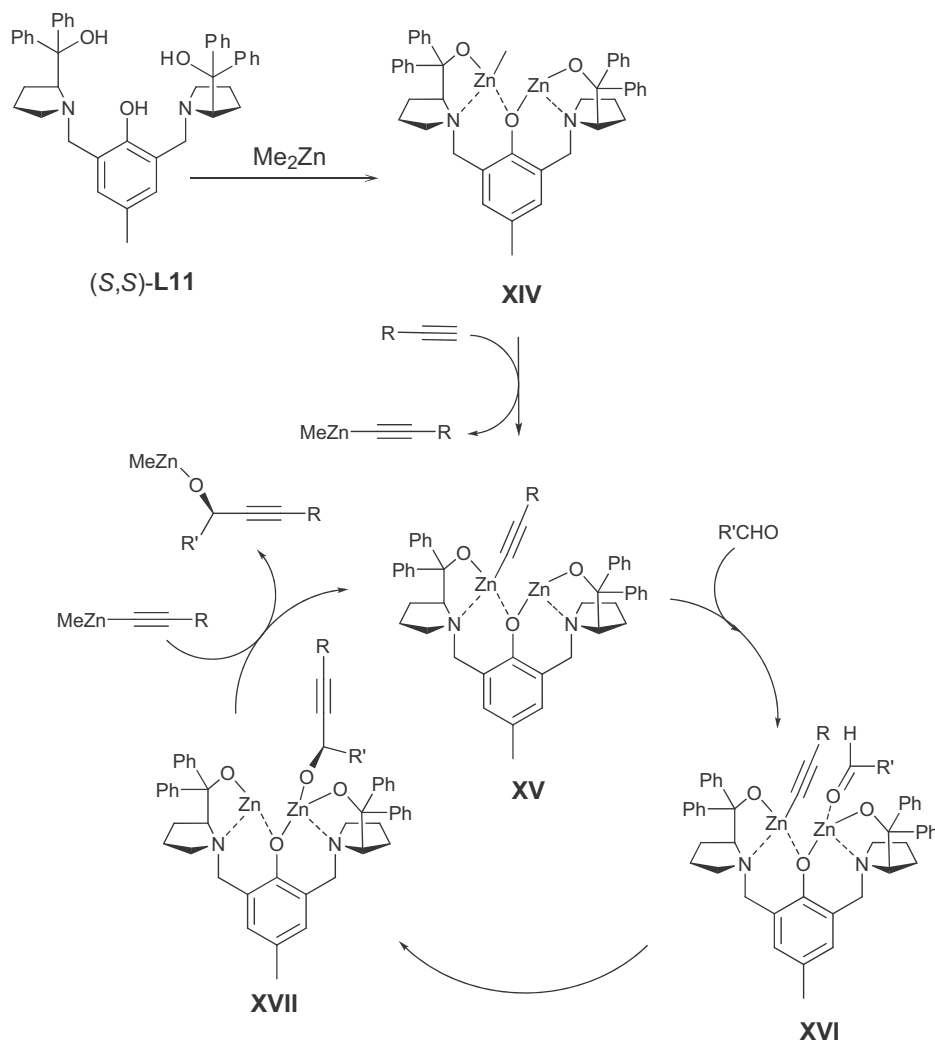
Scheme 8. Addition of methyl propiolate to α,β -unsaturated aldehydes and aliphatic aldehydes catalysed by 10 mol% of ligand (S,S)-**L11**.

bulky 2,6-dimethoxybenzaldehyde (Scheme 7, **33** and **35**). Interestingly, 2,6-dimethylbenzaldehyde reacted in very low yield and enantioselectivity (Scheme 7, **34**), supposedly due to an increased steric encumbrance. The ligand was available in both enantiomeric forms, as (S,S)-**L11** from L-proline and (R,R)-**L11** from D-proline.

The use of **L11** for the alkyne addition underwent continuous optimisation and the preliminary results were modified to include additional results of the addition of methyl propiolate to α,β -unsaturated aldehydes and very challenging enolizable aliphatic aldehydes (Scheme 8 and Supplementary materials – Scheme S1) [79].

The most recent mode of catalysis proposed by Trost and co-workers [79] involves the co-ordination of zinc acetylide and an aldehyde into a chiral pocket of ProPhenol ligand **L11**. Thus, the reaction of **L11** with 2 equiv. of dimethylzinc leads to dinuclear zinc complex **XIV**, followed by the formation of alkynylzinc complex **XV**. The nucleophilic attack of the acetylide on the Si-face of the aldehyde coordinated by the second, Lewis acidic, zinc atom (complex **XVI**) leads to the addition product, which is liberated from the complex **XVII** by metal exchange (Scheme 9).

¹H NMR studies confirmed again the crucial role of an amino alcohol ligand in the zinc acetylide formation. The deprotonation



Scheme 9. Model for the transition state in the presence of (S,S)-L11.

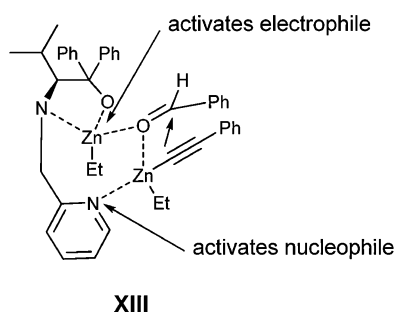


Fig. 1. Transition state model for the addition in the presence of ligand L10.

of methyl propiolate takes place only in the presence of ProPhenol. Moreover, NMR revealed that only 30% of zinc acetylide is formed during the premix step. However, the presence of an excessive amount of dimethylzinc does not lead to the methylation of the carbonyl substrate. In a separate experiment, the researchers proved that methylation in the presence of ProPhenol was a very slow process [79].

Trost developed the first very efficient protocol for the zinc-mediated addition of terminal 1,3-diyne to aldehydes catalysed by ProPhenol L11 [80]. Due to the hyperconjugation effect [81],

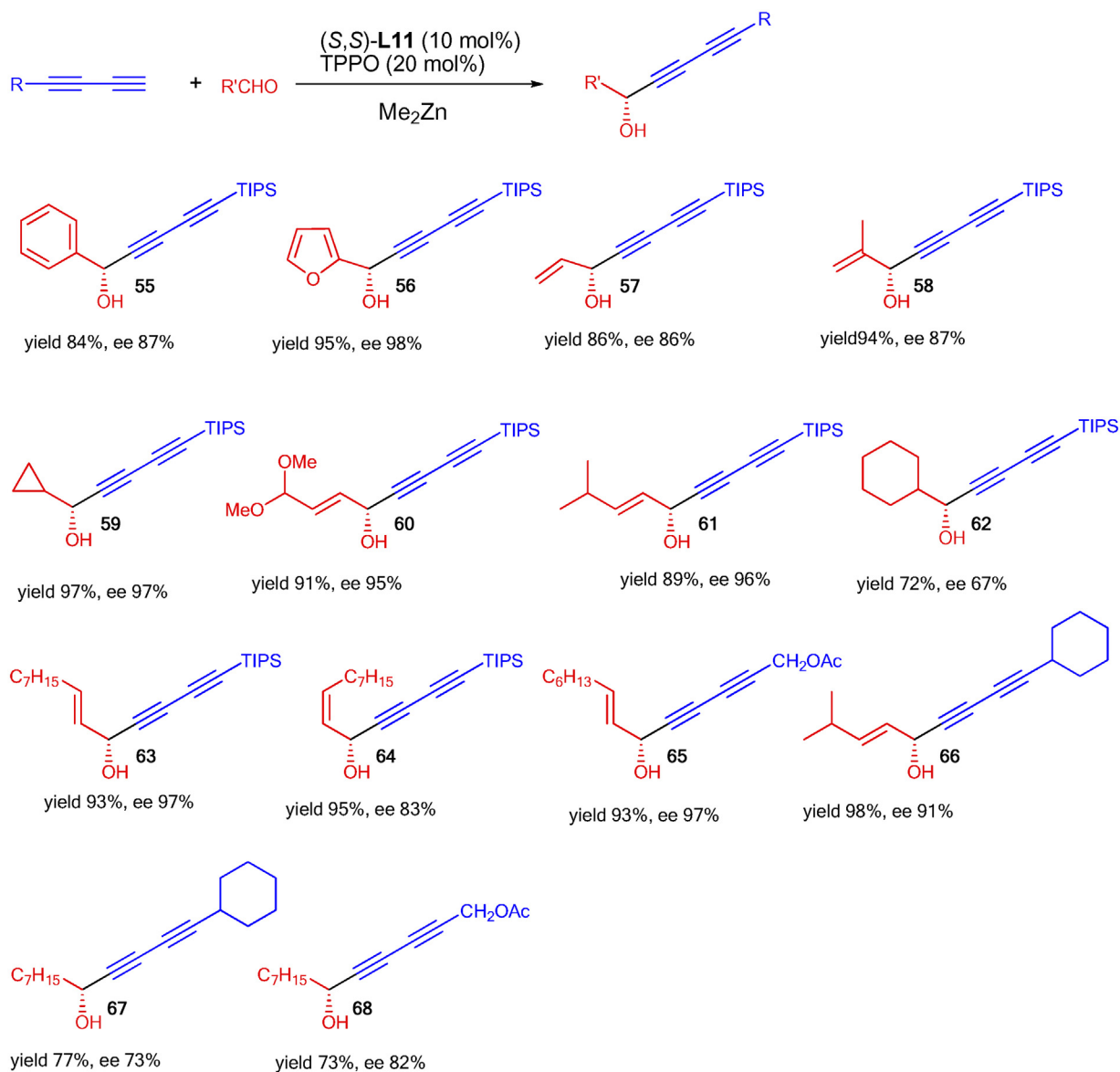
previously established methods for the addition of alkynes to aldehydes could not be used for diynes. The addition of triphenylphosphine oxide (TPPO) as a co-catalyst for this reaction had a beneficial effect on the enantioselectivity.

This new catalytic system gave very good to excellent enantiomeric excesses for various synthetically useful aldehydes and diynes. The selection of examples is shown in Scheme 10.

The proposed transition state XVIII suggested in the article describing the addition of diynes [80] is presented in Fig. 2. This model is supported by the X-ray structure [82], in which each of two zinc atoms is complexed by the THF molecule. However, recent studies on the addition to acetaldehyde have revealed that it is also very plausible that only one molecule of TPPO is bound to the ProPhenol–zinc complex [83].

The exact mode of action of triphenylphosphine oxide still has to be established, but its presence during the alkynylation has a very serious impact on the outcome of the reaction. It leads to a significant improvement of ee values (Table 1, entries 1 and 2) or even makes the addition possible, as in the case of silylated enynes (Table 1, entries 5–7).

The alkynylation of acetaldehyde, the simplest aliphatic aldehyde, is a very attractive goal due to potential synthetic applications of methylcarbinol bearing an alkyne substituent. It is also very challenging because of the self-aldolisation of acetaldehyde. This



Scheme 10. Asymmetric addition of diynes to aldehydes catalysed by (S,S)-L11-TPPO catalytic system.

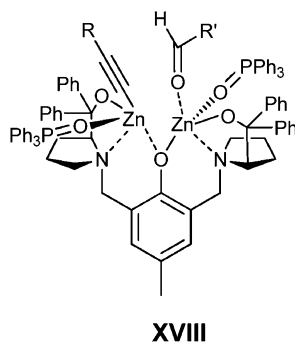


Fig. 2. Suggested transition state for the (S,S)-L11-TPPO catalytic system.

problem has been overcome by a slow addition of the aldehyde to the reaction mixture; good to excellent enantiomeric excesses have been obtained (Scheme 11).

The authors summarise some very important features of ProPhenol-catalysed alkynylations. First, because the alkynylation

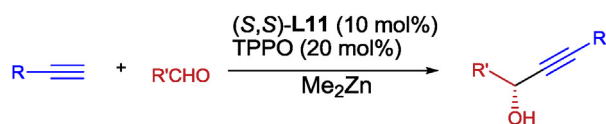
reaction is very fast, while alkylation in the presence of **L11** is much slower, side reactions are limited. Moreover, a slow addition of the aldehyde leads to a significant improvement of yield and enantioselectivity. It is suggested that at low concentration of the aldehyde, only one molecule is bound to the ligand. Thus, the number of diastereoisomeric transition states is reduced, leading to an improved enantiomeric excess. The evaluation of potential non-linear effects shows that the enantiomeric excess is independent of the enantiomeric purity of the ligand. This suggests that a single molecule of ProPhenol is involved in the transition state.

The usefulness of ProPhenol-catalysed additions has been illustrated by several total syntheses of natural products.

The addition of methyl propiolate to aldehyde was used for the total synthesis of asteriscunolide D (**85**) [84]. The asymmetric addition produced propargylic alcohol **83** with 84% enantiomeric excess. The enantiomeric purity of thioether **84** was improved to over 98% after recrystallisation (Scheme 12). Interestingly, the recrystallisation of the final compound **85** did not enhance its enantiomeric purity.

Table 1

Influence of the addition of triphenylphosphine oxide (TPPO) on yield and enantioselectivity.

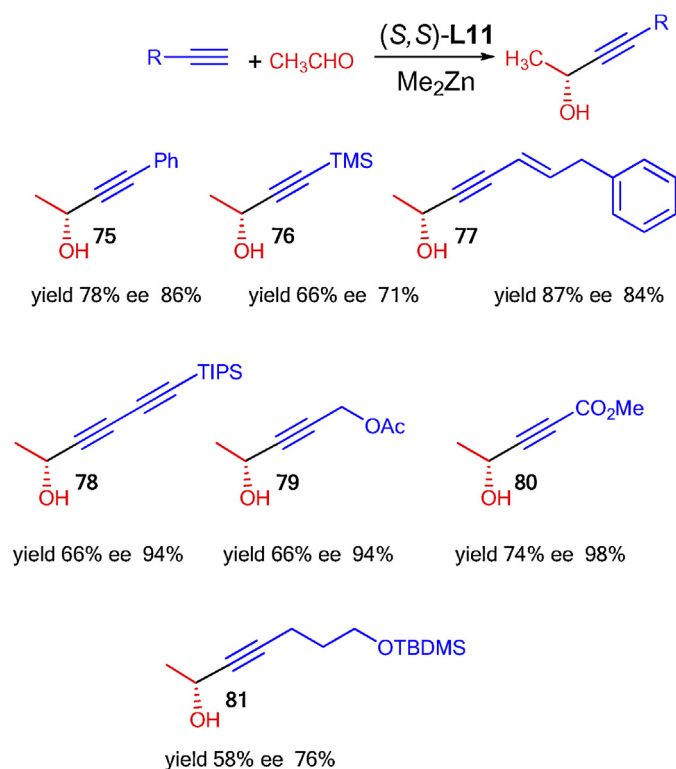


Entry	Product	No TPPO		With TPPO	
		Yield (%)	Ee (%)	Yield (%)	Ee (%)
1		100	77	79	93
2		90	36	79	94
3		71	90	85	95
4		75	85	83	81
5		–	–	85	91
6		–	–	56	66
7		–	–	58	74

Methodological studies have often obtained excellent results with simple aldehydes and alkynes. The same conditions often do not work for more elaborate starting materials required for total syntheses. Substrates needed for an efficient accomplishment of the synthesis give worse results than model compounds, and additional studies are needed on the reaction and careful evaluation of possible substrates.

A very interesting example of such a situation is the total synthesis of aspergillide B (**91**) [85]. Additions of saturated enolizable aldehydes needed for the next steps of this synthesis suffered from low yields caused by the competing aldol reaction. Sometimes this

can be overcome by using superstoichiometric amounts of alkyne, a more reactive alkyne such as methyl propiolate [84] or α,β -unsaturated aldehydes. In this case, the use of superstoichiometric amounts of chiral alkyne **86**, which was obtained in a multi-step synthesis, was economically inefficient. Among α,β -unsaturated aldehydes tested, only fumaraldehyde dimethylacetal **87** gave both the chemical yield and the enantioselectivity on an acceptable level due to the inductive effect of the dimethylacetal moiety. Moreover, a 1:1 mixture of the aldehyde and alkyne in the presence of 10 mol% of ProPhenol **L11** was sufficient to obtain adduct in 19:1 dr and 82% yield. Exactly the same result was obtained using racemic **86**,



Scheme 11. Alkynylation of acetaldehyde in the presence of 20 mol% of (S,S)-L11.

which indicates that the presence of the stereogenic centre far from the reacting triple bond does not influence the stereoselectivity of this reaction. The addition of methyl propiolate to aldehyde **89** was less selective, presumably due to the steric hindrance introduced to the closely located silyl and silyl ether substituents, or due to the acceleration of background reactions by Lewis basic oxygen atoms already present in the carbonyl partner of the reaction (Scheme 13).

Other natural products synthesised using this methodology (Fig. 3) are: adociacetylene **B92** (ee > 99%) [79], stronglylidiols A **93** (62% ee) and B **94** (87% ee), (3R,9R,10R)-panaxytriol **95** (86% ee), an unnamed natural product **96** (8% ee) [80], minquartynoic acid (**97**, 94% ee, formal synthesis), (+)-tetrahydropyrenophorol **98** (98% ee) [83] and (+)-spiroloxine methyl ether **99** (90% ee) [86].

Ligand **L12**, a simplified version of ligand **L11** containing only one molecule of proline and no additional co-ordinating functionalities, has recently been examined with limited success. The ee's did not exceed 85% for the addition of phenylacetylene to aryl aldehydes [87]. An *N*-methyl derivative of prolinol **L13**, the modified version of ligand **L12**, was used with better results for the addition of methyl propiolate to aromatic and aliphatic aldehydes using 1 equiv. of each alkyne and dialkylzinc. The reaction was performed without premixing acetylene with dialkylzinc; no competing alkylation reaction was observed. However, the enantioselectivity reached no more than 89–92%, while using as much as 30 mol% of ligand **L13** [88].

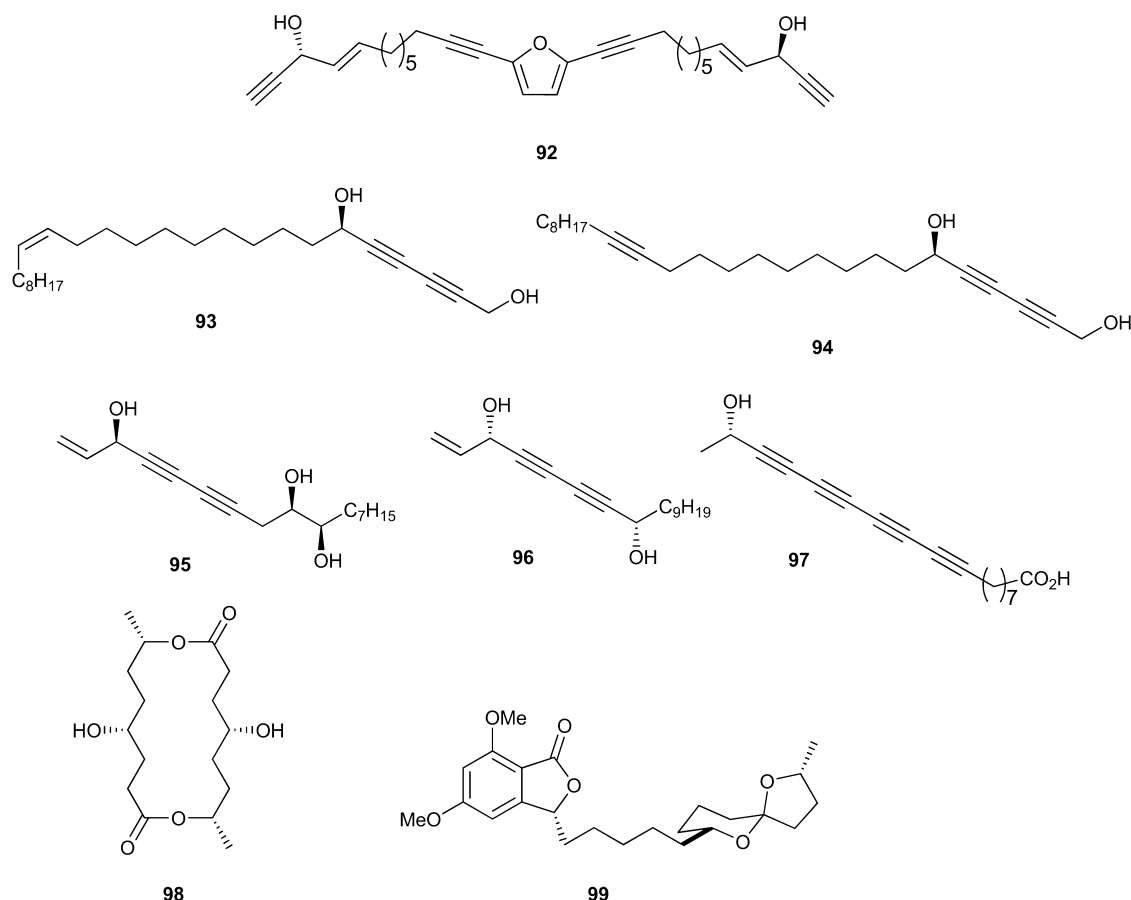
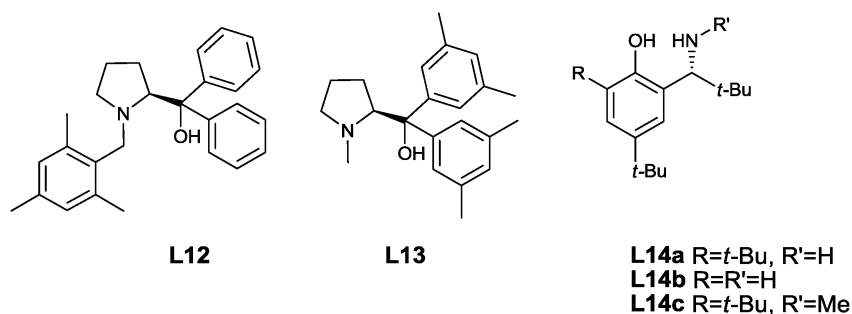


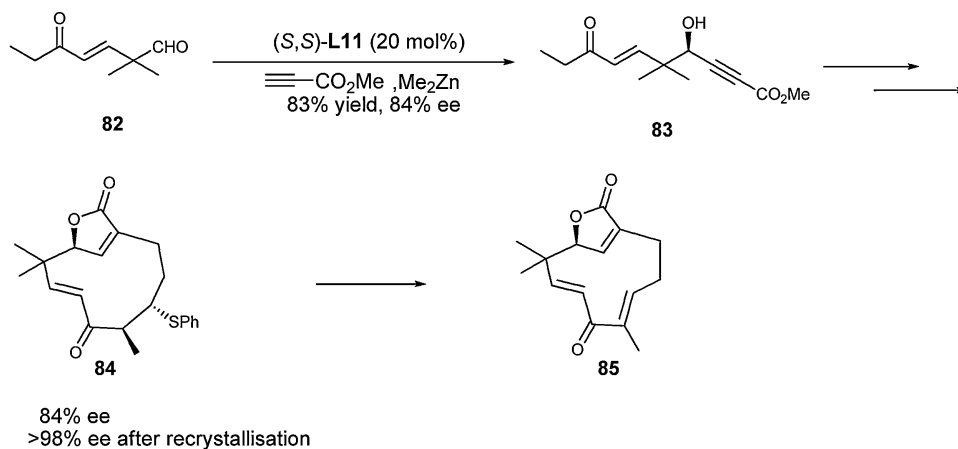
Fig. 3. Natural products obtained by L11-catalysed alkynylations.



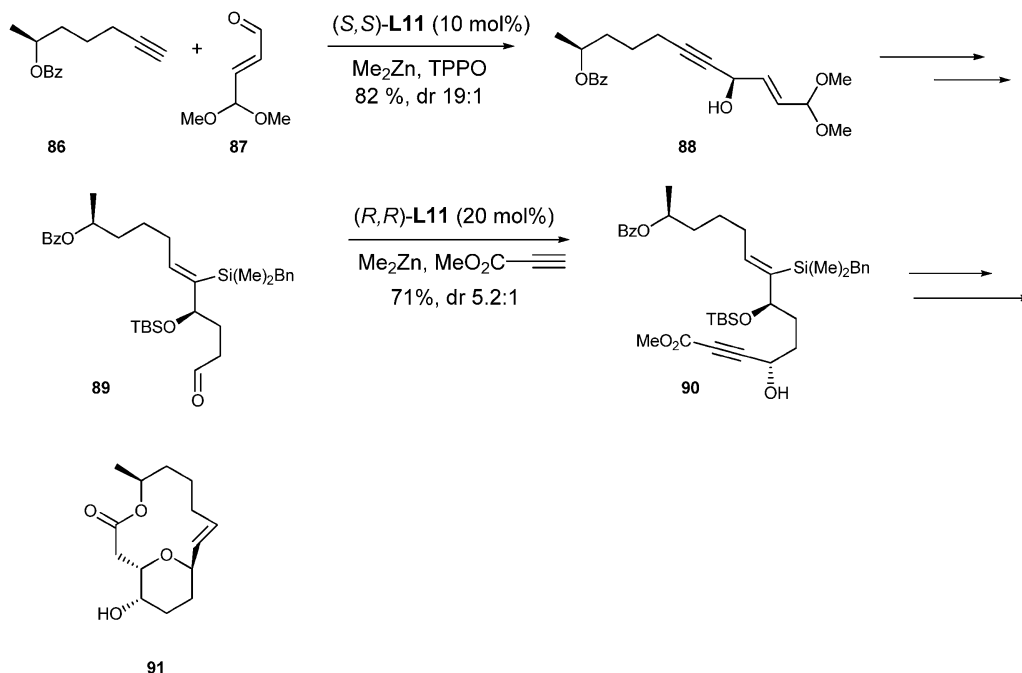
A very interesting 1,3-amino phenol ligand **L14a**, bearing a rarely used primary amino group was reported by Hirose and co-workers. This ligand performed especially well for some *meta*- and *para*-substituted aryl aldehydes providing ee's up to 99% for the addition of phenylacetylene to *m*- and *p*-nitrobenzaldehydes (Supplementary materials – Scheme S2, **25** and **32**) [89]. The enantioselectivity was increased by the addition of 10 mol% of proton sponge.

A good linear correlation of the enantioselectivity vs. Hammett constants σ was observed. The suggested transition states **XIX** and **XX**, based on Noyori's mechanism for diethylzinc addition, are presented in Fig. 4.

The alkynylzinc attack on the less sterically congested *Si* face of the aldehyde (transition state **XIX**) seemed to be the predominating one. According to the semi-empirical PM5



Scheme 12. Synthesis of asteriscunolide D (**85**).



Scheme 13. Synthesis of aspergillide D (**91**).

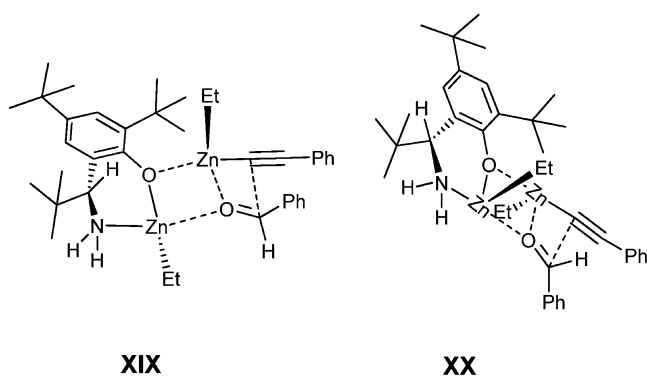


Fig. 4. Suggested transition states for reactions catalysed by **L14a**.

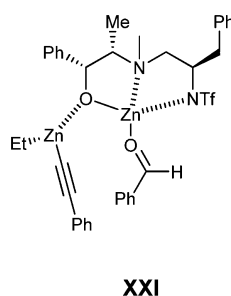
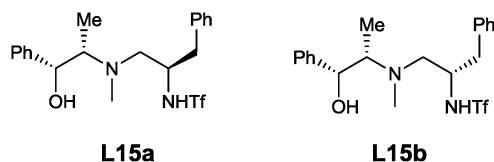


Fig. 5. Possible transition state for the addition catalysed by **L15a**.

calculations, this transition state **XIX** was more stable than **XX** by 14.05 kcal/mol.

The importance of *o*-*tert*-butyl and primary amino groups in the ligand structure was shown. Ligands **L14b** without an *o*-*tert*-butyl group and **L14c** with *N*-methyl amine instead of the primary one gave products with a very low enantiomeric excess.

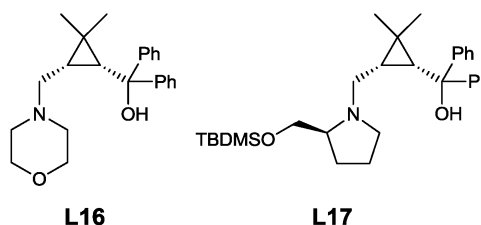


A series of sulphonamide-amino alcohols were prepared from the ephedrine and amino acid-derived amino alcohols. It was checked first in the addition of diethylzinc [90] and then in the reaction of phenylacetylene with benzaldehyde [91]. The *N*-trifluoromethanesulphonamide **L15a** appeared to be an efficient ligand inducing up to 92% ee. The diastereoisomeric ligand **L15b** was much less selective and the addition product to benzaldehyde was obtained with only 45% ee (Supplementary materials – Scheme S3). Interestingly, unlike most reactions catalysed by sulphonamide ligands (see below), this reaction was performed in the absence of titanium tetrakisopropoxide. For the addition of diethylzinc, a transition state **XXI** is proposed, where one of the zinc atoms forms a tridentate complex; the sulphonamide nitrogen serves as the third coordinating site (Fig. 5). This hypothesis could also be valid for the alkynylation reaction. However, no experimental data were provided to confirm it.

Chiral cyclopropane-based amino alcohol **L16**, previously successfully used for alkyl- and phenylzinc addition to aldehydes [92,93], was used for dimethylzinc-mediated addition of phenylacetylene and methyl propiolate to aldehydes with a very high enantiomeric excess [94]. Initially, the observed asymmetric induction for phenylacetylene addition to benzaldehyde was merely good (84% ee) but upon optimisation, it increased to 94% ee and 99% yield (this could be further improved to 96% ee, though

in low chemical yield). The presence of 10% of dimethyl ether of (poly)ethylene glycol of MW 2000, DiMPEG, highly influenced the outcome of the reaction possibly due to the suppression of a parallel uncatalysed reaction (for a more detailed discussion of the role of (poly)ethylene glycol, see arylation reactions in Section 3.1.1); the enantioselectivity of the substituted aromatic aldehydes was in the range of 92–94% (Supplementary materials – Scheme S4). The enantioselectivity was practically constant and independent of the nature of the aromatic aldehyde, though enantiomeric excesses below 90% were obtained for aliphatic aldehydes.

The same paper reported the addition of methyl propiolate to aldehydes. In this case, the beneficial effect of DiMPEG was not observed; moreover, the presence of this compound completely inhibited the addition of the nucleophile. A modification of the reaction conditions gave 90–93% ee for aromatic aldehydes, in moderate to good chemical yield (Supplementary materials – Scheme S4).



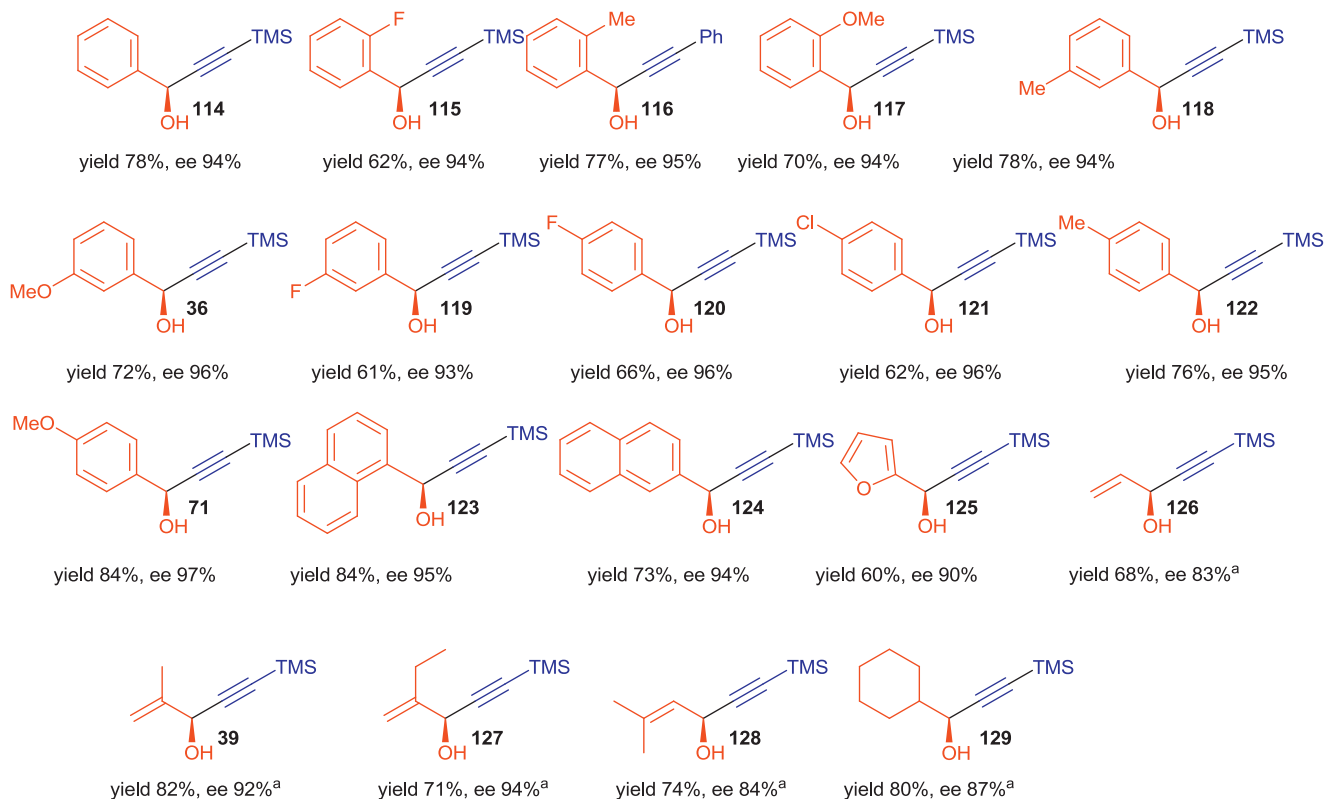
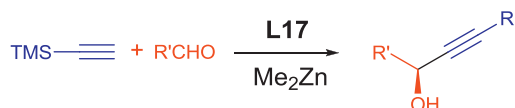
The structure of ligand **L16** was modified later [97a]. The replacement of the morpholinyl substituent by the (*S*)-prolinol-derived one gave new ligand **L17**, which was used for the addition of TMS-acetylene to aldehydes with good to excellent enantioselectivities. The presence of an additional stereogenic centre and its absolute configuration were proved to be crucial for a high asymmetric induction. All tested aromatic aldehydes, including 2-furaldehyde, reacted with at least 90% enantioselectivity (Scheme 14). Upon increasing the amount of ligand **L17** to 20 mol%, also α,β -unsaturated aldehydes gave adducts with enantioselectivities in the range of 76–94% ee (Scheme 14, **39** and **126–129**), where the highest values were obtained for α -branched aldehydes (Scheme 14, **39** and **127**).

The efficacy of this ligand was shown by a short total synthesis of marine alkynol **132**. The enantiomeric purity of the final product could be increased up to 97% by recrystallisation from hexane (Scheme 15). Unfortunately, this approach lacked generality. The total synthesis of (3*S*,8*S*)-falcariindiol (**135**), using the addition of TMS-acetylene to acrolein gave a final product with 83% ee, due to the oily nature of the final product.

Ligand **L17** was used later for the addition of phenylacetylene [97b]. Similarly to TMS-acetylene additions, all tested aryl aldehydes gave ee's in the range of 93–98%. Only cyclohexanecarbaldehyde and isobutyraldehyde furnished products with 84 and 88%, respectively.

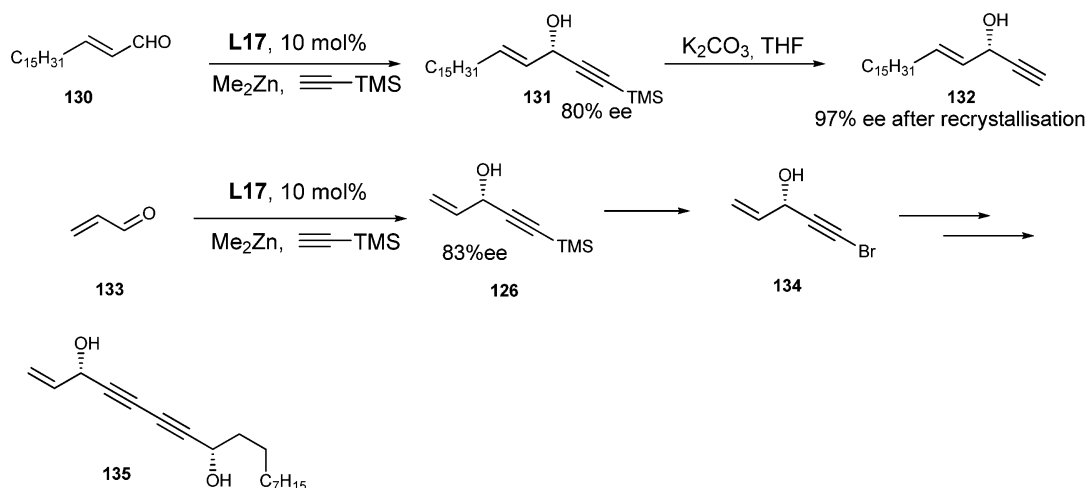
Ligand **L16** provided better enantioselectivities than **L17** for the additions of diynes to aldehydes [96]. Both buta-1,3-diynyltriisopropylsilane and penta-2,4-diynyl acetate reacted with aromatic aldehydes with very good enantioselectivities of over 90%; slightly lower values were obtained for α -branched α,β -unsaturated aldehydes. Almost all reaction with acetoxy-substituted diyne were slightly more selective (Scheme 16, $R = CH_2OAc$). Additions of phenyl, cyclohexenyl or hexyl substituted diynes were less selective, but their enantioselectivities still exceeded 80%. This was the first procedure for the addition of diynes to aldehydes, in which a high enantioselectivity was achieved in the absence of any additives. The reactions proceeded well in the presence of 10 mol% of the ligand, but 3 equiv. of Me_2Zn and diyne were required for efficient additions.

Ligands **L18** and **L19** contain the azetidino ring. Ligand **L18** bears the ferrocenyl moiety on the ring's nitrogen [97], while ligand **L19**

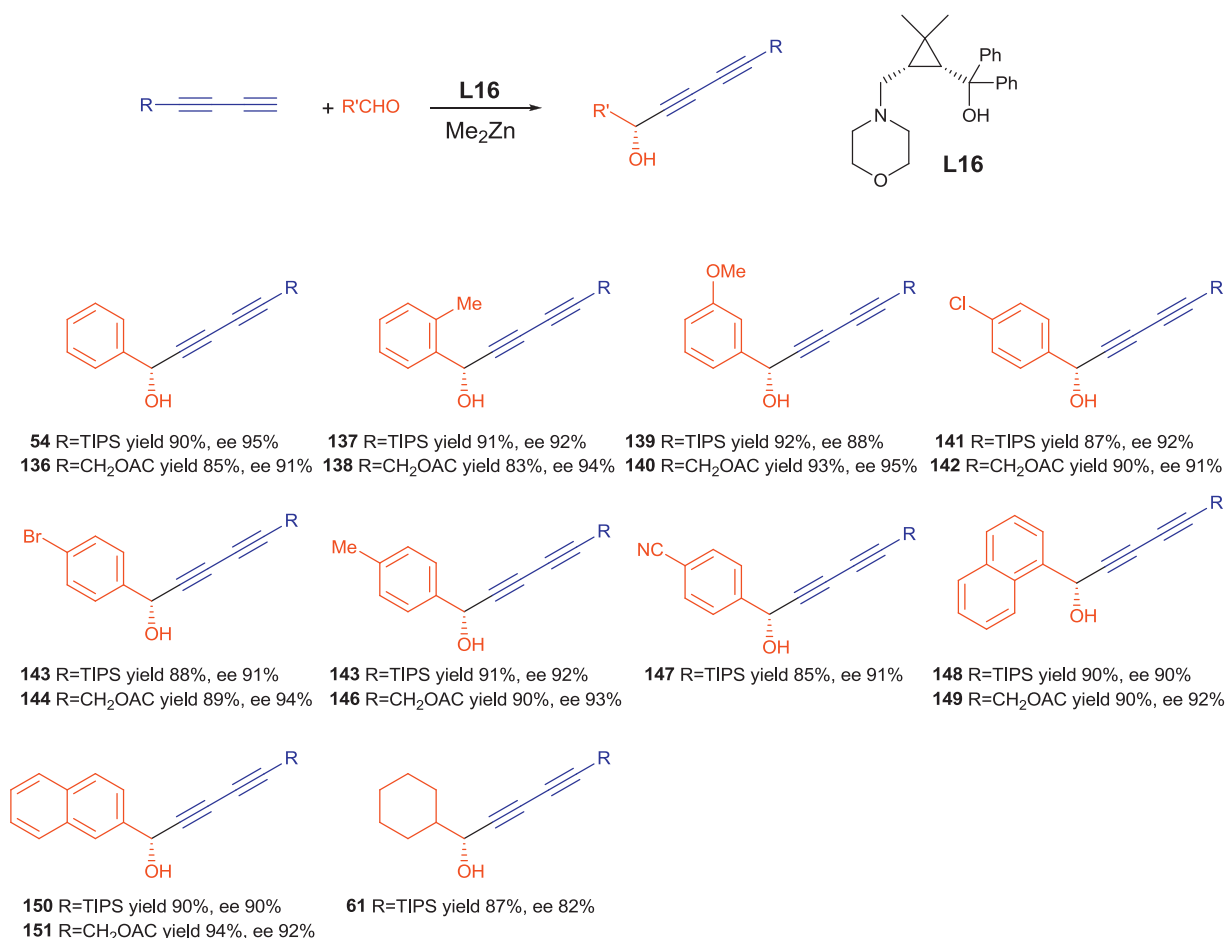


^a in the presence of 20 mol% of L17

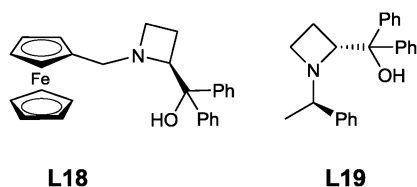
Scheme 14. Addition of TMS-acetylene to aldehydes in the presence of 10 mol% of ligand L17.



Scheme 15. Syntheses of marine alkynol **132** and (3*S*,8*S*)-falcariindiol **135**.

Scheme 16. Addition of diynes in the presence of 10 mol% of ligand **L16**.

has an additional stereogenic centre at this place [98]. The addition of phenylacetylene to aromatic aldehydes in the presence of 10 mol% of either ligand proceeded with moderate to good enantioselectivity and chemical yield, also in the presence of DiMPEG. However, even large-scale synthesis of both ligands required the use of preparative TLC during purification steps.



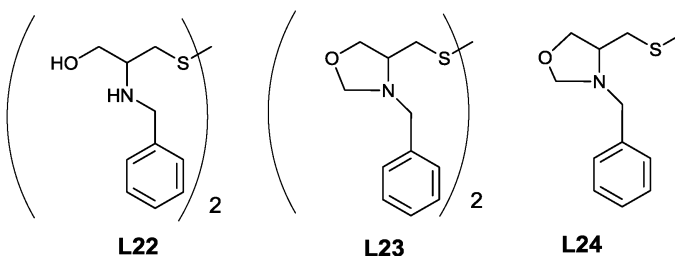
A very interesting ligand **L20** was synthesised by Wolf and Liu (Scheme 17) [99].

The (bis)oxazolidine ligand **L20** was effective in the addition of phenylacetylene to aromatic aldehydes and some alkyl- and cycloalkyl-substituted acetylenes to benzaldehyde; ee's over 90% were generally achieved. The addition of phenylacetylene to *tert*-butyl and cyclohexanecarbaldehyde was less selective and the observed enantioselectivity was close to 80%. Interestingly, high ee's were obtained for silyl-substituted acetylenes (Supplementary materials – Scheme S5).

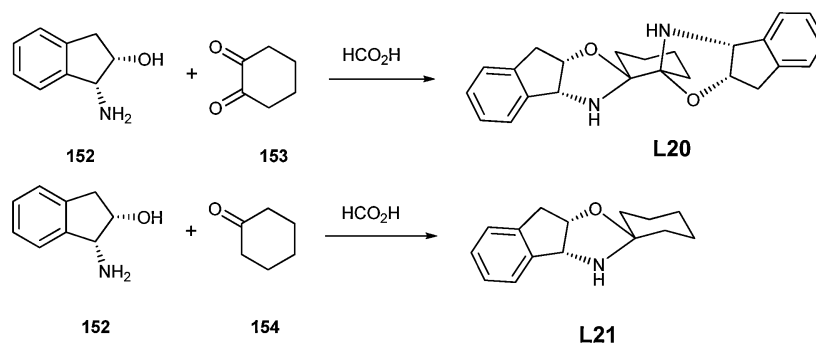
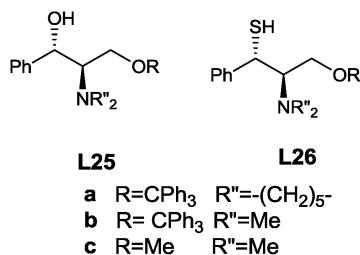
The possible transition state model for this reaction was not discussed, but the importance of C-2 symmetric construction of the ligand was proved experimentally. Compound **L21** showed low catalytic activity and an inferior enantioselectivity. An important feature of this system was the lack of the premixing step. All reagents were simply mixed together and after few minutes an

aldehyde was added. Under these conditions, the reaction gave the highest enantioselectivity and alkynylation:alkylation ratio.

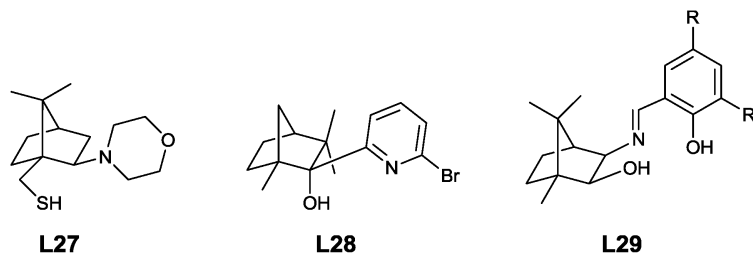
Sulphur-containing ligands could offer an interesting alternative for amino alcohols. While cysteine-derived sulphides **L22–L24** delivered products with low to moderate asymmetric induction [100], the thiol-containing ligands were more promising. Amino thiols could be better ligands than amino alcohols for organozinc addition to aldehydes [101]. Pericàs et al. presented direct comparison of otherwise structurally identical amino alcohols and amino thiols [102]. Ligands **L25** and **L26** were compared in the addition reaction of phenylacetylene to benzaldehyde; these results clearly showed superiority of amino thiols **L26**. However, the enantioselectivity still remained below 90% ee.



Ligand **L26a** gave good chemical yields and enantioselectivities around 80% for combinations of aryl-substituted acetylene with aryl aldehyde, except for sterically demanding 2,6-dimethoxybenzaldehyde (Supplementary materials – Scheme S6, **33**). In either reaction partner, the substituents other than aryl led to diminished yield and asymmetric induction.

Scheme 17. Synthesis of (bis)oxazolidines **L20** and **L21**.

Another amino thiol, 2-*exo*-morpholinoisobornane-10-thiol (MITH) **L27**, previously used as a ligand for the addition of various organozincs to aldehydes [103–105], was also applied for the alkynylation reaction [106]. The best results were obtained for mixed solvents toluene-THF, -20°C , and 2.5 mol% of ligand **L27**. This system was sensitive to substituents at the phenyl ring of aromatic aldehydes and the type of alkyne. The maximum 87% ee was obtained for the addition of phenylacetylene to benzaldehyde, but for all the other aldehydes and alkynes, the enantioselectivity did not reach this level (Supplementary materials – Scheme S7). However, the high activity of this ligand is remarkable – only 2.5 mol% is required for an efficient catalysis.



Terpene-derived amino alcohols were less efficient than amino thiol **L27**. A series of ligands derived from limonene and carvene were tested as ligands but the asymmetric induction was only low to moderate [107].

Pyridine-containing ligands based on (–)-fenchone, (–)-menthol, and (–)-8-phenylmenthol were screened for activity in phenylacetylene addition to 2-bromobenzaldehyde [108]. Only fenchone-based ligand **L28** gave products with a good enantioselectivity. After an optimisation of the reaction conditions, up to 85% ee was achieved for the reaction with 4-NO₂-benzaldehyde. Various α,β -unsaturated aldehydes and substituted acetylenes like ethyl propiolate and trimethylsilylacetylene were reacted under optimised conditions to give products with a moderate to good enantioselectivity (31–81%).

Tridentate camphor-based ligands **L29** was recently synthesised and examined [109]. The presence of alkyl substituents in positions 3 and 5 of the phenyl ring was disadvantageous and the best results were obtained for the ligand based on salicylaldehyde (R = H). Aromatic and heteroaromatic (**166**, **169**) propargylic alcohols were

synthesised with a high enantioselectivity (Supplementary materials – Scheme S8). The selectivity was only slightly dependent on the position and type of the aldehyde's phenyl ring substituent. Ligand **L29** was less efficient for α,β -unsaturated and alkyl aldehydes.

The authors clarified very carefully the misleading information available in existing literature on the relationship between the absolute configuration and the sign of rotation of two propargylic alcohols: 4-methyl-1-phenyl-pent-1-yn-1-ol and 4,4-dimethyl-1-phenyl-pent-1-yn-1-ol. A possible transition state **XXII** for this addition is presented in Fig. 6.

The paracyclophane-based imine ligand **L30** was examined by Dahmen [110]. After a tedious optimisation of the reaction conditions, the product of the addition of phenylacetylene to benzaldehyde was obtained in 90% yield with 92% ee (Supplementary materials – Scheme S9); for other aromatic aldehydes, the enantioselectivity was generally high – over 98% ee was achieved for 2-bromobenzaldehyde. A beneficial effect on the outcome of the reaction was noted for the additives, mono- and dimethyl ethers of (poly)ethylene glycol MW 2000, possibly due to suppressing the parallel uncatalysed reaction (for a more detailed discussion of the role of (poly)ethylene glycol, see arylation reactions in

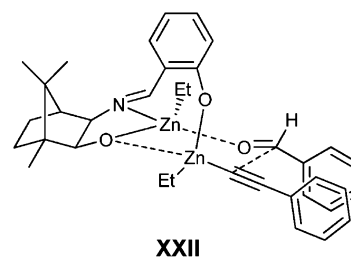
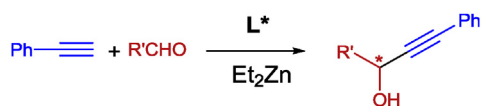
Fig. 6. Transition state for the addition catalysed by ligand **L29**.

Table 2
Comparison of alkynylation results in the presence of ligands **L36**, **L37** and **L38**.



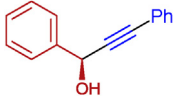
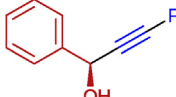
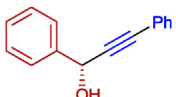
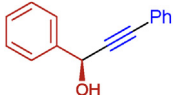
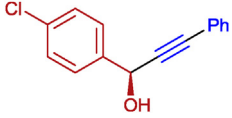
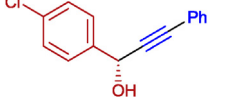
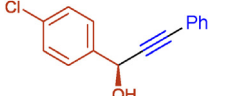
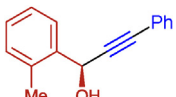
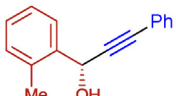
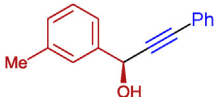
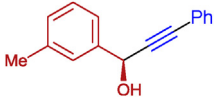
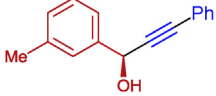
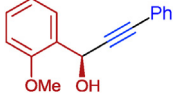
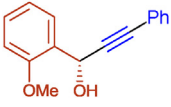
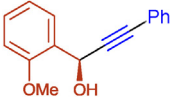
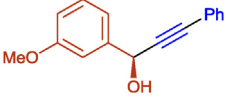
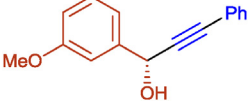
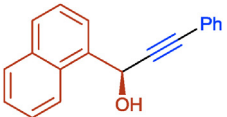
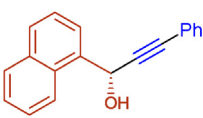
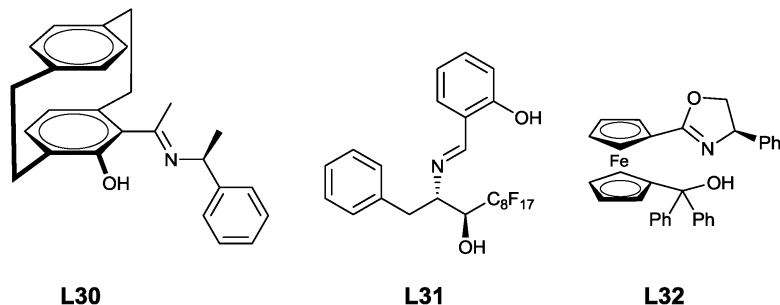
Entry	Propargylic alcohol	L* (mol%)	Temp.	Yield (%)	Ee (%) (conf.)
1		L36 (10)	RT	75	84 (S)
2		L36 (10)	0	48	92 (S)
3		L37 (30)	0	81	91 (R)
4		L38 (20) ^a	0	85	80 (S)
5		L36 (10)	RT	75	85 (S)
6		L37 (30)	0	75	92 (R)
7		L38 (20) ^a	0	75	57 (S)
8		L36 (10)	RT	72	84 (S)
9		L37 (30)	0	67	90 (R)
10		L36 (10)	RT	63	84 (S)
11		L37 (30)	0	71	90 (R)
12		L38 (20) ^a	0	76	70 (S)
13		L36 (10)	RT	64	91 (S)

Table 2 (Continued)

Entry	Propargylic alcohol	L* (mol%)	Temp.	Yield (%)	Ee (%) (conf.)
14		L37 (30)	0	76	87 (R)
15		L38 (20) ^a	0	76	86 (S)
16		L36 (10)	RT	71	85 (S)
17		L37 (30)	0	75	94 (R)
18		L36 (10)	RT	45	80 (S)
19		L37 (30)	0	70	90 (R)

^a In the presence of 2.0 equiv. of Et₃N.

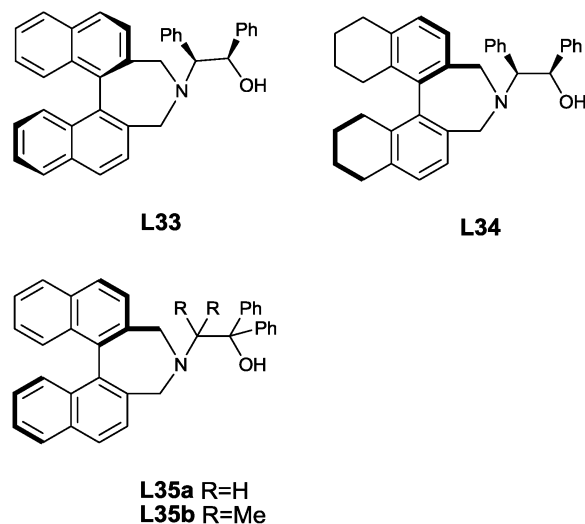
influence of the prolonged mixing step on the enantioselectivity of the reaction. This effect was explained by the formation of dialkynylzinc species reacting with a diminished selectivity.

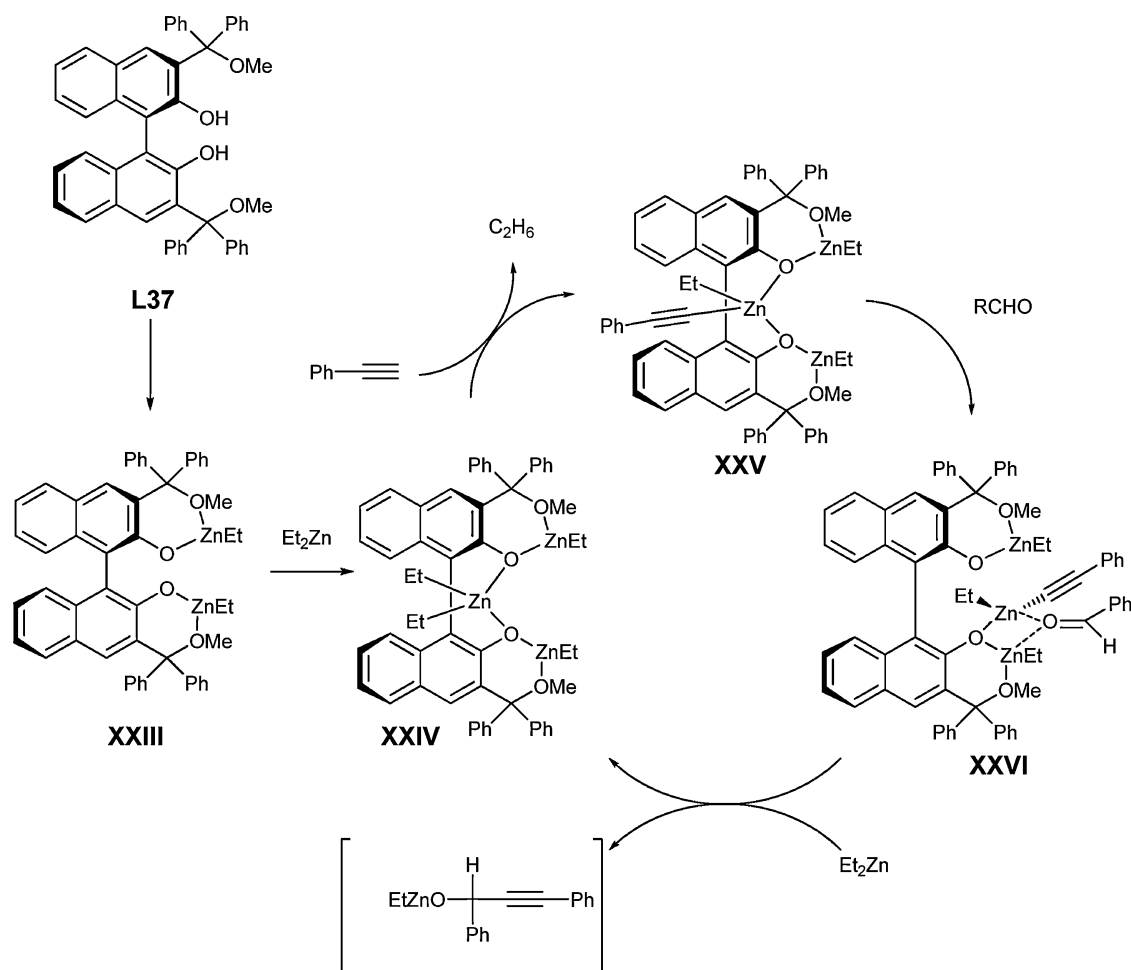


A series of ligands having C=N double bond moiety was reported [111]. An additional feature of these ligands, namely the perfluoroalkyl chain, was earlier successfully introduced into the addition of diethylzinc.

For the best ligand in this series, namely ligand **L31**, the products of the reaction of phenylacetylene with aldehydes were obtained in excellent chemical yields but only with a moderate enantioselectivity. The ferrocenyl ligand **L32** performed the best in the investigated group of structurally similar ligands. An enantioselectivity of over 80% ee was obtained for aromatic aldehydes; α,β -unsaturated and aliphatic aldehydes performed much worse [112]. This reaction could give higher ee's for increased amounts of the ligand, as indicated by experiments with 20 mol% of **L32** (Supplementary materials – Scheme S10).

The widely used BINOL and its derivatives were also applied for the alkylation reaction. Unsubstituted BINOL was successfully used in the titanium tetraisopropoxide-mediated reactions, while the substituted compounds were mostly used for reactions in the absence of a Lewis acid.



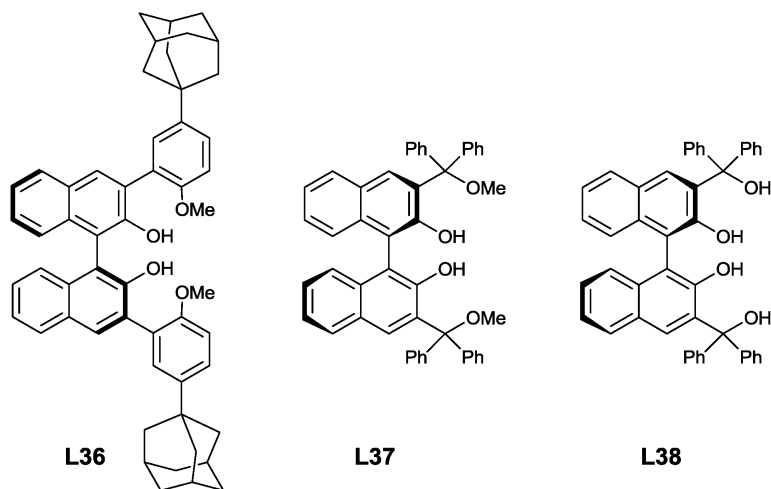


Scheme 18. Suggested mechanism for alkylation of aldehydes in the presence of **L37**.

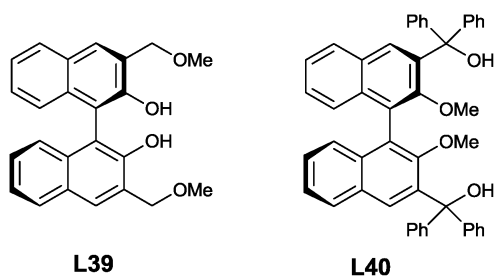
Chan et al. synthesised a series of diastereoisomeric binaphthoazepines; among them, ligand **L33** (10 mol%) performed the best. Additions to benzaldehyde, 2-methyl- and 2-methoxybenzaldehydes were only moderately enantioselective; 2-halo and 2-nitro derivatives reacted with ee's between 85 and 90% [113]. The structurally relevant H_8 -binaphthyl-based ligand **L34** gave exactly the same results as ligand **L33** [114], while ligands **L35** gave a worse selectivity not exceeding 70% ee [115].

Pu and co-workers examined two BINOL derivatives, **L36** [116] and **L37** [117], with bulky substituents at positions 3,3'. Ligand **L37** was synthesised in order to overcome drawbacks of ligand **L36** investigated earlier and gave much better results for most

aromatic aldehydes studied, but only when a high amount of the ligand (30 mol%) was used. As a result, the mass of the ligand was twice that of e.g. benzaldehyde. The reaction was strongly influenced by the solvent. The addition of Lewis bases caused a substantial increase in the asymmetric induction. The best results were obtained for ligand **L38** (20 mol%) in the presence of 2 equiv. of triethylamine; however, the enantioselectivity still remained below 90% and did not exceed that of ligand **L37**. An opposite configuration of the newly formed stereogenic centres observed in the reactions catalysed by **L37** and **L38** indicated completely different transition states for these reactions. The results obtained for ligands **L36**–**L38** are presented in Table 2.



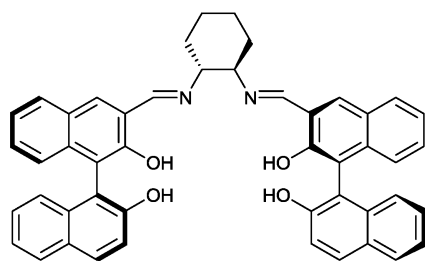
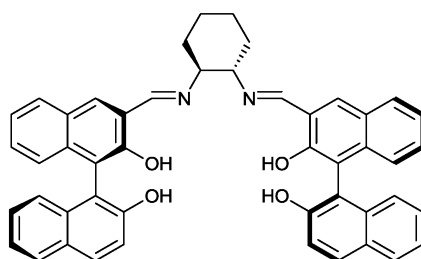
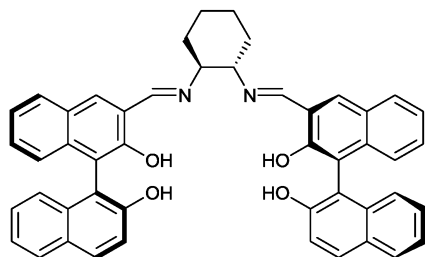
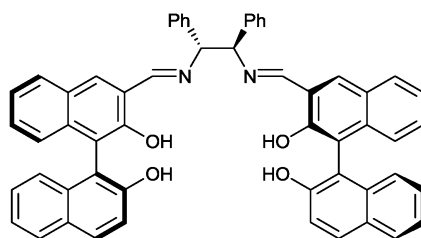
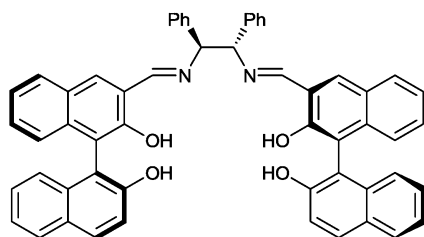
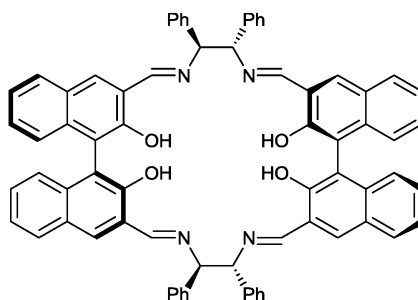
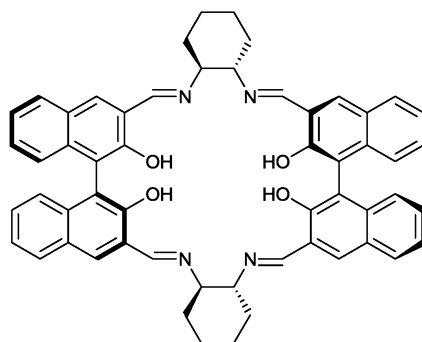
Modified ligands **L39** and **L40** were synthesised and used for the addition of phenylacetylene to benzaldehyde under analogous conditions [117]. The enantioselectivity was very low for **L39**, and the importance of steric congestion introduced by bulky diphenyl moieties present in 3,3'-bis(methoxymethyl) substituents in ligand **L37** was demonstrated. No reaction was observed for ligand **L40**. Probably a high distance between free hydroxy groups impeded the activity of the catalyst.



On the basis of the data collected for ligands **L36–L40**, the mechanism for the reaction in the presence of **L37** was proposed (Scheme 18).

Pu and co-workers prepared salen derivatives of BINOL and diamine (**L41–L47**) and tested them in the model reaction of phenylacetylene with benzaldehyde. Only ligand **L41** (15 mol%) gave an acceptable enantioselectivity (79% ee), while rest of the ligands gave results not exceeding 17% ee. An optimisation of the reaction conditions for **L41** led to a substantially increased enantiomeric excess – 92% ee was obtained for benzaldehyde at room temperature in toluene with Me_2Zn as a source of alkynylzinc [118]. Even better results were reported for the reaction of phenylacetylene with 2-methylprop-2-ene and 1-naphtaldehyde–94% and 97% ee, respectively (Supplementary materials – Scheme S11, **171** and **26**), and the reaction of 1-chloropent-4-yne with aromatic aldehydes – 95–96% ee (Supplementary materials – Scheme S11, **172–174**).

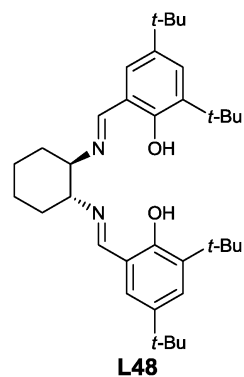
Macrocycles **L46** and **L47** were later used as ligands for the addition of aliphatic and vinyl aldehydes with excellent results [119]. Ligand **L47** (20 mol%) gave an enantioselectivity over 90% in most cases, but only phenylacetylene was used for this reaction (Supplementary materials – Scheme S12). An important drawback of this system was the high mass of the ligand itself, 841 Da. In the case of acroleine, 20 mol% of this ligand weighs 4 times more than the substrate used for the reaction.

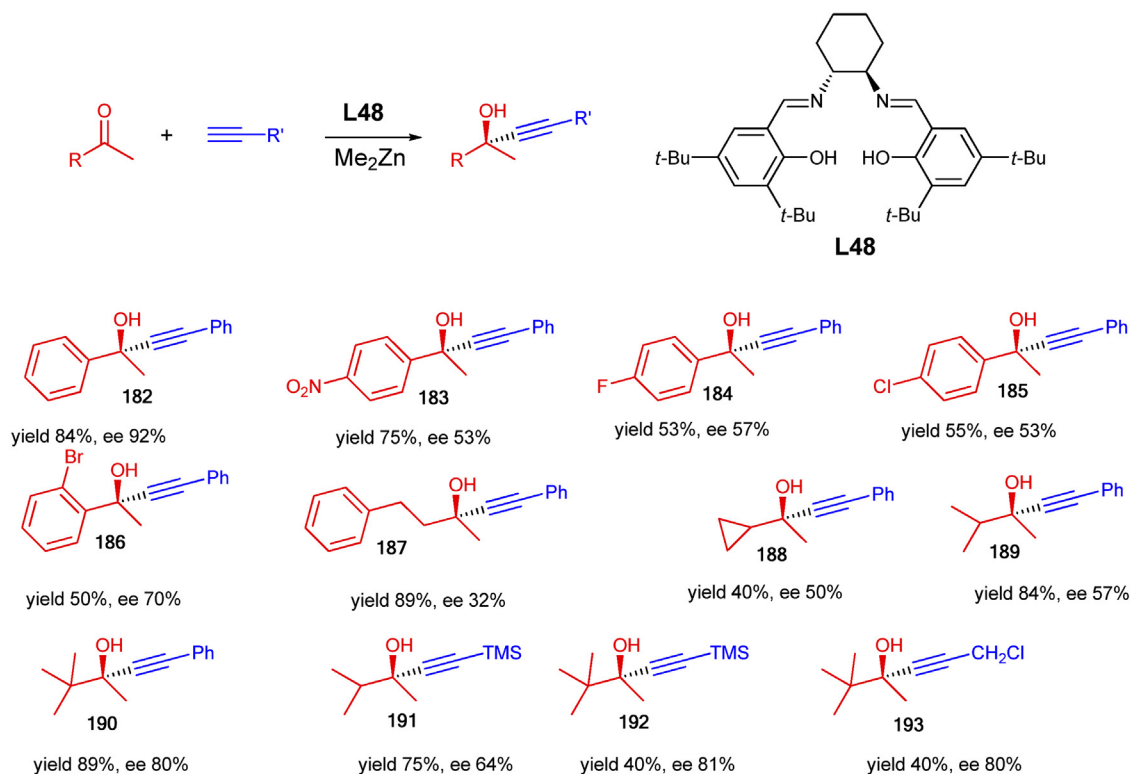
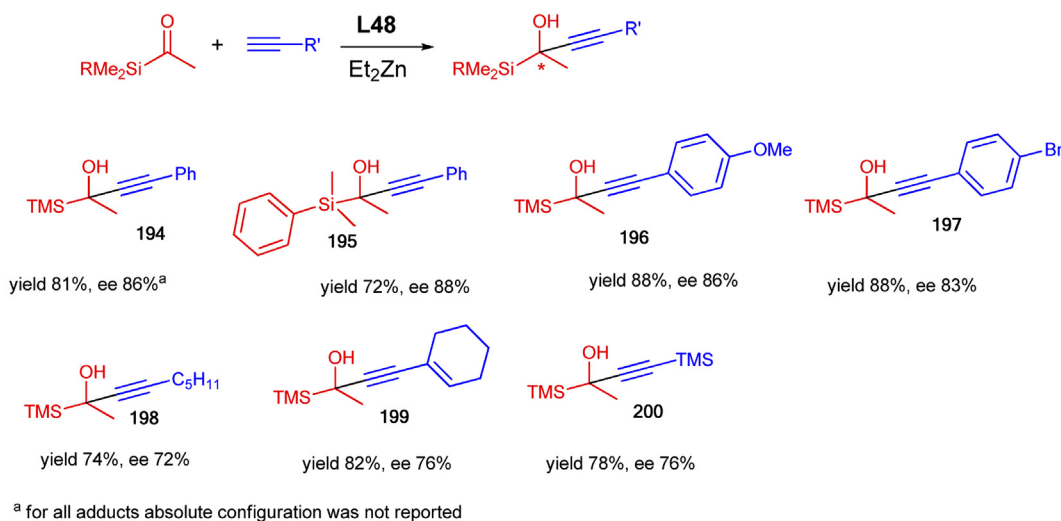
**L41****L42****L43****L44****L45****L46****L47**

2.1.2. Addition to ketones

The addition of terminal acetylenes to ketones is a valuable method for the synthesis of tertiary propargylic alcohols that are otherwise difficult to access.

The asymmetric alkynylation of ketones is much more difficult from the point of view of reactivity and enantioselectivity than the respective alkynylation of aldehydes. Cozzi introduced salen **L48** as the first efficient ligand for the addition of acetylenes to various aromatic and aliphatic ketones [120].

**L48**

Scheme 19. Alkynylation of methyl ketones in the presence of 20 mol% of salen **L48**.Scheme 20. Alkynylation of acylsilanes in the presence of 20 mol% of salen **L48**.

He reasoned that the zinc-salen complex could be an efficient bifunctional catalyst increasing the reactivity of ketones towards the attack of alkynes (Fig. 7).

In comparison to current standards, the enantioselectivities were only moderate but the ligand demonstrated a good functional group selectivity. More sterically hindered ketones reacted with a better selectivity (Scheme 19, **187** vs. **189** vs. **190**).

The linear correlation between ee and the enantiomeric purity of **L48** indicates that only one molecule of the ligand was involved in the transition state.

In the paper published in 2005, Cozzi proved that the ketone itself could act as a “ligand” for the dimethylzinc-promoted addition of alkynes to ketones [121]. The assumption that the coordination of the oxygen's lone pairs to dimethylzinc activated it and

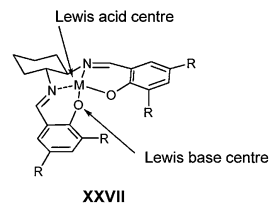
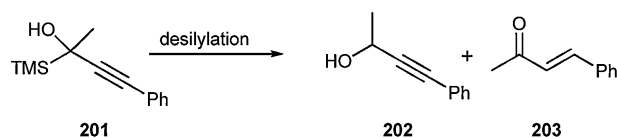


Fig. 7. Bifunctional salen catalyst.

allowed the formation of acetylide was confirmed by the calculations at the B3LYP/LACVP* level. These calculations revealed that energy barriers were indeed substantially lowered, facilitating the formation of zinc acetylide and its addition to the carbonyl group.



Scheme 21. Desilylation of 2-(trimethylsilyl)-4-phenyl-3-buten-2-ol **201**.

This study dealt only with the reactions in the achiral environment, but such a reaction could run in the background during an asymmetric synthesis, hence lowering the enantioselectivity.

Salen **L48** was used for the addition of acetylenes to acylsilanes [122]. Silylated tertiary propargylic alcohols were obtained with a good selectivity (Scheme 20).

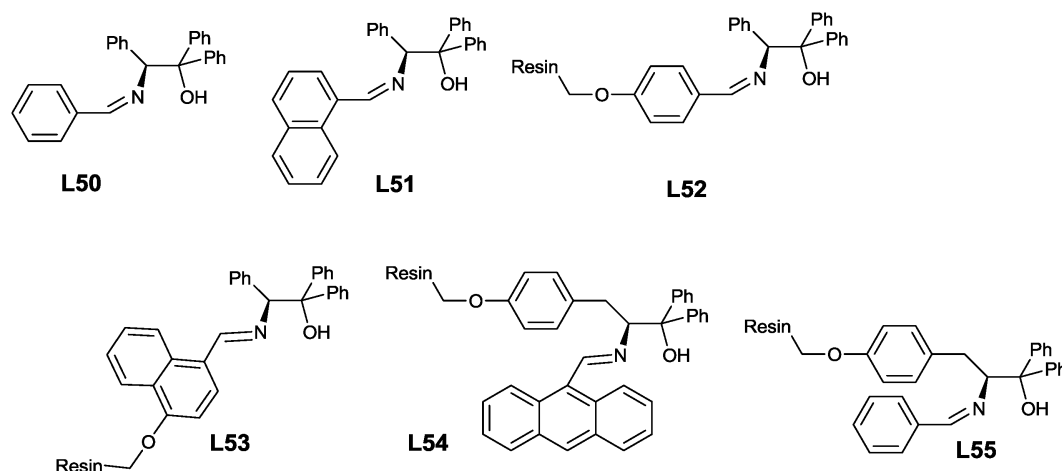
Acylsilanes could be treated as synthetic equivalent of aldehydes and after desilylation, addition products obtained in the above presented reaction could be equivalent to the products obtained in the reaction of the challenging acetaldehyde (Scheme 21).

Unfortunately, in the presence of TBAF, CaF_2 and K_2CO_3 , only the rearrangement product was obtained. NaOMe led to a 16:86 mixture of **202** and **203**. The use of KF led to the desired product **202**, although with racemisation.

A series of salen ligands with various diamine backbones and binaphthyl substituents were synthesised by Saito and Katsuki. The best results were obtained for ligand **L49** [123]. Again, the steric hindrance imposed by the ketone played a crucial role and enantioselectivities over 90% were obtained only for *tert*-butyl methyl ketone (Supplementary materials – Scheme S13).

Imino alcohol ligands **L50** and **L51** were introduced successfully by Wang [124]. Ligand **L51** was very efficient for aromatic ketones, giving high enantioselectivities at -18°C and a very low catalyst load. Only 1 mol% was enough to obtain 95% ee for 2'-naphthacetophenone, and 84% ee was obtained in the presence of as few as 0.1 mol% (Scheme 22, 213). The same activity was shown for the addition of 1-ethynylcyclohexene to aromatic and heteroaromatic ketones, but with a lower enantioselectivity (67–83% ee) [125].

On the basis of the above-described results, four polymer-supported ligands were prepared by grafting respective amino alcohols on a Merrifield resin and investigated [126]. The best results were obtained for 10 mol% of ligand **L55**. The enantioselectivity for aromatic ketones was lower than that obtained with ligand **L51** and did not exceed 90% (Scheme 22).



The catalyst could be reused up to 4 times without any loss of activity or selectivity. Further recycling led to a diminished enantioselectivity and the reactions became very slow.

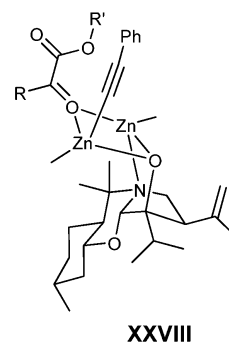
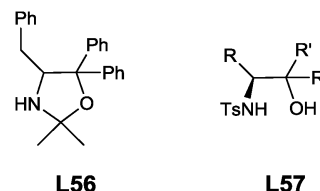


Fig. 8. Transition state for **L58a**-catalysed alkynylations.

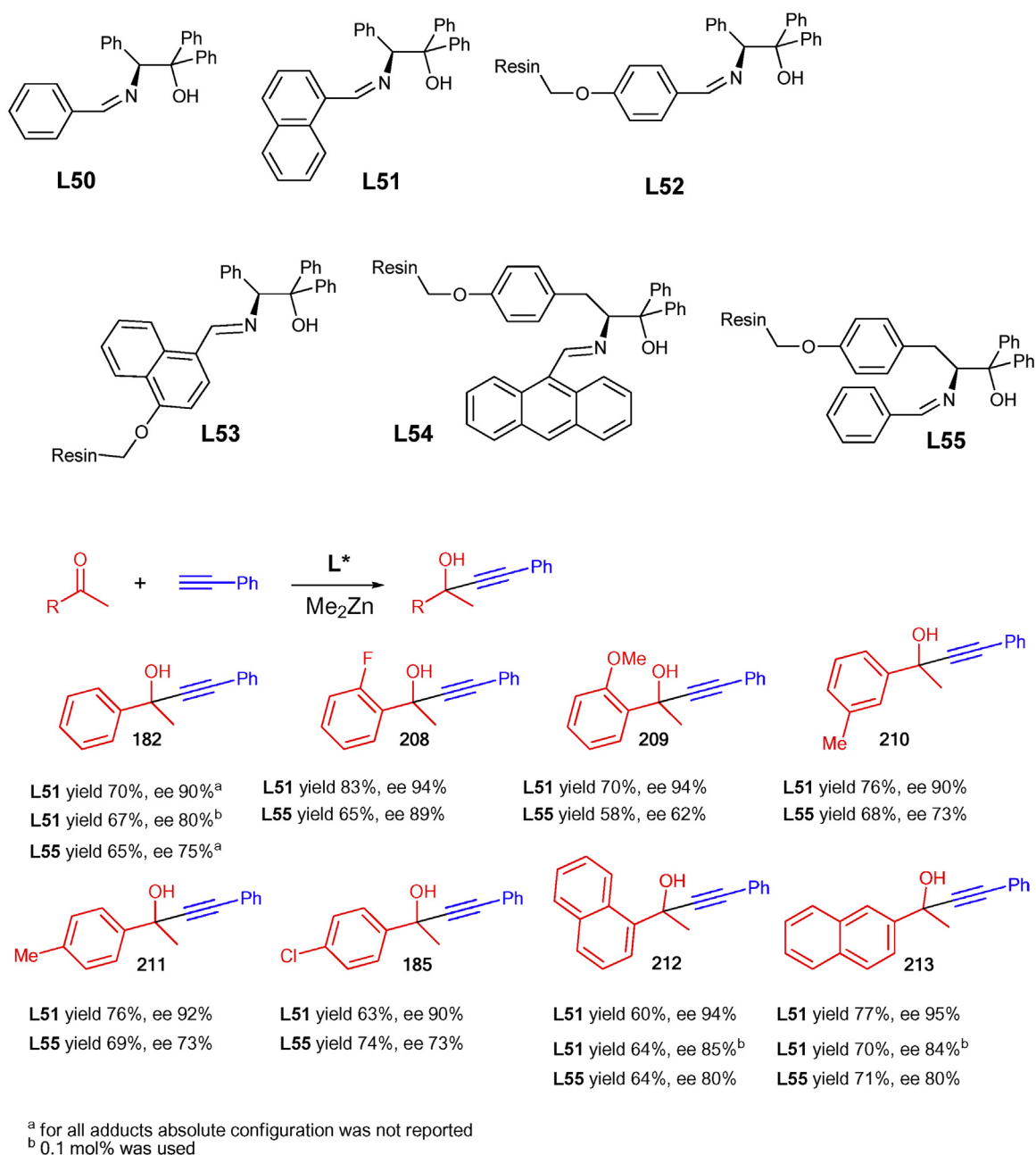
The structurally relevant oxazolidine **L56** was much less efficient and enantioselectivities from 72 to 88% ee were obtained for various acetophenones in the presence of 20 mol% of the ligand [127].



Hydroxysulphonamides of type **L57** were used as ligands for the addition of phenylacetylene to acetophenone, giving a moderate enantioselectivity, but only for ligands with $\text{R}' = \text{Me}$. The introduction of any more bulky substituents, including $\text{R}' = \text{Et}$ led to very low asymmetric induction. The size of R group only slightly influenced the enantiomeric excess [128].

The low reactivity of ketones vs. alkynes could be overcome by using more reactive α -ketoesters. However, a higher reactivity of α -ketoesters could possibly be disadvantageous due to a competing background reaction. Infante et al. investigated such an addition catalysed by a series of perhydro-1,3-benzooxazines. The best tested ligand was **L58a**, which bore a sterically demanding isopropyl group [129].

An optimisation of the reaction conditions revealed that the best results were obtained for the addition of phenylacetylene in the presence of 10 mol% of **L58a** and in the absence of any additives,



Scheme 22. Addition of phenylacetylene to ketones in the presence of 1 mol% of **L51** and 10 mol% of **L55**.

as DiMPEG or $\text{Ti}(\text{OiPr})_4$. The addition was very selective in the case of aromatic α -ketoesters leading to products with an ee exceeding 90%, up to 96%, except the *o*-methyl-substituted **217** (Scheme 23). The addition to enolizable aliphatic ketoesters was less selective (Scheme 23, **223–225**).

The transition state model **XXVIII** based on the classical Noyori's model for diethylzinc addition (see above) was proposed (Fig. 8).

2.2. Alkynylation in the presence of Lewis acid additives

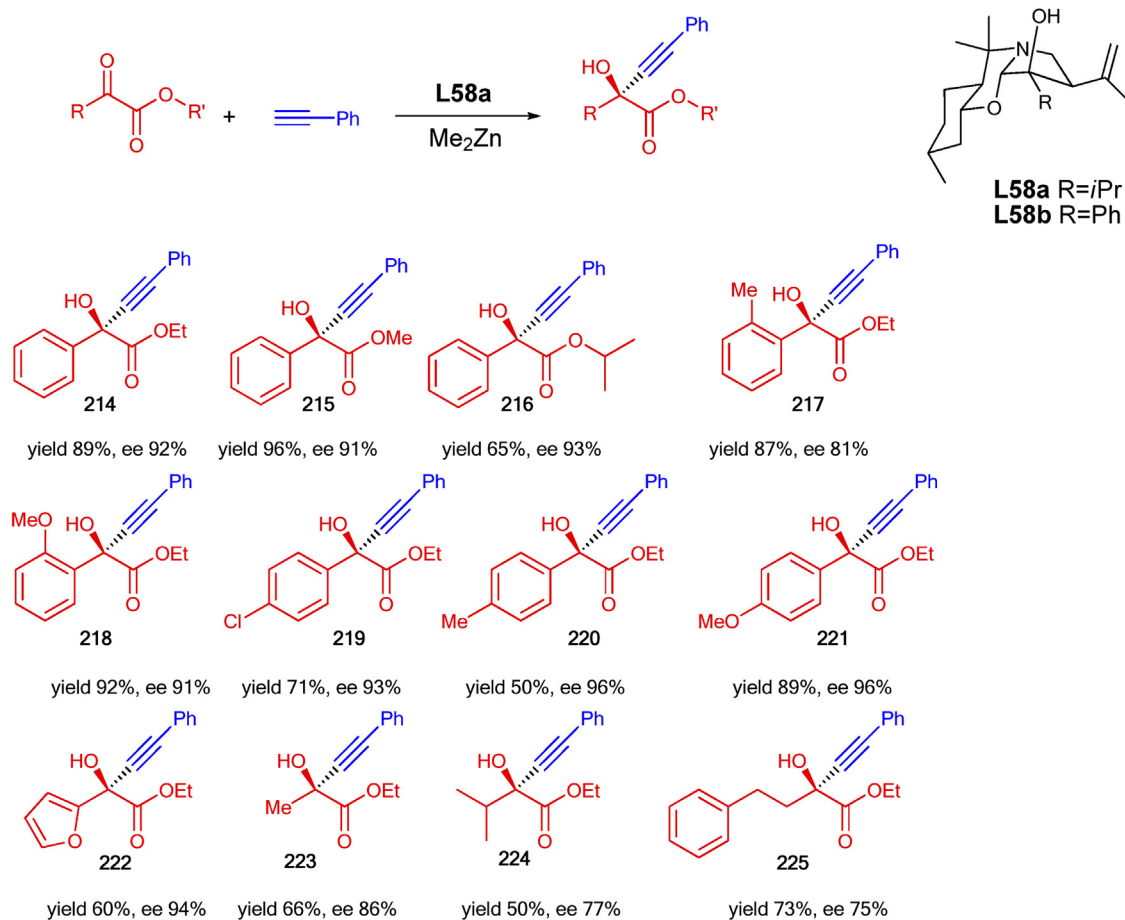
The addition of alkynes to carbonyl groups can be significantly improved for many ligands by performing the reaction in the presence of a Lewis acid, mainly $\text{Ti}(\text{OiPr})_4$ and in some cases also copper (II) triflate or trialkylaluminum. The mechanism for titanium-mediated additions of alkynylation reactions can be similar to

dialkylzinc additions discussed in Section I, but no experimental evidence has been published so far. A hypothetical model, **XXIX**, has been proposed by Pu (Fig. 9) [130].

The role of copper or aluminium has not been investigated, but most probably their role is to activate carbonyl compound during the addition process.

2.2.1. Addition to aldehydes in the presence of $\text{Ti}(\text{OiPr})_4$

As mentioned above, the addition of acetylenes to aldehydes in the presence of a Lewis acid, mostly titanium tetraisopropoxide, is another efficient strategy for the synthesis of propargylic alcohols. This method was developed independently by Chan [131] and by Pu [132]. As a ligand, they used (*S*)-BINOL (*S*)-**L59** (Pu) and (*R*)-BINOL (*R*)-**L59** and (*R*)-H₈-BINOL **L60** (Chan).



Scheme 23. Addition of phenylacetylene to α -ketoesters in the presence of 10 mol% of **L58a**.

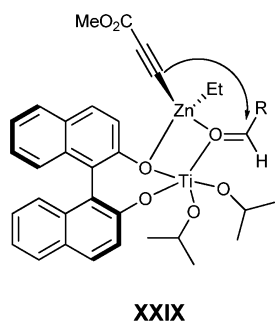
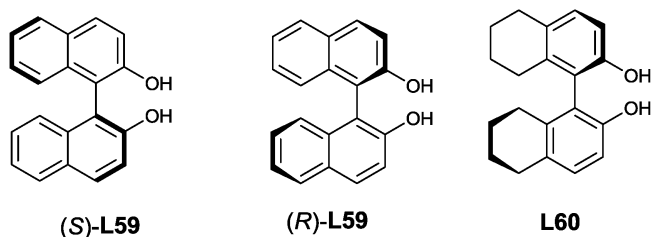


Fig. 9. Working model for the BINOL-catalysed alkynylations.



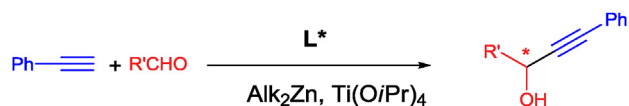
Both approaches differed in their methodology. Chan made the acetylide by the reaction of phenylacetylene with dimethylzinc at 0 °C, while Pu refluxed a toluene solution of diethylzinc and phenylacetylene for 5 h in the first step, followed by a reaction with aldehyde at RT. The elevated temperature is necessary for the deprotonation of acetylene with Et_2Zn in the absence of a ligand.

Performing the first step at a lower temperature (room temperature or refluxing dichloromethane) led to the ethyl addition product being the major one.

These differences were obviously responsible for different ee's for enantiomeric ligands (*S*)-**L59** and (*R*)-**L59** (Table 3). Under these conditions, lower enantioselectivities for aliphatic aldehydes were obtained [131,133]. A further modification of Pu's procedure allowed for the addition to aliphatic and α,β -unsaturated aldehydes with very high enantioselectivities (91–97% ee), albeit using a high catalyst loading [133].

The presence of a Lewis base can facilitate deprotonation of terminal acetylenes, thus improving the formation of zinc acetylide. Chan modified his procedure by performing reactions in the presence of additives. Among a series of phenolic additives, which were tested, a simple phenol appeared to be the best one. This additive allowed also for an improvement of the enantioselectivity for aliphatic aldehydes [134]. Pu applied 2 equiv. of carcinogenic HMPA as an additive and observed an increased enantioselectivity, but only in the presence of 40 mol% of (*S*)-BINOL (*S*)-**L59** and 4 equiv. of both diethylzinc and alkyne (Supplementary materials – Scheme S14) [135]. Supposedly, the Lewis acidic zinc atom in Et_2Zn coordinates to the Lewis basic oxygen atom in HMPA, increasing the negative charge on alkyl groups and thus increasing the basicity of the diethylzinc reagent, making the deprotonation of the alkyne possible (Scheme 24).

In the presence of DMF and DMSO, the enantioselectivity decreased. The same method was applied to the first enantioselective addition of methyl propiolate to aromatic and α,β -unsaturated aldehydes [136]. While products were obtained in a very good yield

Table 3Alkynylation of aromatic aldehydes in the presence of 20 mol% of enantiomeric binaphthols (*S*)-**L59**, (*R*)-**L59** and **L60**.

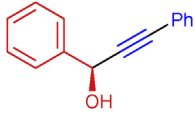
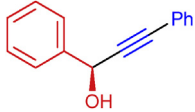
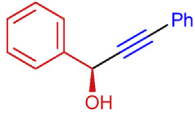
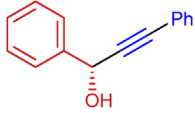
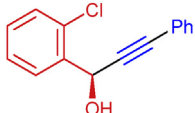
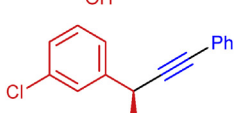
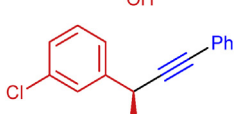
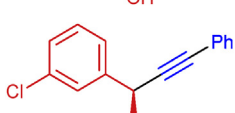
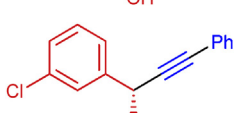
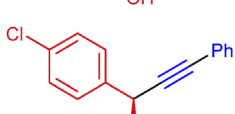
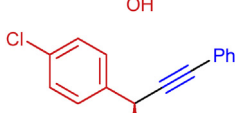
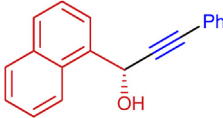
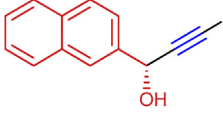
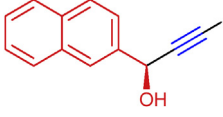
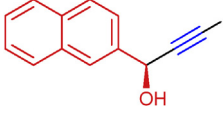
Entry	Product	L*	Alk ₂ Zn	Yield (%)	Ee (%) (abs. config.)	Ref.
1		L60	Me ₂ Zn	85	90 (<i>S</i>)	[131]
2		(<i>R</i>)- L59	Me ₂ Zn	84	92 (<i>S</i>)	[131]
3		(<i>R</i>)- L59	Me ₂ Zn	85	96 (<i>S</i>)	[134] ^a
4		(<i>S</i>)- L59	Et ₂ Zn	77	96 (<i>R</i>)	[132]
5		L60	Me ₂ Zn	90	76 (<i>S</i>)	[131]
6		(<i>R</i>)- L59	Me ₂ Zn	88	64 (<i>S</i>)	[131]
7		(<i>R</i>)- L59	Me ₂ Zn	82	88 (<i>S</i>)	[134] ^a
8		L60	Me ₂ Zn	87	95 (<i>S</i>)	[131]
9		(<i>R</i>)- L59	Me ₂ Zn	88	92 (<i>S</i>)	[131]
10		(<i>R</i>)- L59	Me ₂ Zn	84	95 (<i>S</i>)	[134] ^a
11		(<i>S</i>)- L59	Et ₂ Zn	79	92 (<i>R</i>)	[132]
12		L60	Me ₂ Zn	91	94 (<i>S</i>)	[131]
13		(<i>R</i>)- L59	Me ₂ Zn	87	92 (<i>S</i>)	[131]

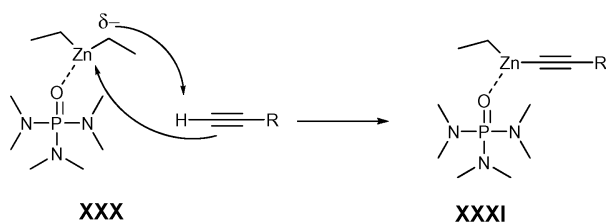
Table 3 (Continued)

Entry	Product	L*	Alk ₂ Zn	Yield (%)	Ee (%) (abs. config.)	Ref.
14		(R)- L59	Me ₂ Zn	83	95 (S)	[134] ^a
15		(S)- L59	Et ₂ Zn	81	92 (R)	[132]
16		(S)- L59	Et ₂ Zn	81	96 (R)	[132]
17		(S)- L59	Et ₂ Zn	77	94 (R)	[132]
18		L60	Me ₂ Zn	84	86 (S)	[131]
19		(R)- L59	Me ₂ Zn	83	86 (S)	[131]
20		(S)- L59	Et ₂ Zn	73	93 (R)	[132]
21		(S)- L59	Et ₂ Zn	78	93 (R)	[132]
22		L60	Me ₂ Zn	82	87 (S)	[131]
23		(S)- L59	Et ₂ Zn	74	96 (R)	[132]
24		L60	Me ₂ Zn	88	95 (S)	[131]
25		L60	Me ₂ Zn	89	96 (S)	[131]
26		(S)- L59	Et ₂ Zn	79	97 (R)	[132]

Table 3 (Continued)

Entry	Product	L*	Alk ₂ Zn	Yield (%)	Ee (%) (abs. config.)	Ref.
27		(<i>S</i>)- L59	Et ₂ Zn	71	92 (<i>R</i>)	[131]
28		(<i>S</i>)- L59	Et ₂ Zn	77	98 (<i>R</i>)	[132]
29		L60	Me ₂ Zn	75	80 (<i>S</i>)	[131]
30		(<i>R</i>)- L59	Me ₂ Zn	63	94 (<i>S</i>)	[134] ^a

^a In the presence of phenol as an additive.



Scheme 24. Formation of zinc acetylide in the presence of HMPA.

and ee (generally above 90%), the major drawback of this approach, namely a very high catalyst loading (40%), remained unchanged. The method was further extended, albeit with an enantioselectivity below 90%, to aliphatic and cycloaliphatic aldehydes and after regiospecific hydration, the products of this reaction were converted to optically active tetronic acids [130].

A more efficient additive, *N*-methylimidazol (NMI), was introduced by You and co-workers [137]. Excellent results were obtained in a reaction co-catalysed by (*R*)-**L59** (10 mol%) and NMI (5 mol%) for aromatic aldehydes and a series of substituted acetylenes. Substitution changes in the aromatic aldehydes influence the chemical yield, but not the enantioselectivity, which remains at the same high level (Supplementary materials – Scheme S15). The system is also compatible with some acetylenes known as “challenging” – methyl propiolate, 1-cyclohexylacetylene, TMS-acetylene. Products of these reactions are obtained in very good chemical yields and an asymmetric induction in the range of 88–94% ee (Supplementary materials – Scheme S15).

Neither HMPA, nor NMI could improve the enantioselectivity for the addition of linear alkynes to aliphatic aldehydes. Pu investigated a broad spectrum of imidazole derivatives in order to find a compound more suitable than NMI. Then he turned his attention to other Lewis bases and obtained the best results in the presence of dicyclohexylamine (Cy₂NH) as an additive [138]. The combination of (*S*)-BINOL (20 mol%) and Cy₂NH (5 mol%) was used for the addition of long-chain acetylenes to aliphatic aldehydes with enantioselectivities in the range of 81–89% ee (Scheme 25).

The addition of linear alkynes, namely hexyne-1 and octyne-1, to benzaldehyde and various substituted aromatic aldehydes in

the presence of (*R*)-**L59** proceeded in good chemical yield and an excellent enantioselectivity up to 94% without any additives [139].

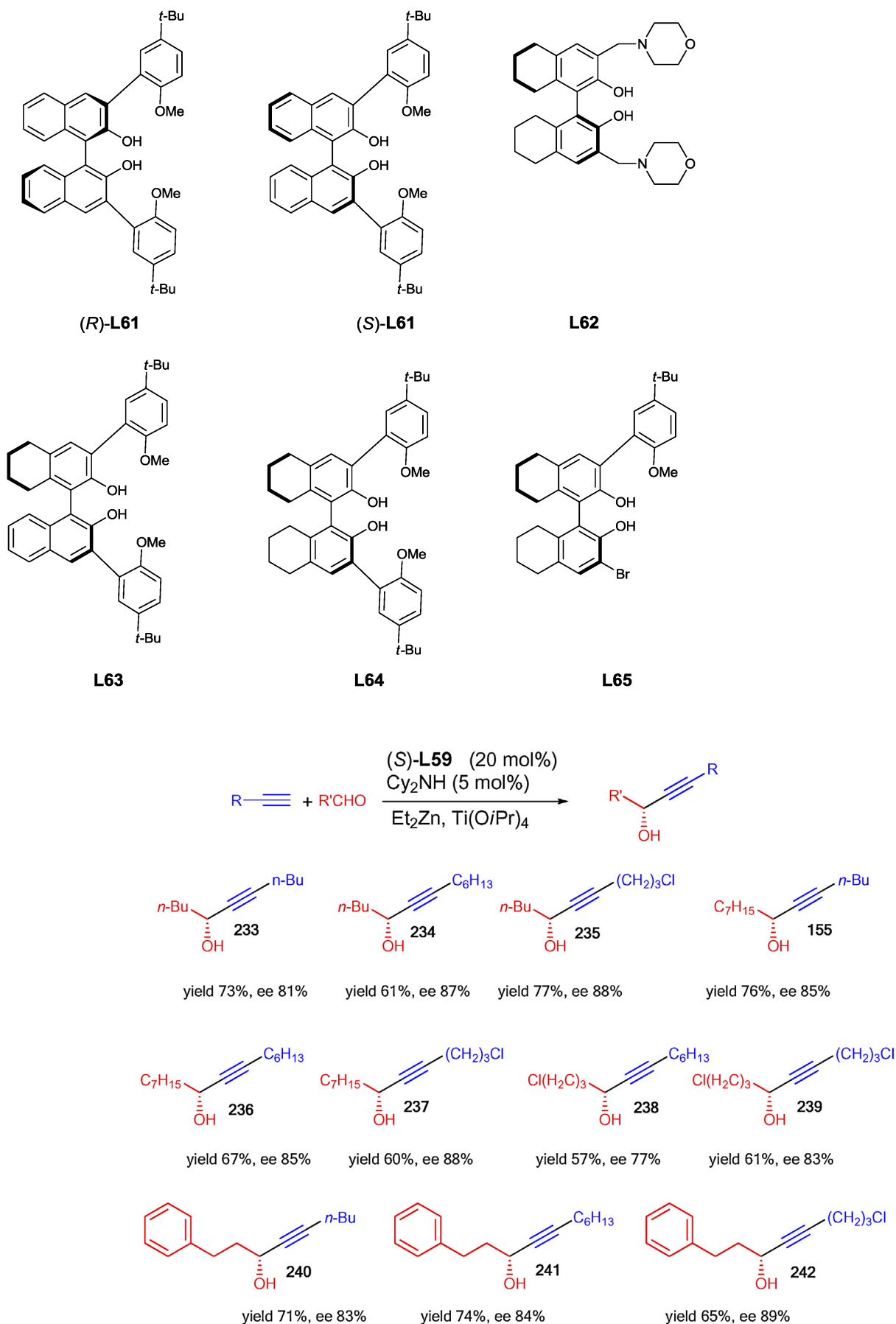
The combination of (*S*)-**L59**-Cy₂NH used for the addition of linear alkynes was applied for the addition of hepta-4,6-diynylbenzene to the variety of aromatic and aliphatic aldehydes [140]. The addition to aliphatic aldehydes and enals proceeded with a very good enantioselectivity only in the presence of 40 mol% of (*S*)-**L59** (Scheme 26).

The highly effective method for the diyne addition to aldehydes presented above provided access to a diverse set of substrates for further synthesis. In the same paper, Pauson–Khand and Diels–Alder reactions, as well as ring-closing enyne metathesis were successfully used for the stereoselective synthesis of polycyclic compounds.

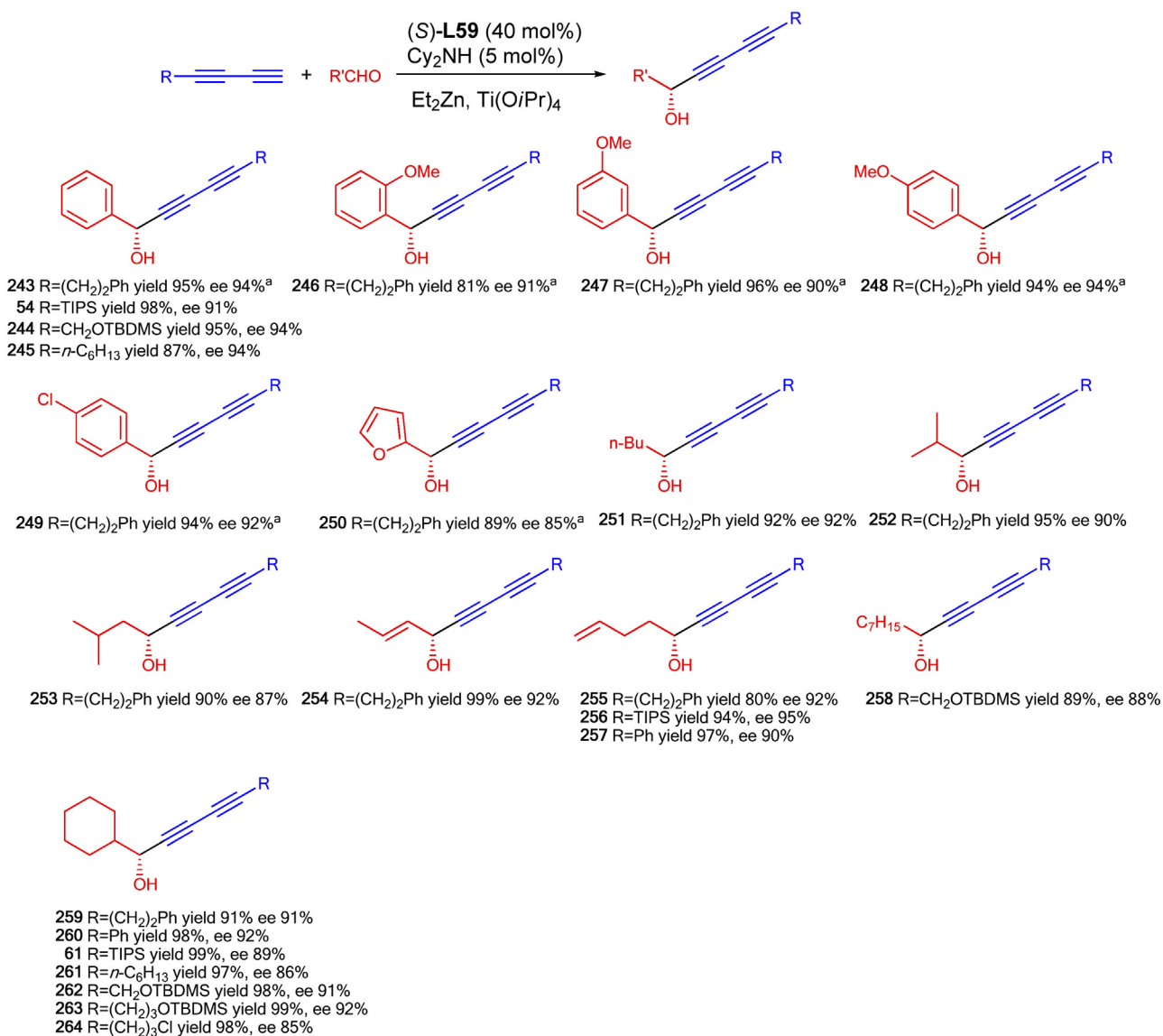
Optically active α -hydroxy- β,γ -unsaturated acid esters, which can be used as a side chains of new cerebrosides, were prepared using (*S*)-**L59**-catalysed addition. Thus obtained alcohol **126** was converted to 3,5-dinitrobenzoate **265** and recrystallisation increased the enantiomeric purity of **265** to 99%. Upon debenzoylation and desilylation, six α -acetoxyesters (R = C₁₀H₂₁ to C₁₅H₃₁) were synthesised (Scheme 27) [141].

To extend the catalytic properties of BINOL, Pu and co-workers synthesised the 3,3'-anisyl substituted BINOLs containing various electron-withdrawing groups, but all of them had a deteriorating effect on the enantioselectivity. An enhanced steric hindrance in the *para* position of the anisyl ring led to an improved enantioselectivity; *tert*-butyl derivatives (*R*)-**L61** and (*S*)-**L61** performed the best [142]. An additional modification of the reaction procedure, especially conducting all steps at RT, allowed for a further increase in the enantioselectivity. Under these conditions, phenylacetylene reacted with benzaldehyde, giving 93% ee; substituted benzaldehydes gave 80–89% ee.

3,3'-Bismorpholinylmethyl H₈-BINOL **L62** was synthesised by Pu's group [143,144] in order to improve the asymmetric induction observed by Chan for unsubstituted H₈-BINOL [131]. This new ligand gave excellent ee's for the addition of phenylacetylene to arylaldehydes substituted with electron-withdrawing group at *ortho*-position (up to 98% ee). In the case of other aldehydes, asymmetric inductions were equal or even lower than for unsubstituted **L60** or BINOL (**L59**) itself.

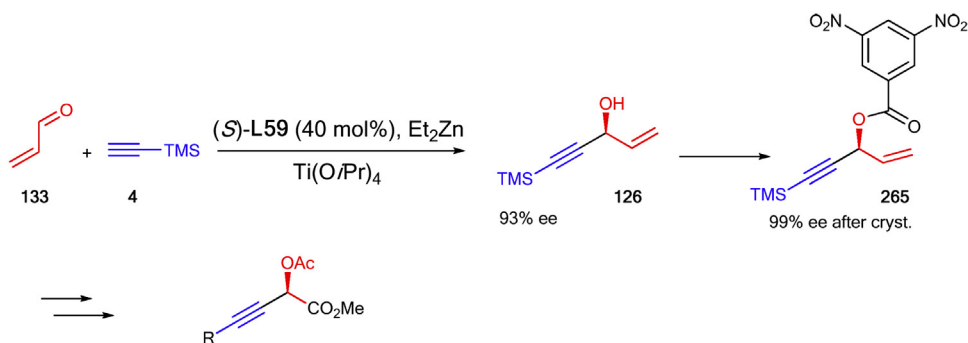


Scheme 25. Addition of alkyl substituted acetylenes to aliphatic aldehydes in the presence of ligand **(S)-L59** and Cy_2NH .



^a in the presence of 10 mol% of L59

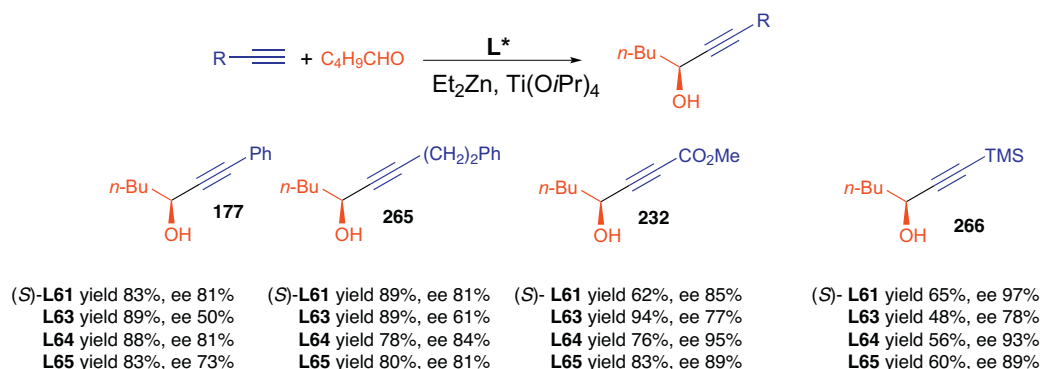
Scheme 26. Addition of diynes to aromatic and aliphatic aldehydes in the presence of ligand (S)-L59.



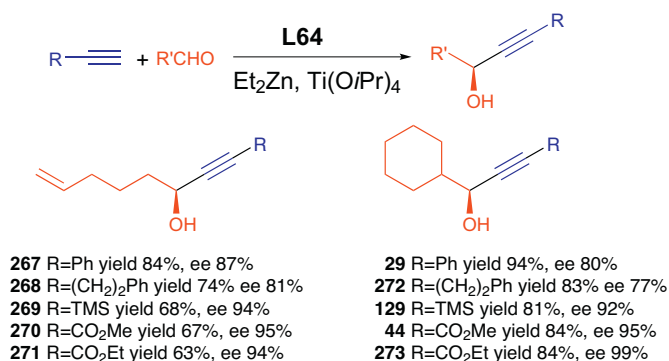
Scheme 27. Synthesis of α -acetoxy esters.

Positive features of ligands **L61** and **L62** were combined in anisyl substituted H₄- and H₈-BINOL ligands **L63–L65** [145]. New ligands were applied for the reaction of *n*-pentyl aldehyde with substituted acetylenes. The H₈-BINOL ligand **L65** performed clearly much

better than other ligands in the addition of various acetylenes except for the addition of TMS-acetylene, which was more efficient in the presence of the fully aromatic compound (S)-**L61** (Scheme 28).



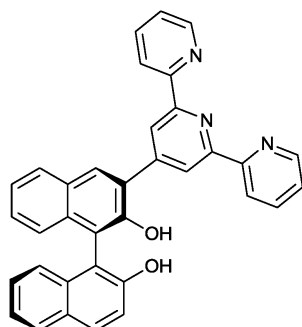
Scheme 28. Alkynylation in the presence of 20 mol% of substituted BINOL-s (**S**)-**L61** and **L63–65**.



Scheme 29. Alkynylation of aliphatic aldehydes and cyclohexanecarbaldehyde in the presence of 20 mol% of **L64**.

The efficacy of ligand **L64** was then examined in the reaction between unsaturated aldehydes and various acetylenes. The addition to pent-4-enal and hex-5-enal proceeded with an excellent selectivity, similar for both aldehydes (92–95% ee). Products of these reactions were used as substrates in the ring-closing metathesis (RCM) and tandem RCM-hydrogenation. An interesting feature of this system is the much lower asymmetric induction observed for the addition of acetylenes containing a phenyl ring, namely phenylacetylene and phenylbutyne. An excellent enantioselectivity was observed for the reaction of propiolate esters with cyclohexanecarbaldehyde (**Scheme 29**).

Another efficient ligand for the addition of methyl propiolate to aldehydes was recently reported by Pu et al. [146]. Ligand **L66** was very effective only for aromatic aldehydes; enantioselectivities of up to 98% ee were obtained.

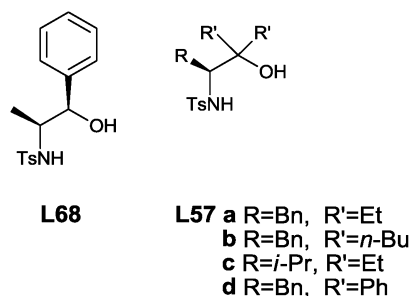


L66

Personal accounts presenting in details the BINOL-catalysed addition of aldehydes [147] and the transformations of thus-obtained propargylic alcohols [10] were recently published.

Very good results were obtained for the soluble polymer **L67** (10 mol%) [148]. The ligand was soluble in three common solvents: toluene, THF, and CH₂Cl₂, it precipitated easily upon the addition of methanol, and gave a very high enantioselectivity for the addition of phenylacetylene to some tested aromatic and aliphatic aldehydes (**Scheme 30**). However, the enantioselectivity depended on the solvent used. Only a few ligands have been able to give such a high enantioselectivity for simple aliphatic aldehydes, but it would be interesting to see the results for a broader range of aldehydes as well as other acetylenes.

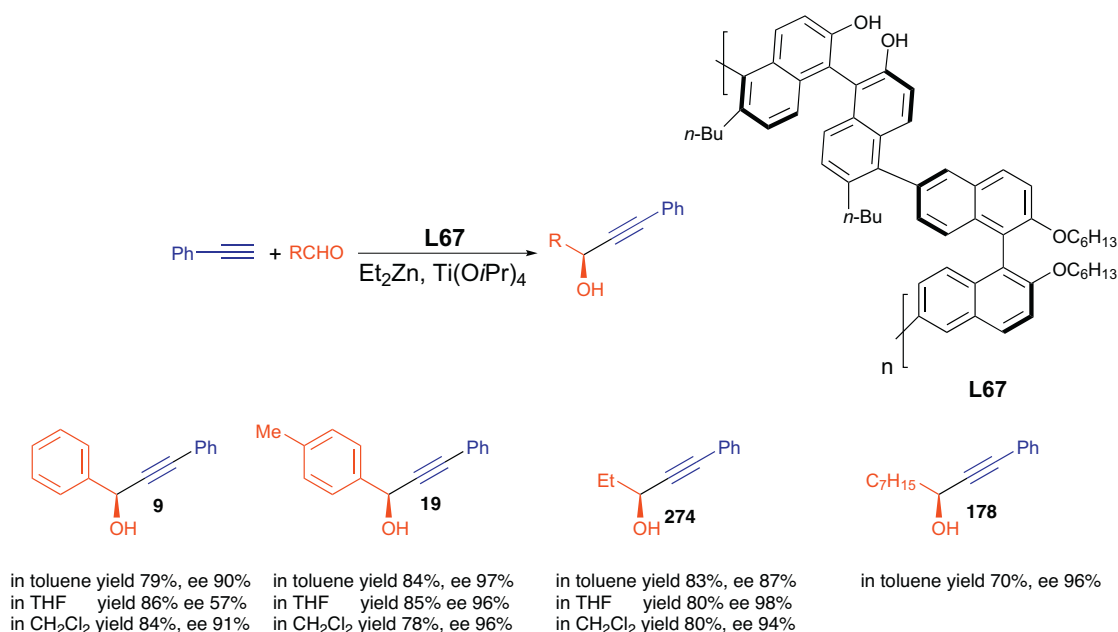
Mikami's concept of the self-assembly of chiral ligands [149] was applied by Chan and co-workers. (*S*)-BINOL (*S*)-**L59** was combined with *N*-tosylated amino alcohols **L68** and **L57a–c** derived from norephedrine [150] or natural amino acids [151], respectively, and used for the addition of phenylacetylene to various aromatic aldehydes.



The simultaneous use of both ligands mixed in a 1:1 ratio gave higher ee's than those obtained for each ligand separately. Ligands **L57** were more active without any significant loss of enantioselectivity. The most effective ligand **L57b** (much better than **L57a** chosen by Wang) gave ee's from 95 to >99% for the tested aromatic aldehydes, and over 80% for aliphatic aldehydes.

Ligands **L57a** and **L57d** were examined earlier by Wang et al. [152], and used as sole ligands for the addition of phenylacetylene. While Chan's group performed experiments in THF, Wang used various solvents, finding toluene to be the best one. In the case of ligand **L57a**, a change of the solvent was sufficient to increase the enantioselectivity to, and even above, the level obtained for mixed ligands. Different alkylzinc derivatives used by both groups could also influence the results. For the comparison of results obtained by both methods, see **Table 4**.

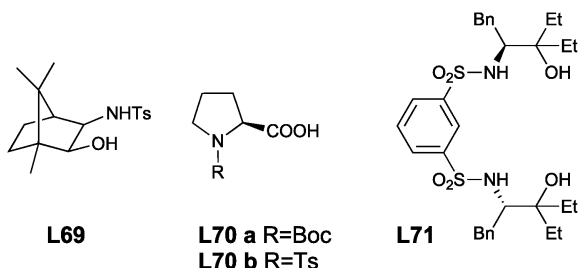
Subsequently, Wang's group presented a full account on this subject [153]. They tested the influence of alcohols as additives in the addition of phenylacetylene to benzaldehyde performed in the presence of ligand **57a**, and observed very interesting results. The addition of MeOH or EtOH slightly increased the enantioselectivity from 81% to 92 and 85% ee, respectively (**Table 5**, entries 1, 2 and 3), while all other alcohols, including BINOLs, led to a decrease of



Scheme 30. Alkynylation in the presence of 10 mol% of soluble polymer **L67**.

the selectivity. In the case of (*R*)-BINOL, a reversed selectivity was observed (Table 5, entry 10).

Excellent results were obtained for the camphor-derived ligand **L69** (10 mol%) [154]. Under mild conditions, with no need for a separate step of alkynylzinc preparation, the enantioselectivities of over 90% were obtained for the addition of phenylacetylene to aromatic aldehydes. Ligand **L69** was less efficient for butyl- and isobutylaldehyde, affording 85% and 75% ee, respectively. The addition of phenylacetylene in the presence of ligands derived from L-proline gave no satisfactory results. Both *N*-Boc **L70a** [155] and *N*-tosyl **L70b** [156] derivatives furnished addition products with a moderate selectivity.



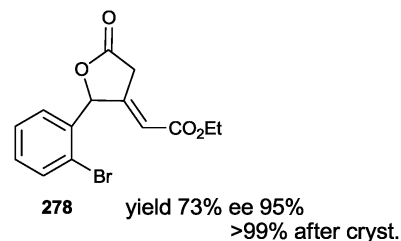
The C₂-symmetrical ligand, **L71**, is one of the most active sulphonamide ligands developed for the phenylacetylene additions to aromatic aldehydes [157]. **L71** shows a very high efficacy in the addition to benzaldehyde at a loading of 4 mol%. A similar level of enantioselectivity (83–97%) was observed for many tested aromatic aldehydes, but not for cinnamyl and isobutyl aldehydes (Supplementary materials – Scheme S16, **30** and **28**), which gave 71 and 57% ee, respectively. Interestingly, a high ee of 93% obtained for the addition to benzaldehyde in the presence of 10 mol% of ligand **L71** dropped to 90% in the presence of 15 mol% and increased to 95% when the amount of the ligand was decreased to 4 mol%. Even in the presence of 1.5 mol% of **L71**, 90% ee was achieved.

The previously reported simplified version of ligand **L71**, the tosyl derivative **L57a**, was successfully applied to the challenging reactions of aldehydes with methyl propiolate [158]. Similarly to the BINOL ligands studied earlier, the presence of 1,2-dimethoxyethane (DME), an achiral Lewis base, allowed for good yields. The reaction proceeded with good to very good

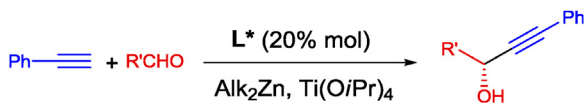
enantioselectivities for aromatic (91–94% ee), α,β-unsaturated (82–85% ee), and cycloaliphatic (79% ee) aldehydes (Supplementary materials – Scheme S17). Unfortunately, no data for aliphatic aldehydes were provided.

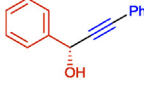
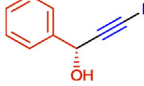
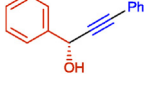
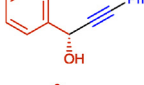
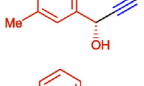
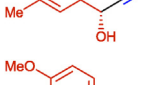
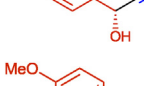
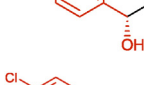
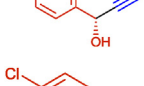
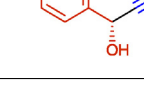
The tentative mechanism of this addition is presented in Scheme 31. The formation of alkynylzinc was facilitated by the intramolecular interaction with Lewis basic sulphonamide oxygen, as well as by the introduction of DME as an additional Lewis base, in cooperation with *N,O*-chelating zinc atom (**XXXII**). The presence of a strong Lewis acid at this stage can hamper the formation of alkynylzinc, so titanium tetraisopropoxide is added in the second step and a titanium Lewis acidic centre is formed (**XXXIV**). The aldehyde is bound to the titanium atom (**XXXV**) and alkynyl transfer takes place (**XXXVI**).

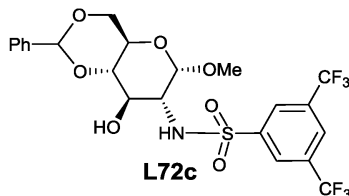
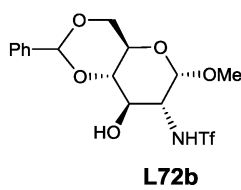
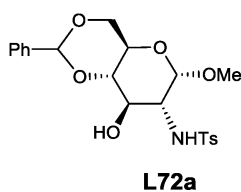
This methodology was successfully applied to a one-pot three-component synthesis of chiral 2-oxazolidinones [159]. For some aldehydes, e.g. 2-bromobenzaldehyde, the enantioselectivity reached 95%, and the purity of the final oxazolidinone **278** could be further improved to >99% ee by recrystallisation.



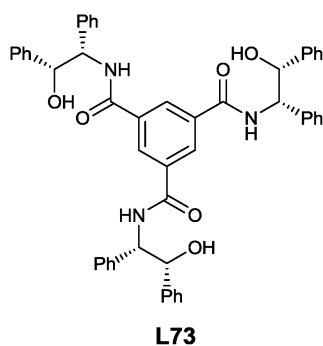
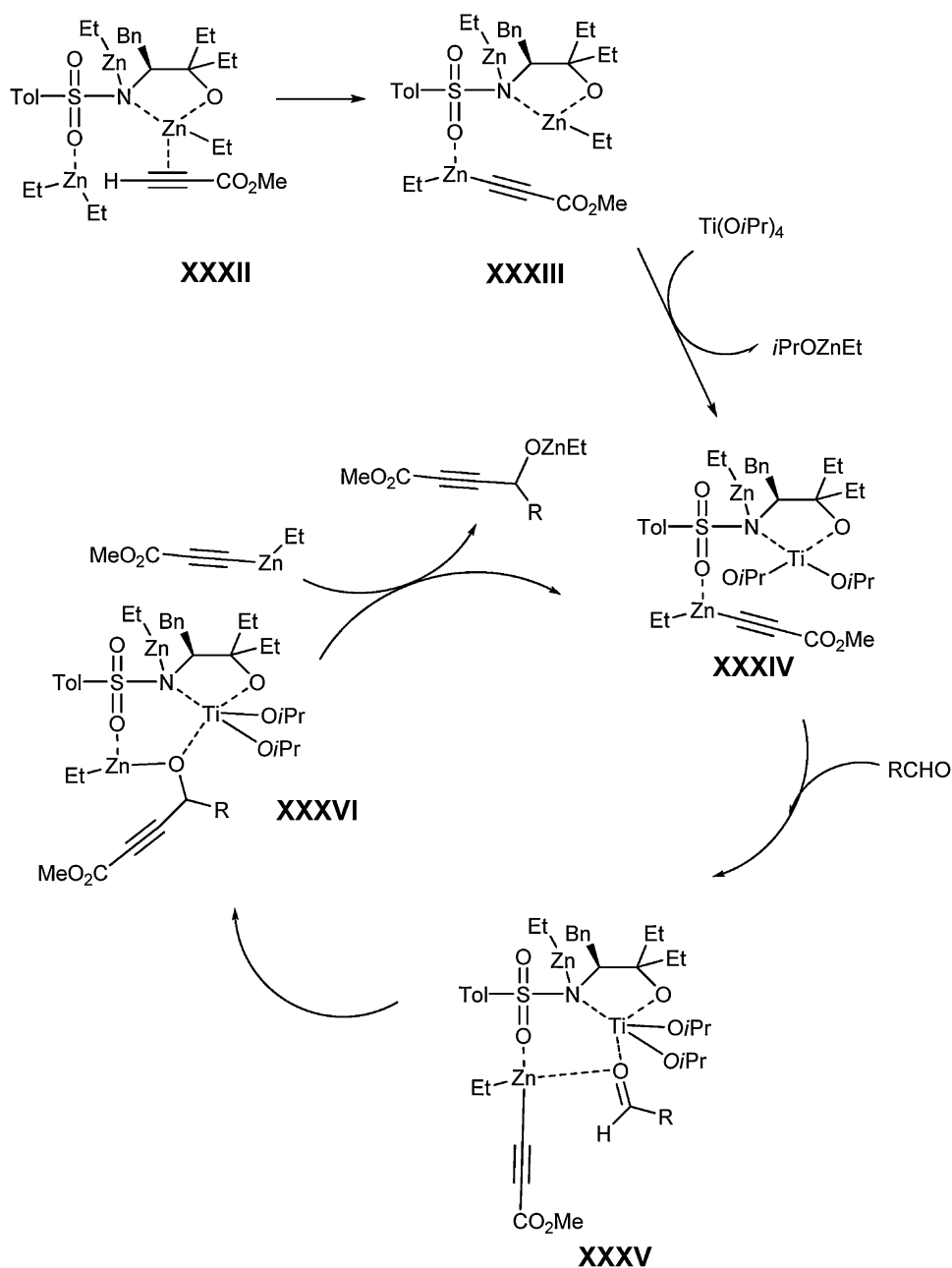
Only one example of monosaccharide-derived ligands **L72** was reported [163a]. Ligands **L72a** and **L72b**, which gave excellent results in the addition of diethylzinc to aldehydes [161,162], were less efficient in additions of phenylacetylene and hexyne-1. The best results were obtained for ligand **L72c** (20 mol%). The enantioselectivity of the reaction was highly dependent on the nature of the aldehyde and the substitution on the phenyl ring. The best results were obtained for the addition of phenylacetylene to 4-nitrobenzaldehyde (96% ee), 4-trifluoromethylbenzaldehyde (89% ee), and 2-fluorobenzaldehyde (92% ee). A good enantioselectivity was observed for cyclohexanecarbaldehyde (85% ee), while hexanal gave only 49% ee [163b].

Table 4Alkynylation in the presence of ligands **L57** and the mixture of **L57** and (S)-**L59**.

Entry	L*	Product	Alk ₂ Zn	Yield (%)	Ee (%)	Ref.
1	L57a		Me ₂ Zn	75	16	[151]
2	L57a		Et ₂ Zn	92	95	[152]
3	L57b		Me ₂ Zn	78	35	[151]
4	L57b + (S)- L59		Me ₂ Zn	85	97	[151]
5	L57a		Et ₂ Zn	89	92	[152]
6	L57b + (S)- L59		Me ₂ Zn	76	91	[151]
7	L57a		Et ₂ Zn	90	91	[152]
8	L57b + (S)- L59		Me ₂ Zn	80	96	[151]
9	L57a		Et ₂ Zn	80	98	[152]
10	L57b + (S)- L59		Me ₂ Zn	88	96	[151]



The tris(β-hydroxy amide) ligand **L73**, having a C₃-symmetry, was synthesised and used (at 20 mol%) with very good results. Enantioselectivities of up to 92% were obtained for the addition of phenylacetylene to aromatic aldehydes, but the ligand was inefficient for aliphatic aldehydes [163].

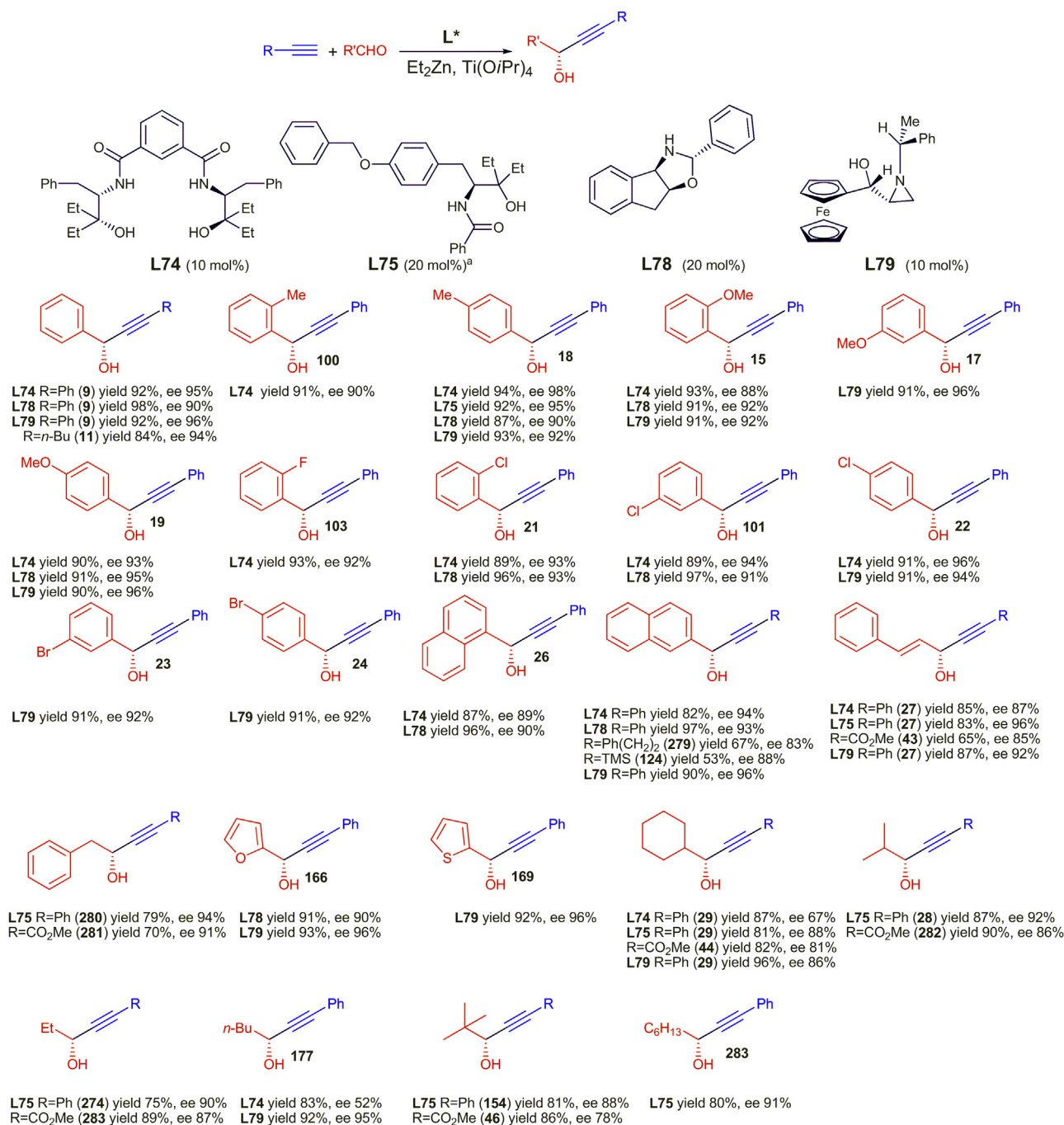


Even better results were obtained for the addition of phenylacetylene in the presence of the C_2 -symmetric bis(β -hydroxy amide) ligand **L74** (10 mol%) [164]. An enantiomeric excess of over 90% was obtained for all the aldehydes tested, except for

Table 5

Use of additives in the **57a**-catalysed asymmetric addition of phenylacetylene to benzaldehyde.

$\text{Ph}-\text{C}\equiv\text{CH} \quad (3) + \text{PhCHO} \quad (1) \xrightarrow[\text{Et}_2\text{Zn, Ti(OiPr)}_4]{\text{L57a (5 mol\%)}} \text{Ph}-\text{CH(OH)}-\text{C}\equiv\text{CH} \quad (9)$			
Entry	Additive	57a /additive	Ee (%)
1	None		81
2	MeOH	1:2	92
3	EtOH	1:2	85
4	<i>t</i> -BuOH	1:2	79
5	DiMPEG	1:2	77
6	PhOH	1:2	74
7	1-Naphtol	1:2	69
8	2-Naphtol	1:2	63
9	(<i>S</i>)-BINOL	1:1	68
10	(<i>R</i>)-BINOL	1:1	–37

Scheme 32. Alkynylation in the presence of ligands **L74**, **L75**, **L78** and **L79**.

2-methoxybenzaldehyde; the best results were achieved for 4-chloro and 4-methylbenzaldehyde; 96% and 98% ee, respectively (Scheme 32, alcohols **22** and **18**). This indicates steric rather than stereoelectronic reasons for the asymmetric induction in the presence of ligand **L74**. Again, aliphatic aldehydes yielded inferior results in this reaction.

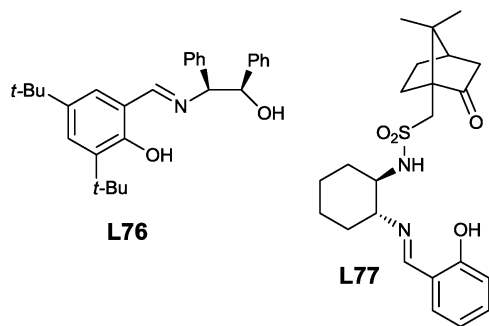
A further modification of β -hydroxyamide ligands led to ligand **L75**, derived from readily available (*S*)-tyrosine. Ligand **L75** (20 mol%) was one of the few ligands giving excellent results in the addition of phenylacetylene to aliphatic and α,β -unsaturated aldehydes, mostly over 90% ee (Scheme 32) [165]. The same ligand was also used for the addition of methyl propiolate,

with a slightly lower, although still high, asymmetric induction. The major drawback was the high catalyst loading (30 mol%) [166].

This work has been expanded to polymer-supported ligands. The best results were obtained for the addition of phenylacetylene catalysed by the polymer-supported version of ligand **L75** (20 mol%). However, the observed asymmetric induction was worse than in the case of homogenous catalysis. Polymer-supported catalysts could be reused, but each subsequent run gave the products with a lower enantioselectivity [167].

Schiff bases **L76** [168] and **L77** [169] were used as ligands for the addition of phenylacetylene to the series of aromatic and

aliphatic aldehydes with an enantiomeric excess not exceeding 90%.

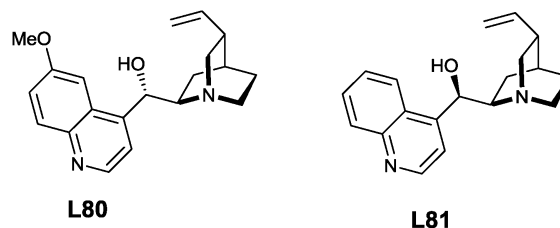


Simple oxazolidine ligand **L78** (20 mol%), derived from (1*R*,2*S*)-*cis*-1-amino-2-indanol, was very efficient for the addition reaction of phenylacetylene to aryl aldehydes with ee's of up to 95% (Scheme 32) [170]. Interestingly, the very similar ligand **L21**, bearing the cyclohexyl substituent of the oxazolidine instead of the phenyl one, gave only 17% ee in the addition of phenylacetylene to benzaldehyde. However, the reaction catalysed by **L21** was performed in the absence of $\text{Ti}(\text{OiPr})_4$, which indicates an entirely different mechanism and transition state for the addition [99]. The addition of but-3-ynylbenzene and TMS-acetylene was somewhat less efficient (Scheme 32, **279** and **124**).

The series of ferrocenyl-substituted aziridinylmethanols was synthesised and used for the alkynylation reactions. Excellent results were obtained for the alkynylation with phenylacetylene in the presence of 10 mol% of the best ligand in the series, ligand **L79** [171]. Enantiomeric excesses in the range of 92–96% and chemical yields of over 90% were obtained for aromatic aldehydes, independently of the substitution of the phenyl ring (Scheme 32). Heteroaromatic aldehydes reacted with a very high enantioselectivity of 96% ee (Scheme 32, **166** and **169**). Ligand **L79** was also very efficient for linear alkynes and 94% ee was obtained

for the addition of hexyne-1 to benzaldehyde (Scheme 32, **11**). Regrettably, the transition state involving this exceptional ligand has not been studied so far.

Singh examined two *Cinchona* alkaloids, quinidine **L80** and cinchonidine **L81**, as ligands for the diethylzinc-mediated addition of phenylacetylene to aldehydes in the presence of $\text{Ti}(\text{OiPr})_4$. Better results were obtained for **L81**. The reaction required as much as 40 mol% of the alkaloid and the enantiomeric excesses were below 90% (Supplementary materials – Scheme S18) [172].

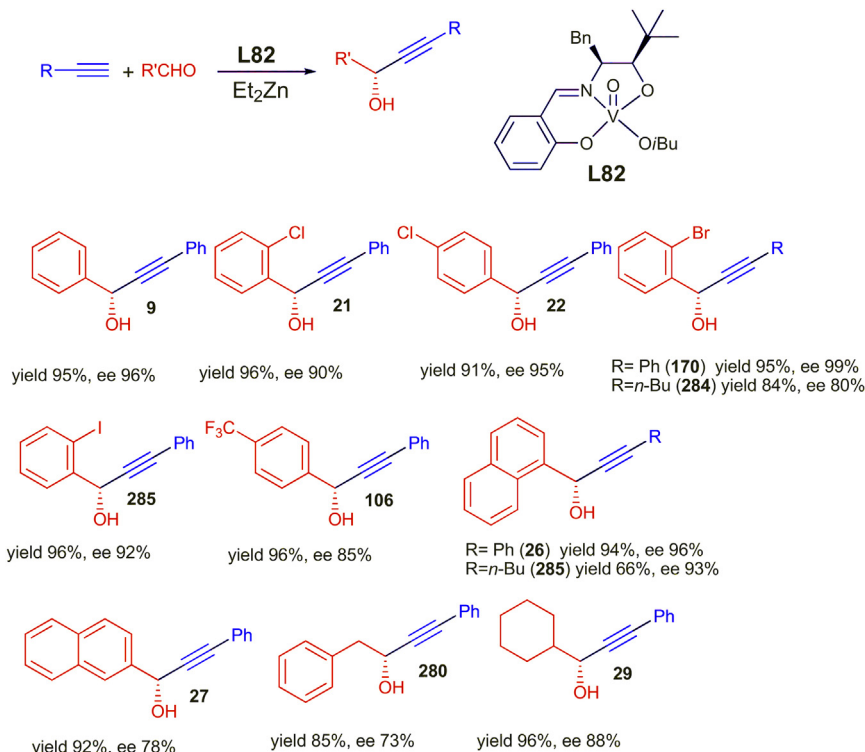


2.2.2. Addition to aldehydes in the presence of a vanadium complex

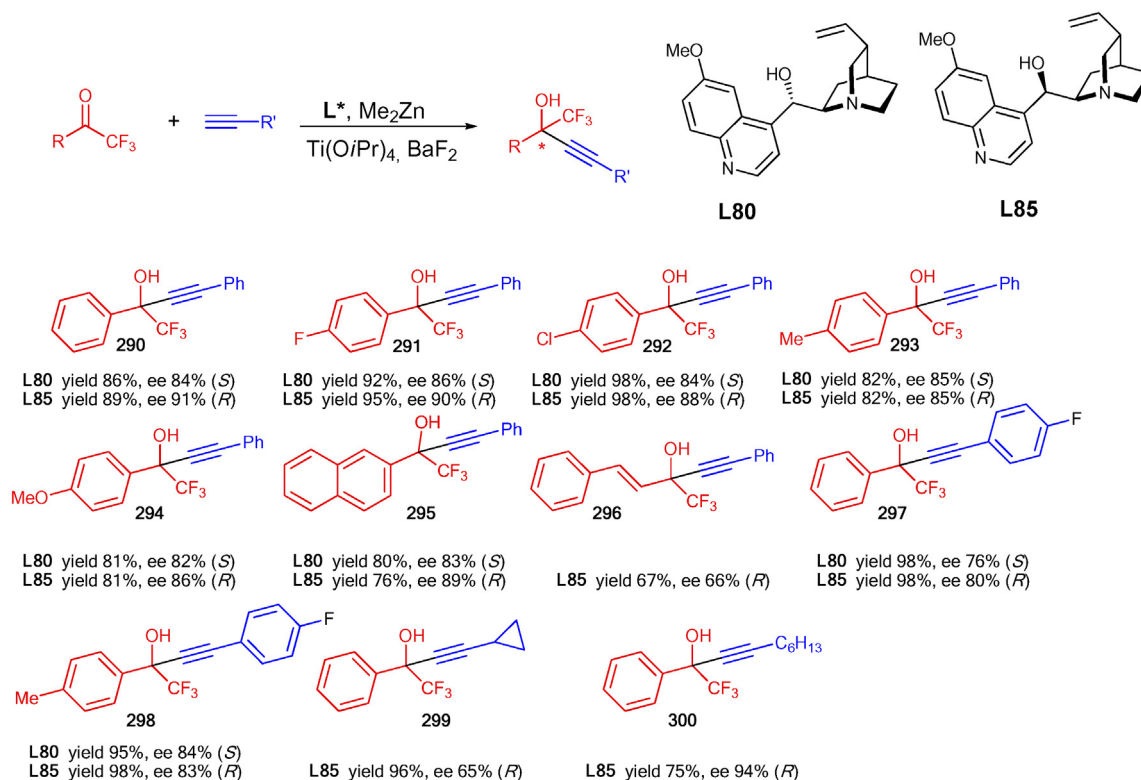
The only example of alkynylation in the presence of a vanadium complex was presented by Hsieh and Gau [173]. The stable complex **L82** was readily prepared by the reaction of a respective Schiff base with triisopropoxyvanadium (V) in isobutanol, and then used as a catalyst (10 mol%) for the addition of phenylacetylene and hexyne-1 to aldehydes with a good to excellent enantioselectivity (Scheme 33). Vanadium (V) behaves as a Lewis acid centre, similarly to titanium (IV) in all reactions described earlier in this section.

2.2.3. Additions to ketones in the presence of $\text{Ti}(\text{OiPr})_4$

After several unsuccessful attempts of other groups, Wang applied (*S*)-**L59** (20 mol%) as a ligand in the titanium-promoted addition of phenylacetylene to aromatic ketones (and to the one



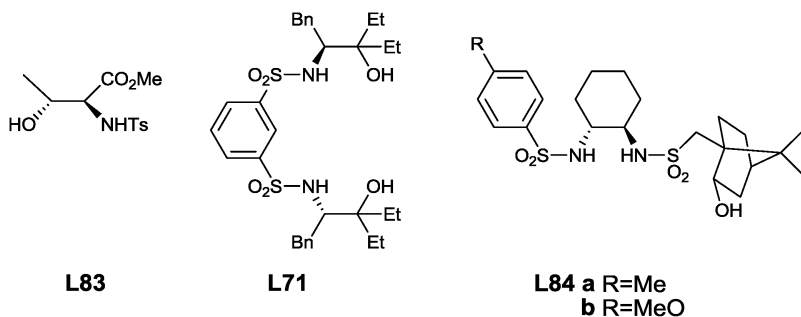
Scheme 33. Alkynylation in the presence of 10 mol% of **L82**.



Scheme 34. Enantioselective alkynylation of trifluoromethyl ketones in the presence of 20 mol% of **L80** and **L85**.

aliphatic ketone) with good results. The obtained ee's ranged from 66% up to 95% (Supplementary materials – Scheme S19) [174].

3,3'-Bismorpholinylmethyl H₈-BINOL **L62**, having been successfully used for some substituted benzaldehydes (see above), was applied for the addition of phenylacetylene to acetophenone, but the asymmetric induction was only moderate (69% ee) [144].



Sulphonamides **L83** [128] and **L71** [157] catalysed the addition of phenylacetylene with a moderate to good enantioselectivity (Supplementary materials – Scheme S19), while very poor ee's were obtained for ligand **L84** [175] (20% for **L84a**, R=Me, and 36% for **L84b**, R=OMe).

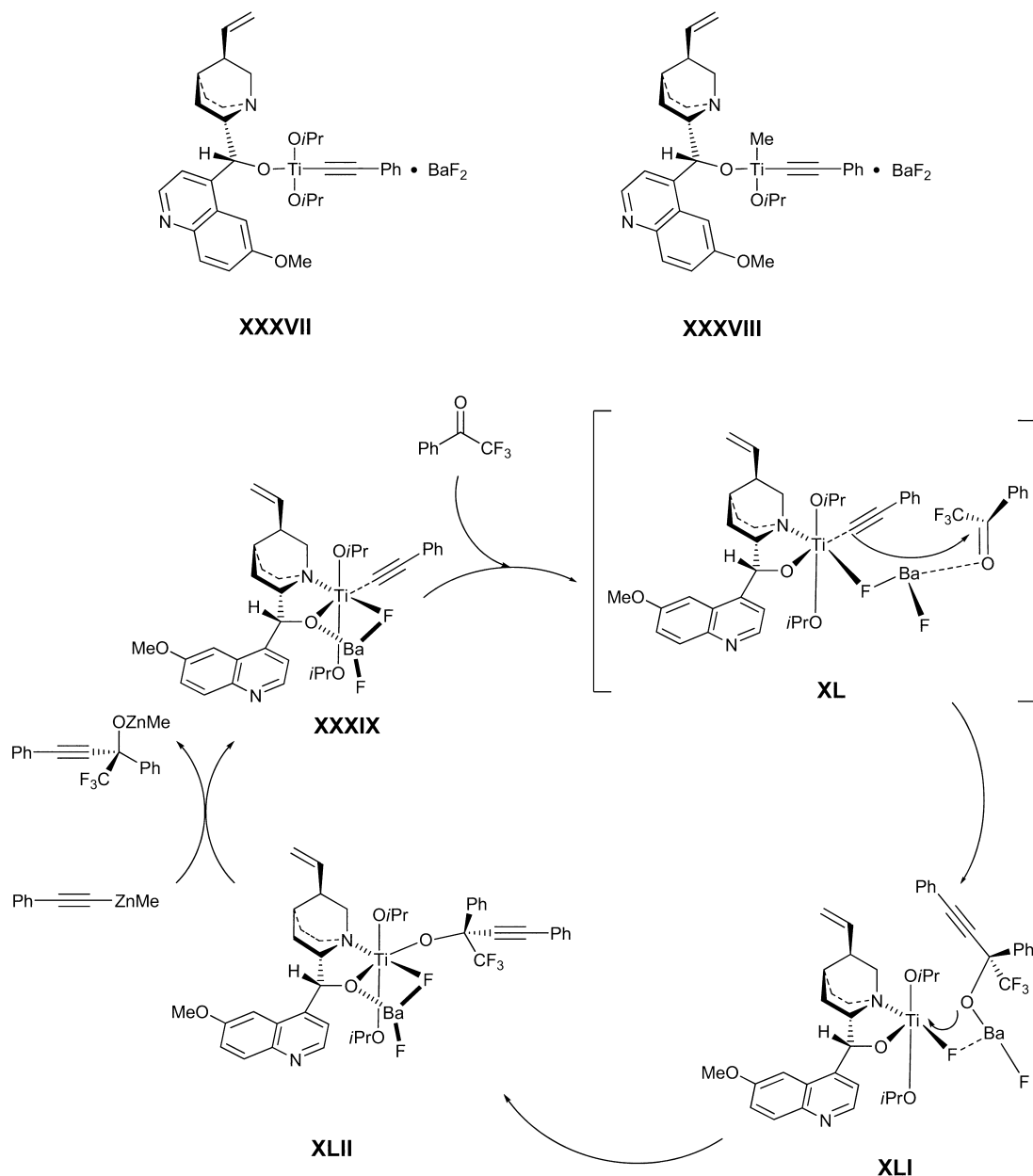
Ma and co-workers developed the first method for a catalytic titanium-promoted diethylzinc-mediated alkynylation of trifluoromethyl ketones [176]. A large number of ligands were screened for the addition of phenylacetylene to 2,2,2-trifluoroacetophenone and pseudoenantiomeric *Cinchona* alkaloids, quinidine **L80** and quinine **L85**, performed the best, inducing opposite absolute configurations of newly formed stereogenic centres.

The enantioselectivity was higher after the replacement of diethylzinc with dimethylzinc. The basic additives as CaH₂ further increased the ee. A high similarity between the hydride anion and the fluoride anion was assumed – both these are hard bases of similar Van der Waals radii (1.2 vs. 1.35 Å, respectively) – giving

a reason to replace calcium hydride with the fluoride salt. The enantioselectivity was highly dependent on the metal counterion and the best results were obtained in the presence of 0.2 equivalent of BaF₂. The suggested role of fluoride was to coordinate to the titanium centre and force the deoligomerisation of the catalyst,

leading to one favourable structure. Almost all presented additions of acetylenes to trifluoromethyl ketones proceeded with ee's in the range of 80–90%, except for the addition to α,β-unsaturated ketone (Scheme 34, 296), and to cyclopropyl-substituted acetylene (Scheme 34, 299). Reactions catalysed by **L80** proceeded in almost all cases with a slightly lower enantioselectivity. The substitution of the phenyl ring of phenylacetylene usually led to higher chemical yield (Scheme 34, e.g. 297 vs. 290, 298 vs. 293). The highest enantiomeric excess was obtained for the linear acetylene octyne-1 (Scheme 34, 300), which makes this catalytic system very useful for practical applications.

The ESI-MS spectra of a mixture of Ti(OiPr)₄, **L85**, phenylacetylene, BaF₂, and Me₂Zn (2.0:0.2:2.5:0.2:3) revealed the presence of titanium species **XXXVII**. An increased amount of Me₂Zn (6 equiv.) led to the formation of methyltitanium complex **XXXVIII** (Scheme 35). DFT calculations, using the B3LYP/6-31G(d) method,



Scheme 35. The suggested mechanism for the additions catalysed by *Cinchona* alkaloids.

were performed for the simplified model of amino alcohol/titanium (IV) complex. The putative mechanism of the reaction was proposed on the basis of ESI-MS studies and DFT calculations.

2.2.4. Addition to ketones in the presence of aluminium and copper complexes

The method for the preparation of alkynylzinc can have serious impact on the outcome of the reaction. Wang used 3 equiv. of Et_2Zn in a successful addition of phenylacetylene to ketones in the presence of (*S*)-**L59**- $\text{Ti}(\text{OiPr})_4$ (see above) [174]. Chan used 1 equivalent of the less reactive Me_2Zn for (*S*)-**L59**- $\text{Ti}(\text{OiPr})_4$ catalyst and observed no reaction. This unsuccessful attempt to use $\text{BINOL-Ti}(\text{OiPr})_4$ led to the development of the first example of dimethylzinc-mediated addition of phenylacetylene to ketones in the presence of the copper salt $\text{Cu}(\text{OTf})_2$ (10 mol%) and ligand **L86**

(10 mol%) [177,178]. Copper salts other than $\text{Cu}(\text{OTf})_2$, as well as zinc and scandium triflates, gave products with very low yield or no product at all. Comparable, but slightly lower values were obtained for **L87**. Camphorsulphonamides **L68**, **L88–L92**, **L94** and various other ligands, e.g. **L33**, **L59**, **L93**, catalysed the addition with low to moderate chemical yield and enantioselectivity. The enantioselectivity was highly dependent on the substituents of the aromatic ring in acetophenones and, in some cases, reached up to 97% ee (Table 6, entries 13, 16, 19, 21). A lower selectivity was observed for alkynes other than phenylacetylene (Table 6, entries 30 and 31). A heteroaromatic ketone reacted in good chemical yield, but the obtained product was racemic (Table 6, entry 32). Supposedly, the mechanism should be similar to the one proposed by Noyori for the sulphonamide-catalysed conjugate addition to enones [179].

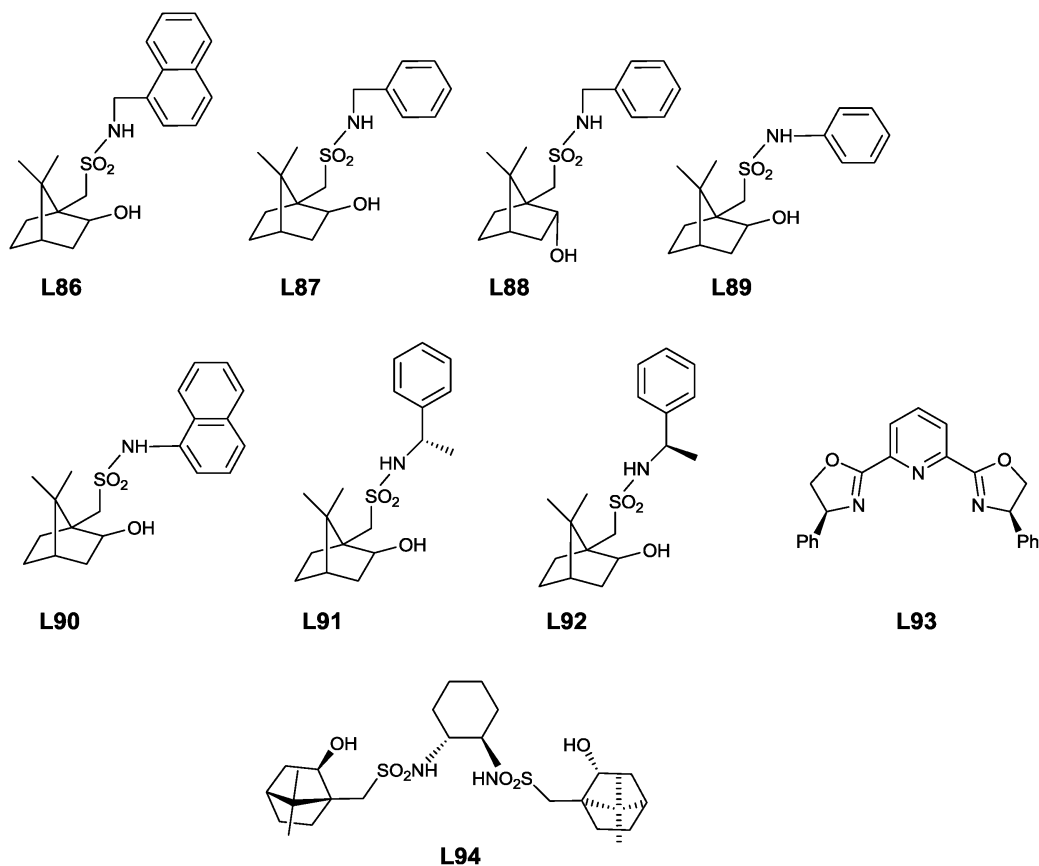
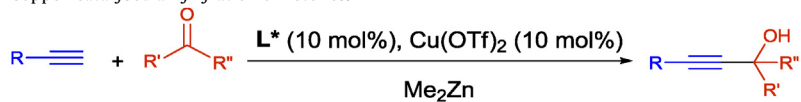
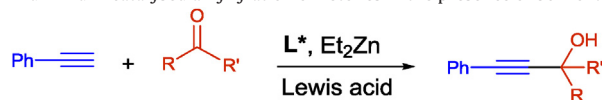


Table 6
Copper-catalysed alkylation of ketones.



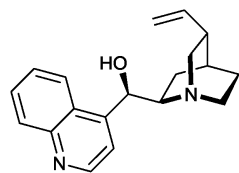
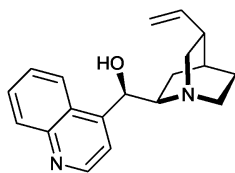
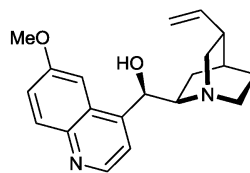
Entry	L*	Alkyne (R=)	Ketone	Yield (%)	Ee (%)
1	L33	Ph	Acetophenone	n.a.	3
2	(R)-L59	Ph	Acetophenone	n.a.	8
3	L68	Ph	Acetophenone	No reaction	–
4	L86	Ph	Acetophenone	92	88
5	L87	Ph	Acetophenone	90	82
6	L88	Ph	Acetophenone	49	13
7	L89	Ph	Acetophenone	32	61
8	L90	Ph	Acetophenone	51	65
9	L91	Ph	Acetophenone	No reaction	–
10	L92	Ph	Acetophenone	No reaction	–
11	L93	Ph	Acetophenone	29	0
12	L94	Ph	Acetophenone	50	13
13	L86	Ph	2'-Me-acetophenone	49	96
14	L86	Ph	3'-Me-acetophenone	83	86
15	L86	Ph	4'-Me-acetophenone	77	92
16	L86	Ph	2'-Br-acetophenone	65	96
17	L86	Ph	3'-Br-acetophenone	80	82
18	L86	Ph	4'-Br-acetophenone	75	91
19	L86	Ph	2'-Cl-acetophenone	94	97
20	L86	Ph	3'-Cl-acetophenone	72	71
21	L86	Ph	2'-F-acetophenone	91	96
22	L86	Ph	3'-CF ₃ -acetophenone	91	70
23	L86	Ph	4'-CF ₃ -acetophenone	82	93
24	L86	Ph	2'-Naphthacetophenone	75	85
25	L86	Ph	2'-(4-Ferrocenyl)-acetophenone	62	90
26	L86	Ph	Propiophenone	57	71
27	L86	Ph	c-C ₃ H ₅ COCH ₃	93	5
28	L86	Ph	(CH ₃) ₂ CHCOCH ₃	90	88
29	L86	Ph	PhCH ₂ CH ₂ COCH ₃	89	35
30	L86	c-C ₃ H ₅	Acetophenone	81	60
31	L86	TMS	Acetophenone	76	54
32	L86	Ph	3-acetylpyridine	81	0

Table 7Aluminium-catalysed alkynylation of ketones in the presence of 80 mol% of *Cinchona* alkaloids.

Entry	L*	Lewis acid	Ketone	Yield (%)	Ee (%)
1	L81	Ti(OiPr) ₄	Acetophenone	No reaction	–
2	L95	Ti(OiPr) ₄	Acetophenone	60	27
3	L85	Ti(OiPr) ₄	Acetophenone	68	40
4	L85	–	Acetophenone	78	5
5	L85	Cu(OTf) ₂	Acetophenone	53	7
6	L85	Al(OiPr) ₃	Acetophenone	59	76
7	L81	Al(OiPr) ₃	Acetophenone	55	73
8	L95	Al(OiPr) ₃	Acetophenone	57	76
9	L85	Et ₃ Al	Acetophenone	70	80
10	L85	Et ₃ Al	Acetophenone	86	61
11	L85	Et ₃ Al	2'-F-acetophenone	63	89
12	L85	Et ₃ Al	4'-F-acetophenone	78	70
13	L85	Et ₃ Al	3'-Br-acetophenone	81	82
14	L85	Et ₃ Al	4'-Cl-acetophenone	80	80
15	L85	Et ₃ Al	3'-Me-acetophenone	65	76
16	L85	Et ₃ Al	4'-Me-acetophenone	60	80
17	L85	Et ₃ Al	3'-MeO-acetophenone	75	84
18	L85	Et ₃ Al	2'-Naphthacetophenone	70	83

The addition of phenylacetylene to 5-methyl-1-*p*-tolylhex-4-en-1-one in the presence of 10 mol% of *ent*-**L86** was less selective and the product was obtained with 62% ee and in 51% chemical yield [180].

The only example of an aluminium-catalysed phenylacetylene addition to ketones in the presence of *Cinchona* alkaloids was studied by Wang [181].

**L81****L95****L85**

Wang attempted to use cinchonidine **L81**, previously used by Singh for the addition to aldehydes [172], for the addition to acetophenone but no product was observed. Two other alkaloids, cinchonine **L95** and quinine **L85**, were also tested and products with low ee's of 27% and 40%, respectively, were obtained (Table 7, entries 1–3). Then, various Lewis acids were investigated; in the absence of any Lewis acid or when Cu(OTf)₂ was added, a very low ee was obtained (Table 7, entries 4 and 5). Very good results were obtained with Al(OiPr)₃ for all alkaloids (Table 7, entries 6–8). Addition in the presence of Et₃Al was even more selective and 80% ee was obtained for **L85** (Table 7, entry 9). Two other alkaloids were less efficient. The reaction was highly sensitive to solvent and the best result of 86% ee for the addition to acetophenone was obtained in diethyl ether (Table 7, entry 10). After the replacement of Al(OiPr)₃ with Et₃Al, the amount of Et₂Zn could be reduced, thus suppressing achiral background reactions. The major drawback of this catalytic system is the high catalyst loading. As much as 80 mol% of **L85** was required for a good enantioselectivity. The results for other ketones are presented in Table 7, entries 11–18.

3. Arylation

Chiral diarylmethanols are important structural elements in many biologically active compounds with interesting properties. For example, clemastine [182] and cetirizine [183] (Zyrtec) are

powerful antihistaminic drugs, and orphenandrine and neobendine [184] have both anticholinergic and antihistaminic properties.

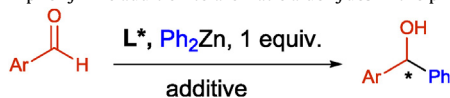
Diarylmethanols are accessible by two methods: an enantioselective reduction of prochiral ketones or an enantioselective aryl addition to aldehydes. The enantioselective reduction of diaryl ketones is a very difficult task due to similar steric and electronic

properties of both aryl groups in the prochiral ketone. However, Corey developed an efficient method for the reduction of some diaryl ketones using chiral oxazaborolidines [185]. Organometallic arylation of aldehydes is a more versatile method which in theory, but as has been shown also in practice, can give access to enantiomerically pure diarylcarbinols. The most studied approach is based on the addition of organozinc compounds to aldehydes. Rhodium [186]–, ruthenium [187]–, nickel [188]–, aluminium/Ti(OiPr)₄ [189]–catalysed enantioselective aryl transfer reactions have also been investigated. While heavy metal-based methods give excellent results, diorganozinc-mediated reaction can be highly advantageous in pharmaceutical applications. In this review, we are presenting achievements based on the diethylzinc-mediated arylation, which is a relatively novel area – the first report was published in 1999.

3.1. Addition to aldehydes

Fu reported the first enantioselective arylation of aldehydes using a diphenylzinc reagent in the presence of ligand **L96** [190]. The major drawback of this methodology is the high activity of diphenylzinc, which reacts with aldehydes without a chiral catalyst, and this background reaction leads to the formation of a racemic product. As a result, the best ee obtained was 57% with excellent chemical yield (Scheme 36). The fact that only one phenyl group is

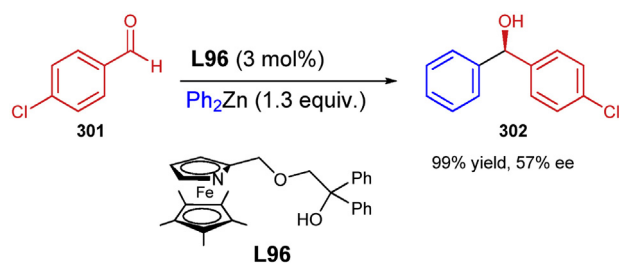
Table 8
Diphenylzinc addition to aromatic aldehydes in the presence of 20 mol% of (R)-**L97** and (S)-**L98**.



Entry	L*	Ar	L* + Additive (mol%)	Aldehyde conc. (mM)	Yield (%)	Ee (%)	Config.
1	(R)- L97	4-MeOC ₆ H ₄	0	50	61	0	R
2	(R)- L97	4-MeOC ₆ H ₄	5 + 0	50	76	54 ^a	R
3	(R)- L97	4-MeOC ₆ H ₄	5 + Et ₂ Zn (10)	50	87	77	R
4	(R)- L97	4-MeOC ₆ H ₄	20 + Et ₂ Zn (40)	50	84	93	R
5	(R)- L97	4-ClC ₆ H ₄	20 + Et ₂ Zn (40)	50	71	57	R
6	(R)- L97	4-ClC ₆ H ₄	20 + Et ₂ Zn (40)	5	86	94	R
7	(R)- L97	2-Naphtyl	20 + Et ₂ Zn (40)	5	66	87	R
8	(R)- L97	(E)-PhCH=CH—	20 + Et ₂ Zn (40)	5	98	50	S
9	(R)- L97	(E)-PhCH=CH—	20 + Et ₂ Zn (80) + MeOH (40)	5	92	77	S
10	(R)- L97	(E)-PhCH=CH—	20 + Et ₂ Zn (80) + MeOH (40)	5	94	83 ^b	S
11	(S)- L98	(E)-PhCH=CH—	20 + Et ₂ Zn (40)	5	82	97	R
12	(S)- L98	4-ClC ₆ H ₄	20 + Et ₂ Zn (40)	5	92	95	S
13	(S)- L98	2-MeC ₆ H ₄	20 + Et ₂ Zn (40)	5	87	91	S
14	(S)- L98	2-Naphtyl	20 + Et ₂ Zn (40)	5	90	88	S

^a 2 Equiv. of Ph₂Zn.

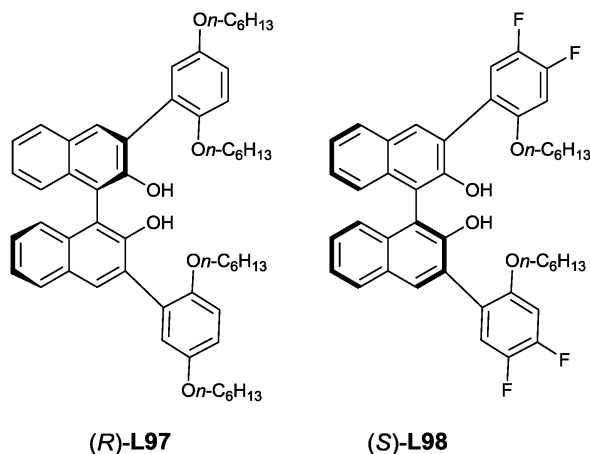
^b In boiling CH₂Cl₂.



Scheme 36. First enantioselective addition of diphenylzinc to aldehyde.

transferred to the product makes such additions less economically viable.

The first highly enantioselective method was developed by Pu [191]. The addition of diphenylzinc to propionaldehyde in the presence of 10 mol% of (R)-**L97** gave (S)-1-phenylpropanol with 87% ee.



This product could be obtained in a much easier way by the addition of Et₂Zn to benzaldehyde, but the reaction was an excellent starting point for further development of the addition to aromatic aldehydes. Indeed, diarylcarbinols were obtained with a comparable or better enantioselectivity, provided that the reaction was performed in the presence of 20 mol% of **L97** (Table 8, entries 2, 3 vs. 4). Unlike Et₂Zn, diphenylzinc reacts easily with aldehydes in the absence of any catalyst (Table 8, entry 1). This background reaction, leading to a racemate, is responsible for the low enantioselectivity

in Ph₂Zn additions. The pre-treatment of **L97** with Et₂Zn before the reaction with Ph₂Zn increases the enantioselectivity, which suggests that thus obtained catalyst is more selective than the one produced directly in the reaction of **L97** with Ph₂Zn. Moreover, a low concentration of the aldehyde has a beneficial effect on the enantioselectivity (Table 8, entries 5 vs. 6), supposedly due to the suppression of the background reaction. The best selectivity was obtained for *trans*-cinnamaldehyde when the reaction was conducted in refluxing methylene chloride (Table 8, entry 10), probably because of the increased activity of the chiral catalyst or because of the formation of another, more active catalytic species.

In order to increase the catalytic activity of the ligand, Pu synthesised a series of fluorinated BINOLs [192]. Only (S)-**L98** could compete effectively with the background reaction and high enantioselectivities were obtained (Table 8, entries 11–14).

Two months later, Bolm published a paper presenting enantioselective addition of pure diphenylzinc to aromatic aldehydes in the presence of ferrocenyl ligand **L99** [193]. In the presence of 5 mol% of the ligand, several diarylcarbinols were obtained with a moderate to good enantioselectivity. As described above, the main obstacle to the high enantiomeric excess was the uncatalysed addition of Ph₂Zn.

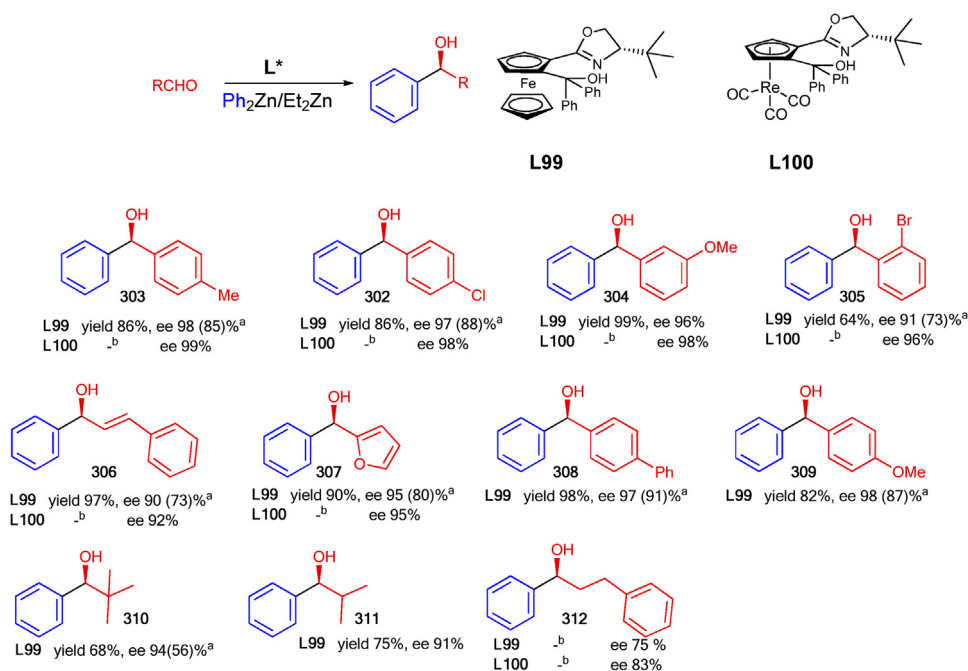
Soon after their first communication, Bolm and co-workers presented an improved approach to **L99**-catalysed addition [194]. They mixed diphenylzinc and diethylzinc in 1:2 (0.65:1.3 equiv.) ratio and obtained excellent enantioselectivity and chemical yield in the reactions catalysed by 10 mol% of ligand **L99**.

An equilibrium of thus formed ethylphenylzinc with diethyl- and diphenylzinc was suggested. This allowed for the complete transfer of the phenyl group to the product, while the ethyl group behaved as a non-transferable one, resulting in much higher yields and ee's than in the case of pure Ph₂Zn. The enantioselectivity for the substituted aryl aldehydes and simple branched alkyl aldehydes was over 90%, reaching 98% for *p*-methyl and *p*-methoxybenzaldehyde (Scheme 37, **303** and **309**).

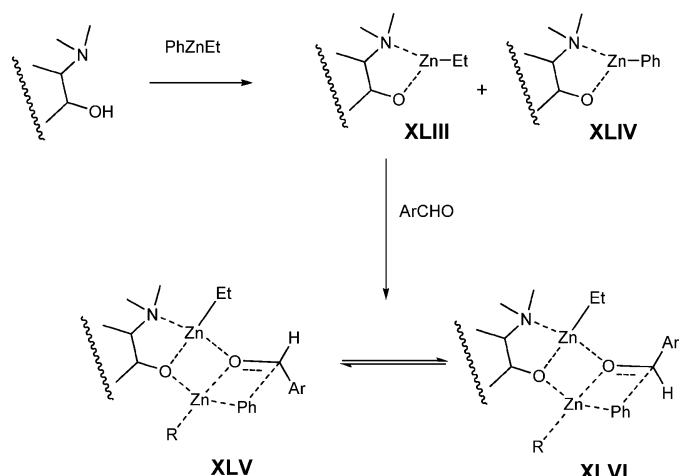
Ligand **L99** was modified to the rhenium complex **L100** in order to increase the activity of the ligand by an enhanced Lewis acidity of the catalyst [195]. Very high enantioselectivities were obtained in most cases (Scheme 37).

Calculations leading to a theoretical explanation of experimental results were presented by Norrby and Bolm [196,197].

The transmetallation step forms the arylation reagent PhZnEt, which reacts with the ligand to form the ethylzinc and phenylzinc species **XLIII** and **XLIV**. The favoured **XLIII** reacts with an aldehyde



Scheme 37. Arylation of aldehydes by $\text{Ph}_2\text{Zn}/\text{Et}_2\text{Zn}$ mixture in the presence of 10 mol% of **L99** and **L100**.



R=Et or Ph

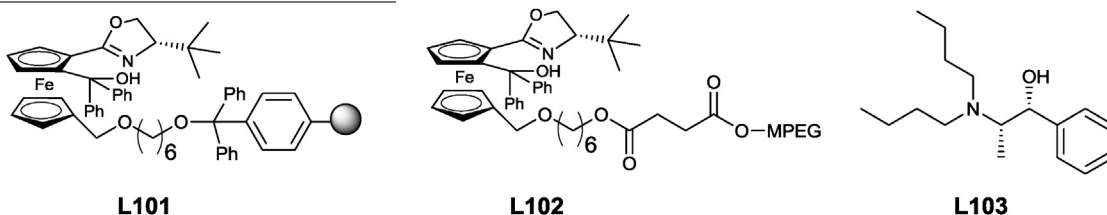
Scheme 38. Transition states of arylation reaction proposed by Norrby and Bolm.

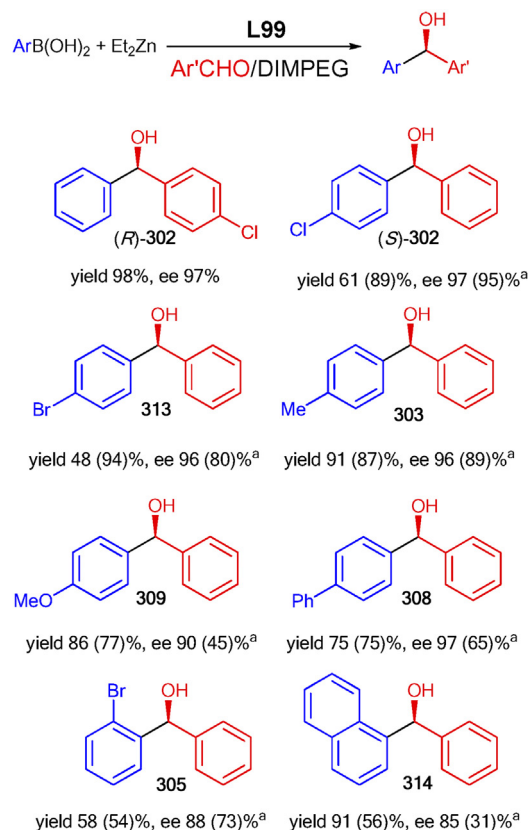
to form tricyclic complexes **XLV** (favoured) and **XLVI**, being in good accordance with Noyori's postulates for diethylzinc addition (**Scheme 38**) [26]. A higher reactivity of the phenyl group, which has to be the transferable one in order to obtain the arylation

product, results from simultaneous overlapping of the phenyl π -system with zinc and the carbonyl group. The energy of such a transition state is significantly lower than that for the ethyl transfer.

The necessity to use expensive diphenylzinc as well as problems with the synthesis of variously substituted diarylzinc compounds remained a serious drawback of the arylation reaction until 2002, when Bolm and Rudolph presented a report on the use of arylboronic acids as a phenyl source [198]. The procedure, which requires the pre-formation of a phenylation reagent over 10 h stirring of PhB(OH)_2 and Et_2Zn at 60°C prior to the addition, worked very well for a broad spectrum of aldehydes, including 2-substituted arylaldehydes, but not for 2-substituted boronic acids. Unfortunately, under these conditions a high excess of diethylzinc (7.2 equiv.) and arylboronic acid (2.4 equiv.) was needed for 1 equivalent of the aldehyde. Arylboronic acid esters were tried without success due to serious problems regarding the purification of the product. An appropriate combination of arylboronic acid and aryl aldehyde allowed for the synthesis of either enantiomer of diarylmethanol. A further improvement of the enantioselectivity, sometimes very impressive, was achieved by the addition of 10 mol% of poly(ethylene glycol) dimethyl ether and very high enantioselectivities of up to 97% and chemical yields were obtained (**Scheme 39**).

Bolm also synthesised soluble polymer-supported ferrocenyl ligands **L101** and **L102** [199].





^a Number in parentheses are for the reaction in the absence of DiMPEG

Scheme 39. Arylation of aldehydes using boronic acids as an aryl source in the presence of 10 mol% of **L99**.

Ligand **L101** was completely inactive in the arylation reaction, while **L102** reacted with an enantioselectivity identical to that of **L99** (10 mol%, 97% ee). The MeO-PEG-supported ferrocene **L102** was recycled 5 times, retaining its excellent activity (95% ee in cycles 3–5). There was no discussion of the influence of the polymeric support on the enantioselectivity but comparable results obtained in the reaction catalysed by **L102** and the reactions in the presence of MPEG (see below) as an additive suggested a great mechanistic similarity of both systems.

Table 9
Influence of aryl source and DiMPEG on asymmetric induction.

Entry	L*	Aryl source	Catalyst loading (mol%)	No additive		DiMPEG as additive (10 mol%)	
				Yield (%)	Ee (%)	Yield (%)	Ee (%)
1	L99	Ph ₂ Zn/Et ₂ Zn	10	93	98 (R)	75	98 (R)
2	L101	Ph ₂ Zn/Et ₂ Zn	10	81	90 (S)	87	95 (S)
3	L99	Ph ₃ B/Et ₂ Zn	10	95	93 (R)	96	96 (R)
4	L101	Ph ₃ B/Et ₂ Zn	10	85	87 (S)	85	93 (S)
5	L99	Ph ₂ Zn/Et ₂ Zn	5	87	95 (R)	74	96 ^a (R)
6	L99	Ph ₂ Zn/Et ₂ Zn	1	84	79 (R)	71	93 ^a (R)
7	L99	Ph ₂ Zn/Et ₂ Zn	0.1	77	3 (R)	72	12 ^a (R)

^a 25 mol% was used.

The “DiMPEG effect” was further studied for two additional arylation systems: (a) the classical Ph₂Zn/Et₂Zn, and (b) a novel one Ph₃B/Et₂Zn, using **L99** and **L103** as catalysts [200]. For the combination of catalyst **L99** (10 mol%) and system (a), a slight decrease of yield was observed and the enantioselectivity remained unaffected; for all the other combinations, a positive effect on both the chemical yield and the asymmetric induction was noted. Furthermore, the addition of DiMPEG allowed for the reduction of the catalyst loading to 5 mol% (Table 9).

The role of DiMPEG in the addition process was discussed. The DFT calculations [196] indicated that Et₂Zn, Ph₂Zn and EtZnPh are at equilibrium and all the species were present in the reaction mixture. Ph₂Zn was the most reactive one and it was involved in the uncatalysed background reaction. The resultant enantioselectivity depended not only on the catalyst ability to stereocontrol the reaction, but also on the ratio of catalysed and background reactions. The presence of Et₂Zn or DiMPEG (or both) in the reaction mixture containing Ph₂Zn shifted the equilibrium towards a mixed organozinc species involved in the catalysed pathway by forming EtZnPh (role of Et₂Zn) or by coordinating the most reactive reagent – Ph₂Zn (the role of MPEG). DiMPEG could be replaced by the monomethyl ether of poly(ethylene glycol) of MW=2000 without the loss of enantioselectivity.

This work was further extended to the synthesis of 1,1'-substituted diarylmethanols (e.g. aryls other than the simple phenyl group) and very good results were obtained [201]. As shown in Scheme 40, an appropriate combination of arylboronic acid and aryl aldehyde allowed for the synthesis of either enantiomer of diarylmethanol using 10 mol% of the same catalyst, **L99**. The applicability of this methodology for the synthesis of some heteroaromatic compounds was demonstrated. Derivatives of thiophene and furan were obtained, but not as the enantiomeric pairs. However, 2-furyl and 3-pirydylyl aldehyde failed to give any desired products.

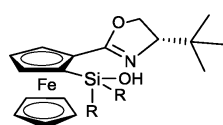
The influence of the amount and type of additives on the enantioselectivity in the arylation reaction was studied using a combinatorial chemistry approach; the predominance of DiMPEG as an additive was confirmed once more [202].

The use of triphenylborane as a source of phenyl in this reaction was fully investigated and reported in another paper [203]. Being commercially available and less expensive than diphenylzinc (albeit more expensive than phenylboronic acid), triphenylborane gave results comparable to the Ph₂Zn/Et₂Zn system. High and very high chemical yields and enantioselectivities were obtained for aromatic and some aliphatic aldehydes in the presence of **L99** (10 mol%), but heteroaromatic aldehydes (except for

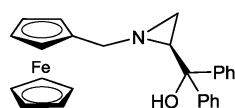
2-thiophenecarbaldehyde) failed to give expected diaryl-methanols. Moreover, triarylboranes other than triphenylborane are difficult to access or require tedious preparations, which narrows the applicability of this methodology to the synthesis of arylphenylmethanols.

Ferrocenes **L104** (10 mol%) substituted with an organosilanol moiety were synthesised and tested as ligands in arylation reactions [204]. The best results were obtained for the arylation of 4-chlorobenzaldehyde with a (1:2) diphenylzinc/diethylzinc mixture in the presence of **L104a** (91% ee) and **L104b** (89% ee). Other ligands, aldehydes and phenyl sources (triphenylborane and phenylboronic acid) gave lower enantioselectivities.

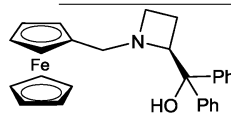
Wang et al. reported the synthesis of ferrocenyl ligands bearing heterocyclic rings in the side chain. Both the aziridino derivative **L105** [205] (10 mol%) and the azetidino ligand **L18** [206] (10 mol%) gave very good to excellent selectivities in the reaction of aryl aldehydes with phenylboronic acid and substituted boronic acids with benzaldehyde (the approach leading to the opposite enantiomer). The use of a proper combination of a substituted boronic acid and a substituted aldehyde allowed for the synthesis of either enantiomer with a high enantioselectivity (Scheme 40). Regrettably, the products were isolated by preparative TLC, being unsuitable for large scale applications.



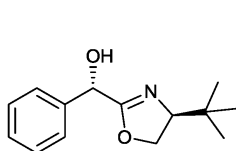
L104 a R=iPr
L104 b R=Ph



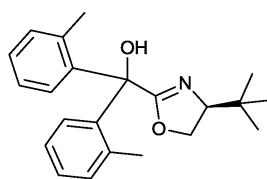
L105



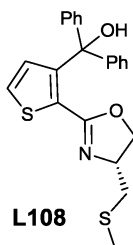
L18



L106



L107



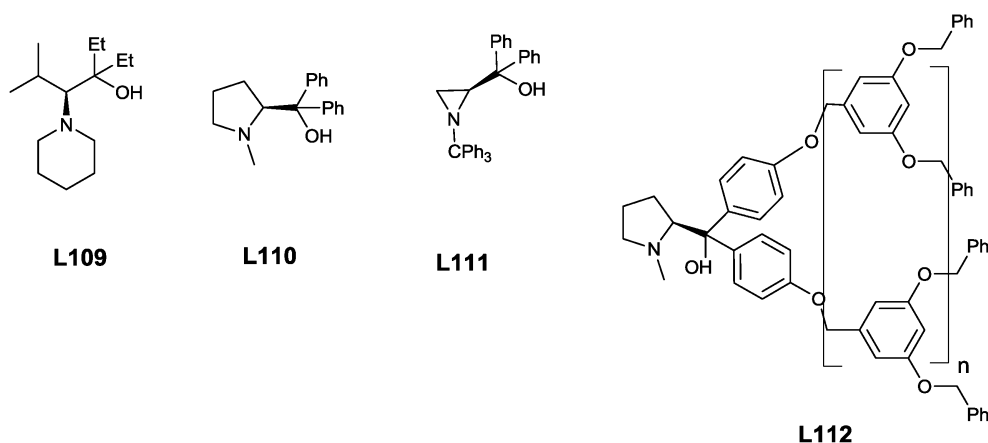
L108

In the early stage of the development of dialkylzinc-mediated arylation reactions, the best enantioselectivities were reported for rather complex ferrocenyl-oxazoline ligands. Bolm attempted to apply less complicated oxazolines **L106** [207] and **L107** [208] with limited success. The L-mandelic acid-derived oxazoline **L106** was very ineffective and the enantioselectivity in the arylation reaction did not exceed 35%. The ligand underwent decomposition during the reaction, probably due to the instability of the benzylic position under the reaction conditions. To prevent this, a series of ligands bearing quaternary hydroxy group were synthesised and tested. The best enantioselectivity (81%) was obtained for the arylation of 4-methoxybenzaldehyde with $\text{Ph}_3\text{B}/\text{Et}_2\text{Zn}$ in the presence of **L107** (10 mol%) and DiMPEG (13 mol%), but this value was still well below the results achieved for ferrocenyl oxazoline **L99**. Zhao synthesised a series of thiophene-derived oxazolines and again, the maximum reported ee was only 82% for the phenylation of 4-chlorobenzaldehyde in the presence of 10 mol% of **L108** [209].

A series of amino alcohol ligands not containing the ferrocenyl moiety was prepared by Braga and co-workers. Ligands **L109–L111** yielded diarylmethanols with good to excellent enantioselectivities. Ligands **L109** [210] and **L110** [211] (20 mol%) gave adducts with enantioselectivities reaching 97–98% ee, having *R* and *S* configurations, respectively. The major drawback of these ligands was

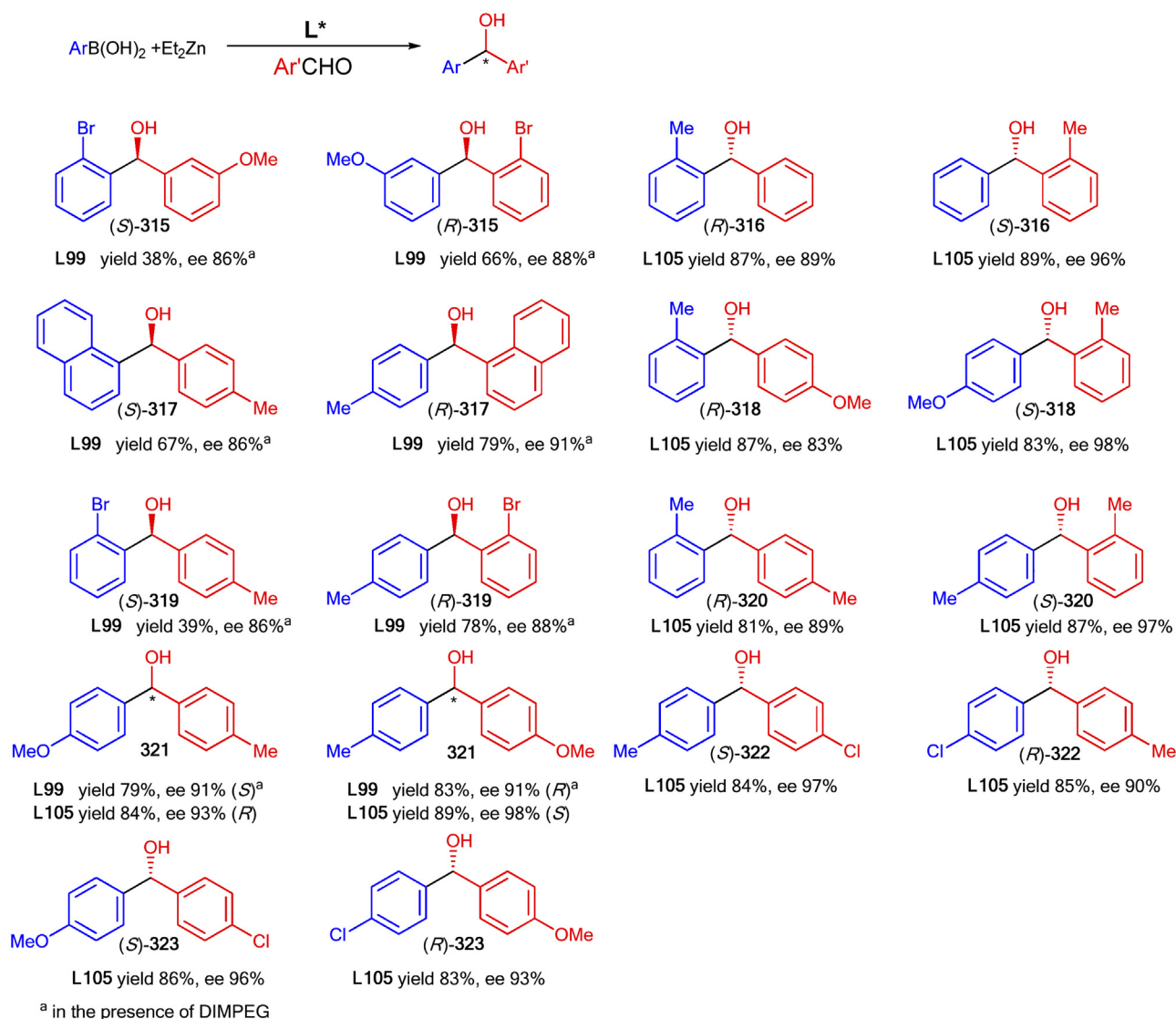
a long reaction time. Therefore, ligand **L111** [212] (10 mol%), having a more rigid structure, was synthesised and the arylation reaction was aided by the use of microwave irradiation. This ligand had inducing capabilities comparable to **L109** and **L110** but under microwave irradiation it gave superior results (Supplementary materials – Scheme S20). Moreover, both stages of the reaction, the phenylethylzinc formation and the aldehyde addition, were accelerated by microwaves, hence the total reaction time was reduced from 24 h to 15 min.

Compound **L110** was independently developed by Zhao's group as an efficient ligand for an aryl transfer to aldehydes. Arylboroxines were introduced as a new aryl source in order to reduce the amount of diethylzinc required for this reaction [213]. Indeed, some improvement was made but still 4 equiv. of Et_2Zn and the preformation of the arylation reagent were required in order to achieve good yield and enantioselectivity. After optimisation, the addition of the phenyl group to a series of substituted aromatic aldehydes proceeded with a very good enantioselectivity of up to 96% ee. Boroxines could easily be synthesised from arylboronic acids, so this method was not restricted only to phenylboroxine. Similar results were obtained for boroxines containing substituted aryl rings.

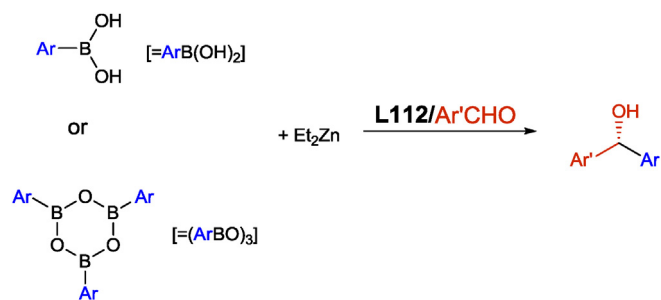


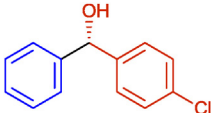
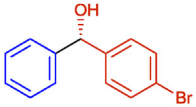
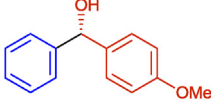
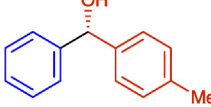
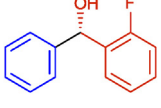
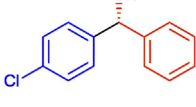
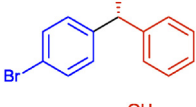
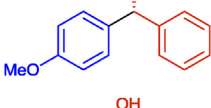
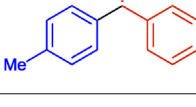
This work was further extended for the application of dendritic ligands based on a very effective pyrrolidinylmethanol ligand, **L112** [214]. For the reaction of 4-chlorobenzaldehyde with phenylboronic acid, the best results were obtained for ligand **L112** (20 mol%) with $n = 2$ (Table 10, entry 3). Lower enantioselectivities were obtained for ligands with $n = 1, 3$ or 4. When phenylboroxine

was used as an aryl source, the reactions proceeded with even higher enantioselectivities (Table 10, entries 6–9). Moreover, ligand **L112** ($n = 2$) could be precipitated with methanol upon the completion of the reaction and reused. The recovered ligand was recycled 5 times with only a slight decrease in the chemical yield and enantioselectivity.

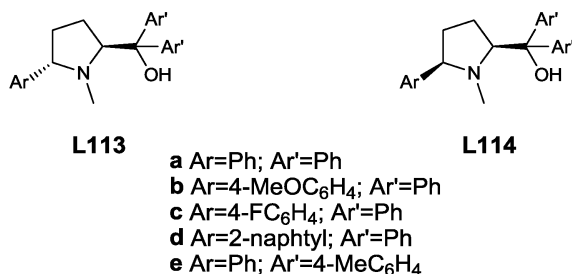


Scheme 40. Synthesis of enantiomeric diarylmethanols by use of proper combination of aldehyde and boronic acid in the presence of 10 mol% of **L99** and **L105**.

Table 10Arylation in the presence of 20 mol% of dendritic ligands **L112**.

Entry	<i>n</i>	Ar source	Diarylmethanol	Yield (%)	Ee (%)
1	0	ArB(OH) ₂		90	88
2	1	ArB(OH) ₂		92	89
3	2	ArB(OH) ₂		98	93
5	3	ArB(OH) ₂		98	91
6	0	(ArBO) ₃		98	94
7	1	(ArBO) ₃		92	97
8	2	(ArBO) ₃		98	98
9	3	(ArBO) ₃		97	96
10	2	(ArBO) ₃		92	98
11	2	(ArBO) ₃		98	98
12	2	(ArBO) ₃		98	93
13	2	(ArBO) ₃		88	95
14	2	(ArBO) ₃		92	98
15	2	(ArBO) ₃		95	95
16	2	(ArBO) ₃		80	89
17	2	(ArBO) ₃		86	91

A series of triarylprolinol ligands **L113a–e** and **L114a–e** was synthesised by a Heck reaction and used for the arylation reaction [215]. All of them gave products with very good enantiomeric excesses; the best ee's were observed for ligands **L113a** and **L114a** bearing unsubstituted phenyl rings.

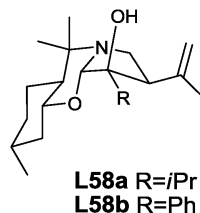


A comparison with the simpler **L110** revealed that a similar level of enantioselection was achieved with only 10 mol% of new ligands. The additional stereogenic centre had no influence on the level of the asymmetric induction; however, lower yields of diaryl-methanols were obtained. The addition of DiMPEG had an adverse effect and its addition led to lower ee's and moderate yields (Supplementary materials – Scheme S21). The outcome of the reaction was highly dependent on the temperature: the maximum was found at 0 °C, while higher and lower temperatures led to decreased ee's and yields.

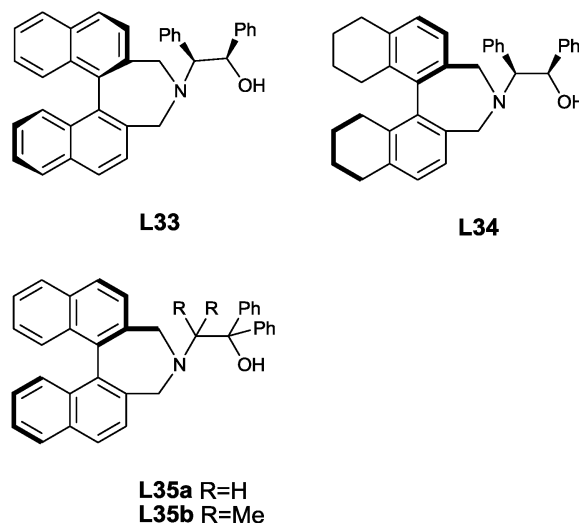
Perhydrobenzoxazines **L58**, successfully used in alkynylation of α -ketoesters [129], are also excellent ligands for the enantioselective arylation of aldehydes; the best results were obtained in the presence of **L58b** [216]. Boroxines were used as a boron source and according to presented procedure, the ratio of boroxine and Et₂Zn to an aldehyde could be reduced to 0.6 and 2.4 equiv., respectively. Moreover, only 30 min of heating at 60 °C was sufficient for the boron-ring exchange and the formation of ArZnEt reagent to take place. This was a substantial improvement to the classical method based on free boronic acids. The optimum temperature of the addition process was 0 °C; at +20 °C and –20 °C the reaction proceeded with a lower enantioselectivity. The arylation reaction in the presence of **L58b** occurred with a high enantiocontrol, with only a slight influence of substituents of the aryl ring. No additives were needed for a high enantioselectivity except for the reaction of 4-MeC₆H₄-substituted boroxine with 4-CF₃C₆H₄CHO. In this case, standard conditions led to the racemic **329**. However, the reaction in the presence of 10 mol% DiMPEG gave the respective diarylmethanol with 95% ee, albeit in low chemical yield for both (*R*)-**329** and (*S*)-**329** (Supplementary materials – Scheme S22).

The arylation of aryl aldehydes in the presence of 1,1'-binaphthylazepine **L35a** proceeded with poor enantioselectivities

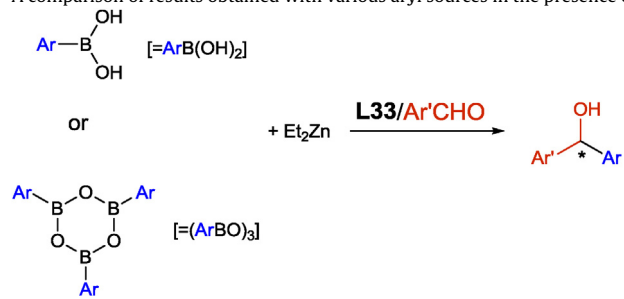
not exceeding 53% ee [115]. Again, and similarly to the alkynylation reaction (see above), the azepines (10 mol%) **L33** [217] and, in some cases, **L34** [114] performed much better. The arylation of various substituted aryl aldehydes with phenylboronic acid gave products with a consistent enantioselectivity of over 90%. Moreover,

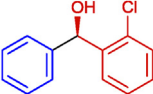
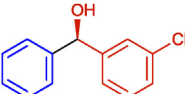
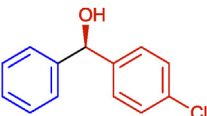
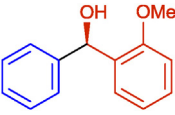
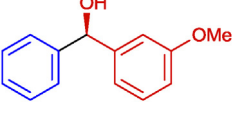
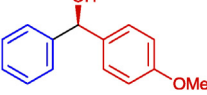
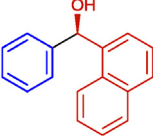
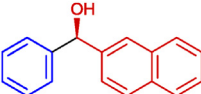
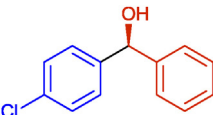
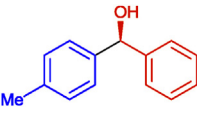


excellent results were obtained also for cyclohexanecarbaldehyde (91% ee), furfural (92% ee), and 2,2-dimethylpropanal (96% ee). While the authors explain the high asymmetric induction achieved by high rigidity of the ligand aided by the chiral wall from the binaphthyl skeleton, a comparison of ligand **L33** to ligand **L35a** clearly shows importance of the 1,2-diphenyl-1-hydroxyethyl substituent at azepine nitrogen as well.

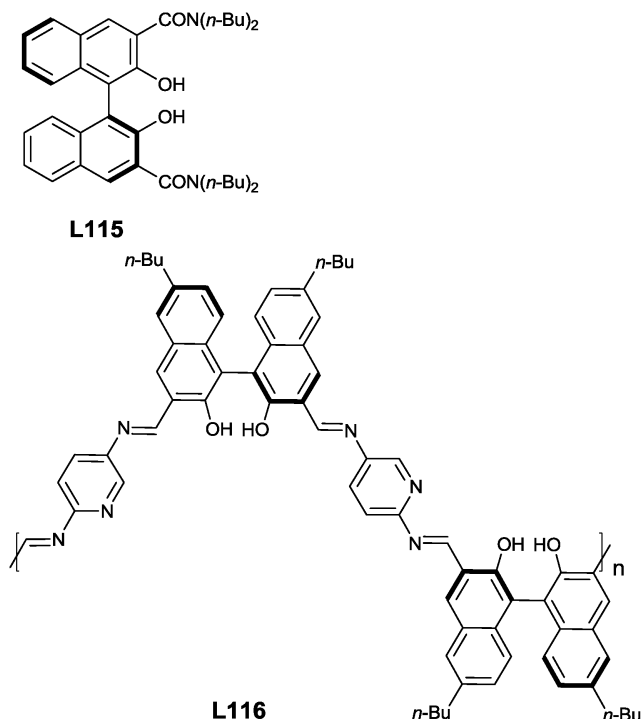


Asymmetric induction in the presence of ligand **L33** was studied again using arylboroxines as an aryl source [218]. A comparison of the results obtained in this study and the previous ones [217] is presented in Table 11. A standard procedure was used for the

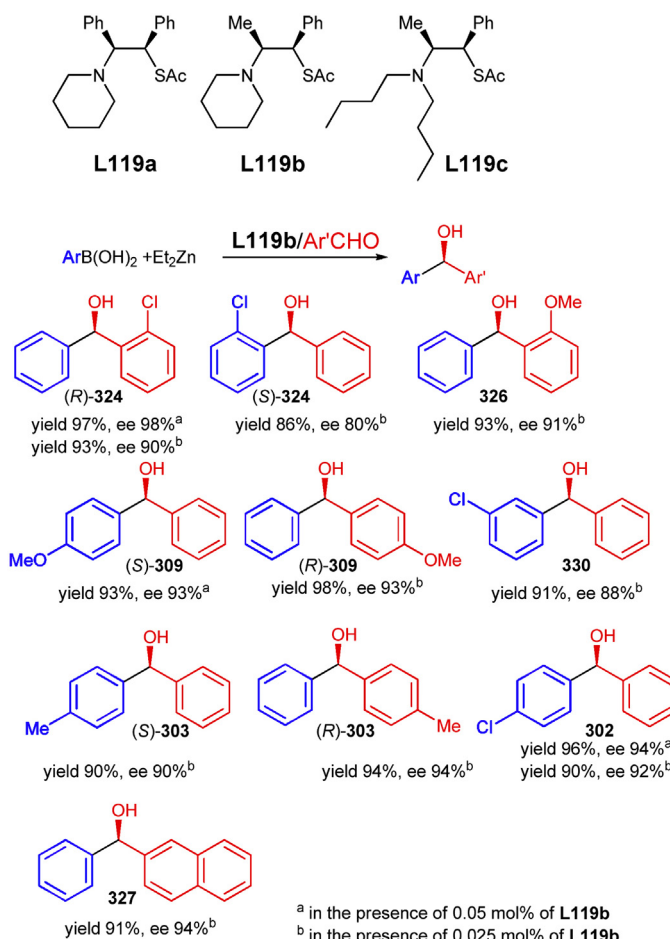
Table 11A comparison of results obtained with various aryl sources in the presence of 10 mol% of **L33**.

Entry	Ar source	Diarylmethanol	Yield (%)	Ee (%)
1	ArB(OH) ₂		96	90 (R)
2	(ArBO) ₃		87	94 (R)
3	ArB(OH) ₂		95	94 (R)
4	(ArBO) ₃		75	95 (R)
5	ArB(OH) ₂		96	96 (R)
6	(ArBO) ₃		87	93 (R)
7	ArB(OH) ₂		96	93 (R)
8	(ArBO) ₃		84	94 (R)
9	ArB(OH) ₂		95	97 (R)
10	(ArBO) ₃		85	95 (R)
11	ArB(OH) ₂		93	98 (R)
12	(ArBO) ₃		80	94 (R)
13	ArB(OH) ₂		91	95 (R)
14	(ArBO) ₃		84	96 (R)
15	ArB(OH) ₂		94	93 (R)
16	(ArBO) ₃		87	89 (R)
17	ArB(OH) ₂		95	87 (S)
18	(ArBO) ₃		85	78 (S)
19	ArB(OH) ₂		95	93 (S)
20	(ArBO) ₃		98	94 (S)

pre-formation of the arylzinc reagent, consisting of heating boronic acid or boroxine with 4 equiv. of Et_2Zn for 10 h.

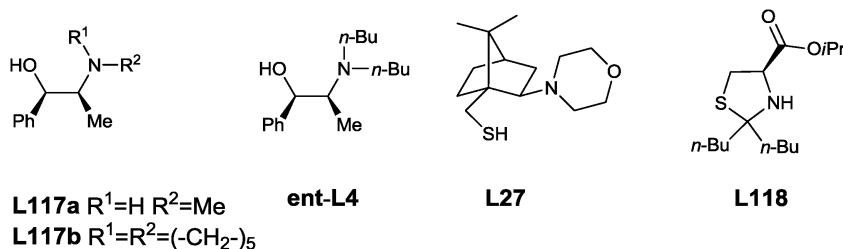


The (*R*)-BINOL-derived diamide **L115** (10 mol%) was applied for the arylation with phenylboronic acid; diarylmethanols were obtained in very good yields and with selectivities of up to 96% ee. The influence of solvents on the enantioselectivity was studied and the addition of diethyl ether or *tert*-butyl methyl ether to toluene was found to enhance the asymmetric induction. Other polar solvents, as well as DiMPEG, had an adverse effect [219]. The soluble polymer **L116** was used for the arylation of aromatic aldehydes with good to very good asymmetric induction (up to 92% ee), but not for the aliphatic ones. **L116** could be recycled and reused twice without a significant loss of enantioselectivity, but its inductive abilities deteriorated quickly in the following cycles [220].

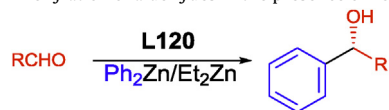


Scheme 41. Arylation in the presence of thioacetate **L119b**.

Ephedrine and norephedrine ligands **L117** and **ent-L4** were examined in the arylation reaction with limited success and a low to moderate asymmetric induction was reported [221].

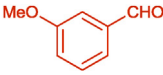
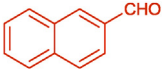


Amino thiol **L27** (10 mol%), having been successfully used in the alkynylation reaction (see above), gave enantioselectivities over 95% for aromatic aldehydes in the arylation reaction [103]. An excellent result, over 99.5% ee and 95% chemical yield, was obtained for 2-methylbenzaldehyde, giving a direct precursor for (*R*)-orphenandrine. Thiazolidine **L118** performed significantly worse and the obtained enantiomeric excesses were low to moderate [222]. Thioacetates **L119** could be placed among the best arylation ligands; excellent enantioselectivities were reported at a very low catalyst loading, 0.025–0.05 mol% (**Scheme 41**) [223].

Table 12Phenylation of aldehydes in the presence of 10 mol% of 2-piperidino-1,1,2-triphenylethanol **L120**.

Entry	Aldehyde	T [°C]	L120 (mol%)	Yield (%)	Ee (%)
1		0	10	91	93
2		RT	10	82	95
3		0	10	61	98
4		0	10	84	98
5		0	10	99	96
6		RT	10	91	95
7		0	10	n.d.	97
8		RT	10	70	96
9		0	10	96	90
10		0	10	84	82 ^a
11		0	10	98	94 ^a
12		0	10	91	97
13		10	1.5	90	95
14		10	1.5	86	99
15		10	1.5	85	94
16		10	1.5	94	91

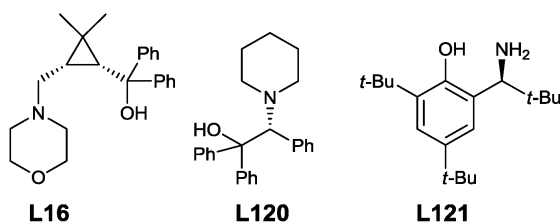
Table 12 (Continued)

Entry	Aldehyde	T [°C]	L120 (mol%)	Yield (%)	Ee (%)
17		10	1.5	79	95
18		10	1.5	65	92

^a In toluene.

Similarly good results were obtained for the addition of phenylboronic acid to aryl aldehydes and arylboronic acids to benzaldehyde. In the latter case, 2- and 3-chlorobenzaldehyde gave slightly lower ee's. Amino thioacetates were much more selective than the corresponding amino alcohols.

As suggested, this enhanced activity may have had its origin in a stronger affinity of zinc to sulphur.



Chiral cyclopropane-based amino alcohol **L16**, applied successfully for phenylacetylene and methyl propiolate additions [94], was less effective for arylation reactions, and enantioselectivities up to 89% ee were obtained [92].

Ligand **120** was introduced into arylation chemistry by Pericàs and co-workers [224]. This catalyst appeared to be extremely active for phenylation reactions and excellent results were obtained with 10 mol% of **L120** (Table 12) using the mixed $\text{Ph}_2\text{Zn}/\text{Et}_2\text{Zn}$ reagent. Hexane appeared to be the solvent of choice for these reactions.

A high selectivity was reported for aromatic aldehydes (Table 12, entries 1–8) as well as the aliphatic ones which had no hydrogen atoms at the α -position to the formyl group (Table 12, entries 11 and 12). Similar enantioselectivities were observed at room temperature and at 0 °C. An investigation of the correlation between the temperature and the enantioselectivity of the phenylation of 4-methylbenzaldehyde using 1 mol% of **L120** revealed that the maximum enantiomeric excess could be obtained at about 10 °C. Performing reactions at higher and lower temperatures led to a reduced enantiodiscrimination. These results can be explained by the isoinversion principle [225]. The catalyst loading could be lowered to 0.5–1.5 mol% of **L120** at 10 °C without a significant loss of enantioselectivity. The results obtained at inversion temperature (10 °C) with 1.5 mol% of ligand are presented in Table 12, entries 13–18.

The very elegant studies based on solution FT-IR and DFT calculations proved experimentally that upon mixing Ph_2Zn and Et_2Zn in 1:2 ratio, PhEtZn species were responsible for the arylation process. The presence of an amino alcohol ligand in the reacting mixture was essential for high enantioselectivity and yield of the

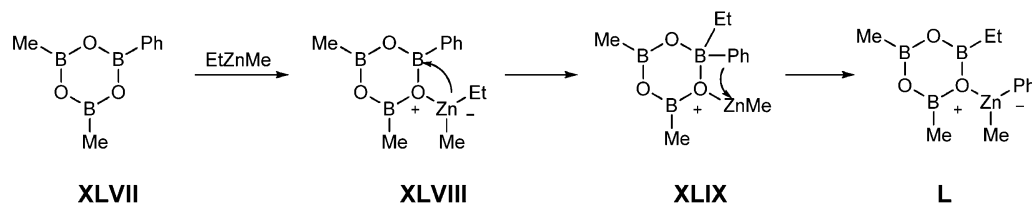
reaction. In the absence of Et_2Zn and an amino alcohol, the addition of Ph_2Zn was rapid even at 0 °C (25 min to completion) and very fast in the presence of 5 mol% of **L120** (5 min). However, there were relatively small differences in the rates of the catalysed and uncatalysed reactions, and the uncatalysed background reaction highly influenced the final result. The situation changed drastically when a 1:2 mixture of $\text{Ph}_2\text{Zn}/\text{Et}_2\text{Zn}$ was used. The rate of the uncatalysed background reaction was very low, while the catalysed one was complete after 15 min. Further experiments showed the absence of a very reactive Ph_2Zn in the reaction mixture and a very low reactivity of PhEtZn in the uncatalysed pathway, which allowed for a high enantioselectivity of the process.

The same ligand was further used in the industrial application aimed at the synthesis of a potential precursor of the mGlu2 receptor [226]. As previously, arylboroxine was used as the most efficient boron source, but 15 mol% of **L120** was required in this case to achieve an enantioselectivity of over 94%. The reaction could be scaled up to several kilograms with essentially the same selectivity. A deep insight into the mechanical aspects of the boroxine-based boron-to-zinc transmetallation was later presented [227]. The theoretical calculations of boron-to-zinc transmetallation using a slightly simplified model indicated the most plausible pathway (Scheme 42). It also became apparent that the formation of ArZnEt should be a very fast reaction.

Microcalorimetric measurements allowed for the calculation of reaction heats and showed that the reaction of triarylboroxines and Et_2Zn leads to the formation of ArZnEt compounds in a fast and reversible manner.

The subsequent experiments fully confirmed these calculations. The transmetallation process was indeed very fast, a reaction time below 30 min was sufficient, as confirmed by ^{11}B NMR of the post-transmetallation mixture, as well as by HPLC and ^1H NMR analysis of the post-reaction mixture. The addition to aldehydes proceeded with a high enantioselectivity (Scheme 43).

In theory, enantiomeric diarylmethanols are readily available by a proper combination of an aldehyde and an arylation reagent. However, as it has already been shown in this review, not every pair of reagents leads to the same level of enantiodiscrimination. Good examples include (*S*)- and (*R*)-**326**, and particularly (*S*)- and (*R*)-**329** in Scheme 43. This behaviour could be attributed to the rates of catalysed vs. uncatalysed reactions. The influence of electron-donating and electron-withdrawing substituents in the aryl groups of both boroxine and aldehyde was calculated. The catalysed reaction of aldehydes bearing electron-withdrawing groups with an arylation reagent bearing electron-donating



Scheme 42. Boron-to-zinc transmetallation.

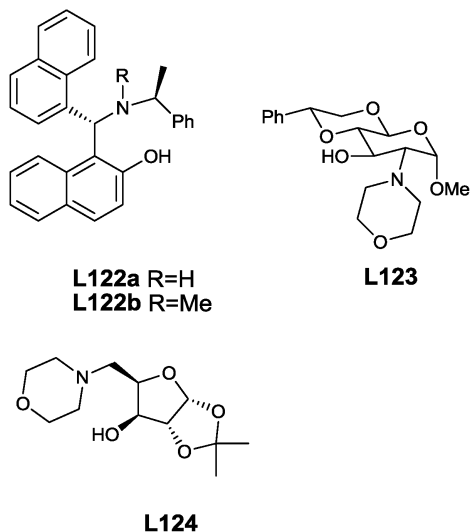
groups was significantly faster than the reaction with the opposite arrangement of substituents.

Another report, being a part of broader studies on the structural optimisation of amino alcohol ligands, was presented later. Again, ligand **L120** (5 mol%) gave the highest asymmetric induction with enantioselectivities in the range of 92–97% for some selected aryl aldehydes, when the $\text{ZnEt}_2/\text{ZnPh}_2$ reagent was used as an aryl source [228].

A series of chiral aminophenols were synthesised and used as ligands for the arylation reaction. The best ligand in the family was compound **L121** (20 mol%) [229].

The enantioselectivity and, in some cases, chemical yield were highly dependent on the substitution of the phenyl ring; over 99% ee was obtained for **316** (Supplementary materials – Scheme S23).

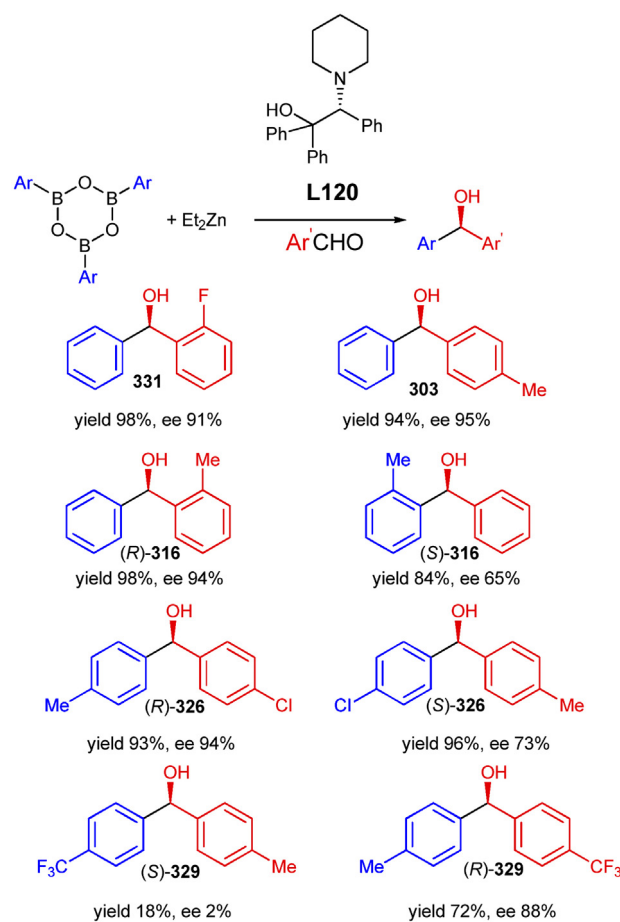
Two ligands, structurally relevant to the Betti's base, were used for the arylation of aldehydes. Chan's ligand **L122b** bearing a tertiary amine gave excellent enantioselectivities for 2-substituted benzaldehydes (Cl-, Br-, F-, Me-, OMe-), all in the range of 94–99%. Three other aldehydes (3-Me-, 4-Me-, 4-Br-) also reacted with 94–95% ee. The initial amount of 10 mol% of the ligand could be reduced to 1% still giving 95% ee (for 2-Me) [230]. Dahmen's ligand **L122a** (5 mol%) was also very selective for these and some other aldehydes [231], albeit the selectivity did not match exactly that of ligand **L122b**.



The same article [231] disclosed a new phenyl source for arylation reactions. The ammonia complex of triphenylborane was much more stable than the parent compound and gave a substantially higher asymmetric induction. A thorough kinetic study on the addition of triphenylborane, its ammonia complex, as well as another potential phenyl source – 2-aminoethyl diphenylborinate, was performed.

The phenylzinc reagent (PhZnEt) was rapidly formed from triphenylborane and remained at a high level during the reaction, which resulted in a low asymmetric induction due to the uncatalysed background reaction. In the case of ammonia complex, the amount of PhZnEt remained at a very low level, below 5%, and the enantioselectivity was very high (97%). The borinate quickly used only the first phenyl group, and then the reaction slowed down substantially, although the enantioselectivity was still very good (87%). The major drawback of this new system is the difficult access to substituted triarylboranes, considerably increasing the limitations of this method.

Derivatives of readily accessible monosaccharides D-glucosamine and D-xylose were investigated as ligands. Ligand **L123** was very ineffective (ee's slightly over 30%), but a simple

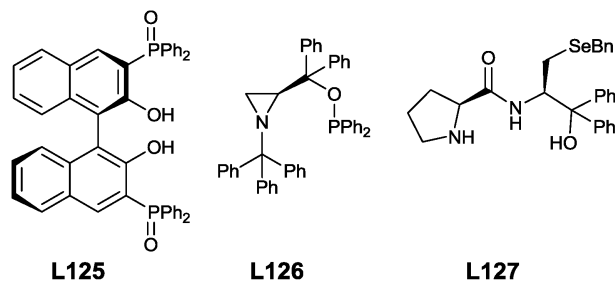


Scheme 43. Arylation of aromatic aldehydes using boroxines and 15 mol% of ligand **L120**.

D-xylose-derived ligand **L124** (20 mol%) secured enantioselectivities in the range of 80–88%, reaching 90% and 96% for 2- and 4-methylbenzaldehyde, respectively. The replacement of the C5 morpholinyl substituent by piperidine in ligand **L124** led to a much lower asymmetric induction [232].

Pericàs's group presented two articles dealing with arylation catalysed by polymer catalysts supported on polymeric resins [233,234]. While the idea of a reusable catalyst is very important, moderate enantioselectivities obtained in both systems make them of a little practical value.

Ishikara developed (R)-3,3'-bis(diphenylphosphinoyl)-BINOL **L125** (10 mol%) for the phenylation of aryl aldehydes with $\text{Ph}_2\text{Zn}/\text{Et}_2\text{Zn}$ mixture; the enantioselectivity did not exceed 88% [235].

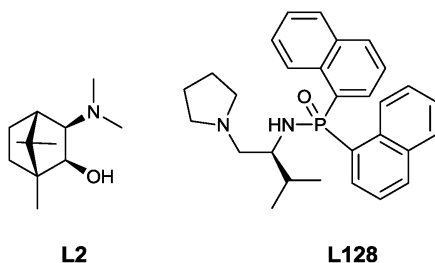


The amino-phosphinite aziridine **L126** (10 mol%) was slightly more efficient and the arylation of aldehydes with phenylboronic acid proceeded with an enantioselectivity of up to 93% for

4-methyl- and 2-bromobenzaldehydes [236]. For the first time, chalcogen peptides were used as ligands for the arylation reaction. The best ligand, **L127**, catalysed the addition of phenylboronic acid to 2-methylbenzaldehyde with 91% ee and in 98% chemical yield (at 10 mol% of the ligand). This was the highest enantioselectivity achieved in this study; the addition to other aldehydes as well as the addition of *para*-substituted phenylboronic acids proceeded with an enantioselectivity not exceeding 80%. The addition of DiMPEG or lowering the temperature had an adverse effect on the enantioselectivity [237].

3.1.1. Addition to ketones in the absence of $\text{Ti}(\text{OiPr})_4$

The enantioselective arylation of ketones is an excellent method for the synthesis of tertiary alcohols. However, this method is also one of the most challenging due to the low reactivity of ketones and problems with the differentiation of both sites of the carbonyl group by catalysts. The first example of a catalytic addition of Ph_2Zn to ketones in the presence of 15 mol% of (+)-DAIB **L2** was presented by Dosa and Fu [238] with a maximum of 91% ee obtained for the phenylation of 1-(3-bromophenyl)propan-1-one.



Since then, only one example of an aryl addition which does not use $\text{Ti}(\text{OiPr})_4$ to facilitate the reaction was published. Ishikara studied the arylation of ketones catalysed by phosphoramidate **L128** using arylboroxines as an aryl source [239]. After optimisation, good to excellent enantioselectivities and chemical yields were obtained for the addition not only to substituted acetophenones, but also to heteroaromatic and alicyclic ketones (Scheme 44).

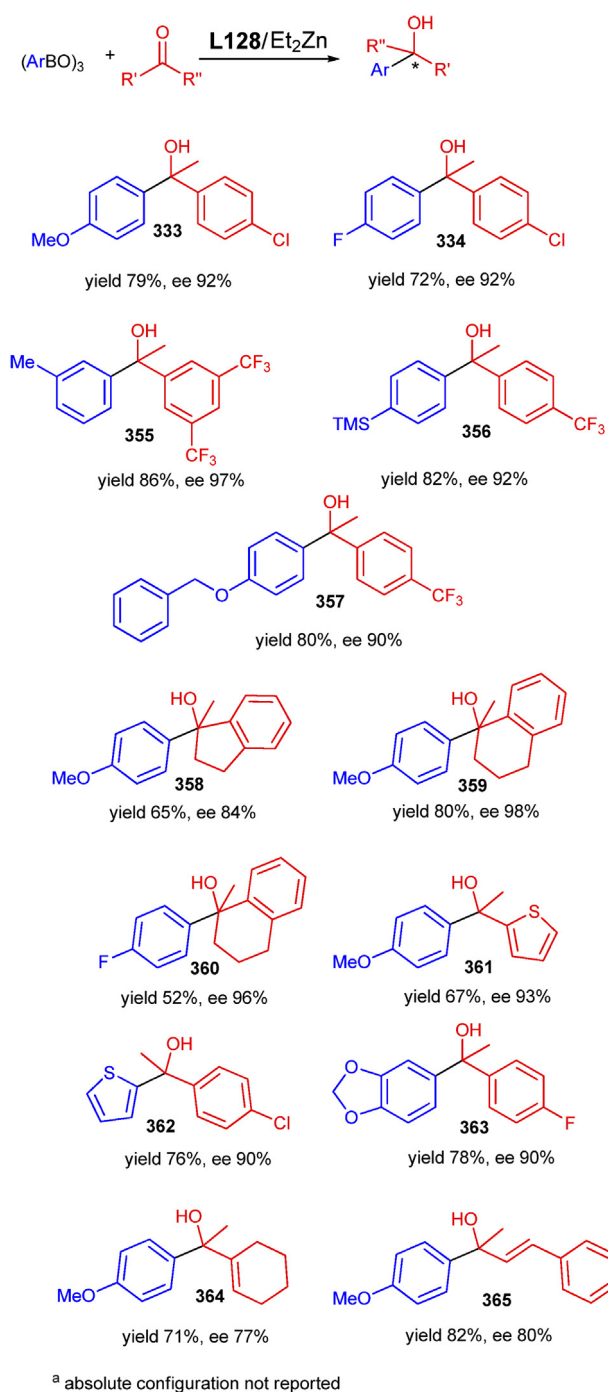
3.1.2. Addition to ketones in the presence of $\text{Ti}(\text{OiPr})_4$

The mechanism of the arylation of ketones promoted by $\text{Ti}(\text{OiPr})_4$ has not been investigated, but one can assume a close analogy to an alkyl addition. The $\text{Ti}(\text{OiPr})_4$ can form a complex with a chiral ligand, and then the aryl transfer from zinc to titanium would occur, followed by the addition to ketone (see Section 1, Scheme 3). Additionally, the co-ordination of the carbonyl group to the Lewis acidic titanium atom would activate the ketone, thus facilitating the addition reaction.

Bis(sulphonamide) ligand **L94** was introduced by Walsh [240] and has been used for highly efficient alkylation and phenylation (by Ph_2Zn) of ketones [241].

The first example of a dialkylzinc-mediated, titanium-promoted arylation of ketones, using boronic acids as an aryl source in the presence of ligand **L94** (5 mol%), was published by Yus et al. [242]. The asymmetric induction for the phenyl addition to ketones was lower than that of the addition of ZnPh_2 as reported in the same paper and earlier, for another set of aldehydes, by Fu and Dosa [238], but – in one case – exceeded 90% ee.

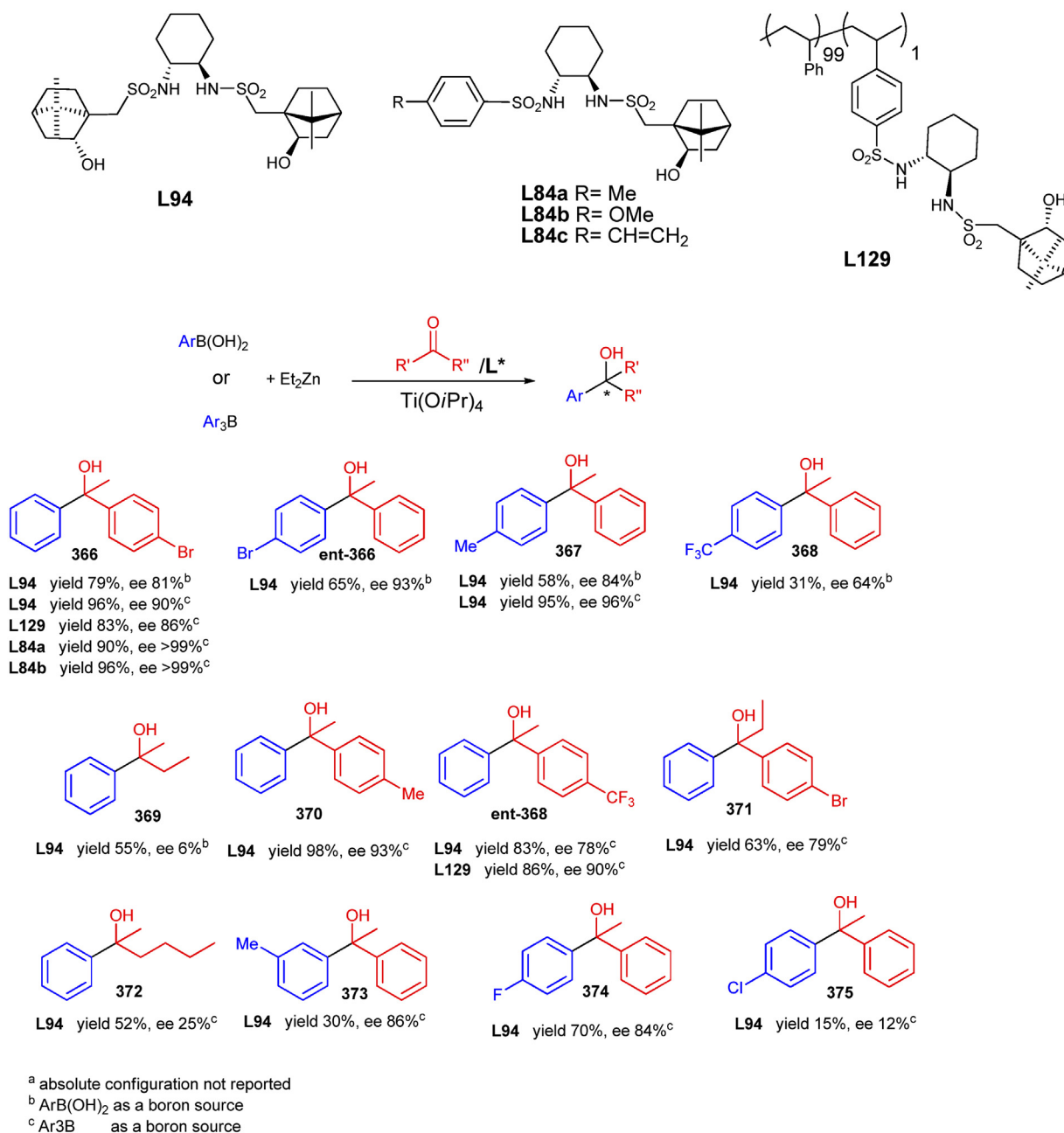
This work was further extended using nonsymmetrical ligands **L84a,b** and symmetrical ligand **L94** (5 mol%) [175,243]. In order to overcome the known disadvantage of phenylboronic acids as aryl sources, namely the necessity to employ a large excess of diethylzinc, Ph_3B was used instead. This modification gave excellent results and an enantioselectivity of over 99% was obtained for some ketones, but only using Ph_3B as an aryl source. The reaction failed to give any products when other commercially available



Scheme 44. Arylation of ketones in the presence of 10 mol% of ligand **L128**.

triarylboranes were used. This problem was only partially solved by the preparation of triarylboranes via the reaction of an appropriate aryl Grignard reagent with trifluoroboron etherate. The summary of results obtained by Yus et al. is shown in Scheme 45.

A styryl derivative, **L84c**, was synthesised and immobilised by the co-polymerisation with styrene and divinylbenzene using various amounts of a cross-linker to give four polymeric ligands [244]. The soluble polymer, **L129**, performed the best, with enantioselectivities over 80%, albeit reported ee's were lower than in the case of **L84** and **L94** (Scheme 45, **366** and **ent-368**). Moreover, the activity of this ligand decreased rapidly after several uses. Similar results were obtained with chiral dendrimers obtained by the radical addition to Frechet-type dendron [245]. Ligands grafted on

Scheme 45. Arylation of ketones in the presence of 5 mol% of ligands **L84** and **L94**.

commercially available polystyrene performed worse and ee's in this case did not exceed 70% [246]. As expected, in all cases, the activity of immobilised ligands was highly dependent on the structure of the polymeric core.

4. Alkenylation

4.1. Alkenylation based on boron–zinc transmetalation

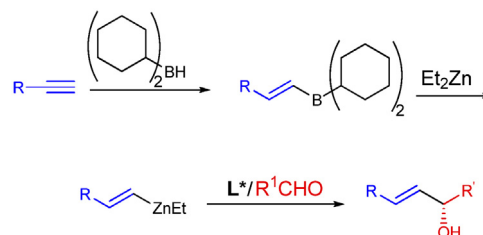
4.1.1. Addition to aldehydes

Chiral, enantiomerically pure secondary allylic alcohols constitute an important starting material for the synthesis of many natural and biologically active compounds.

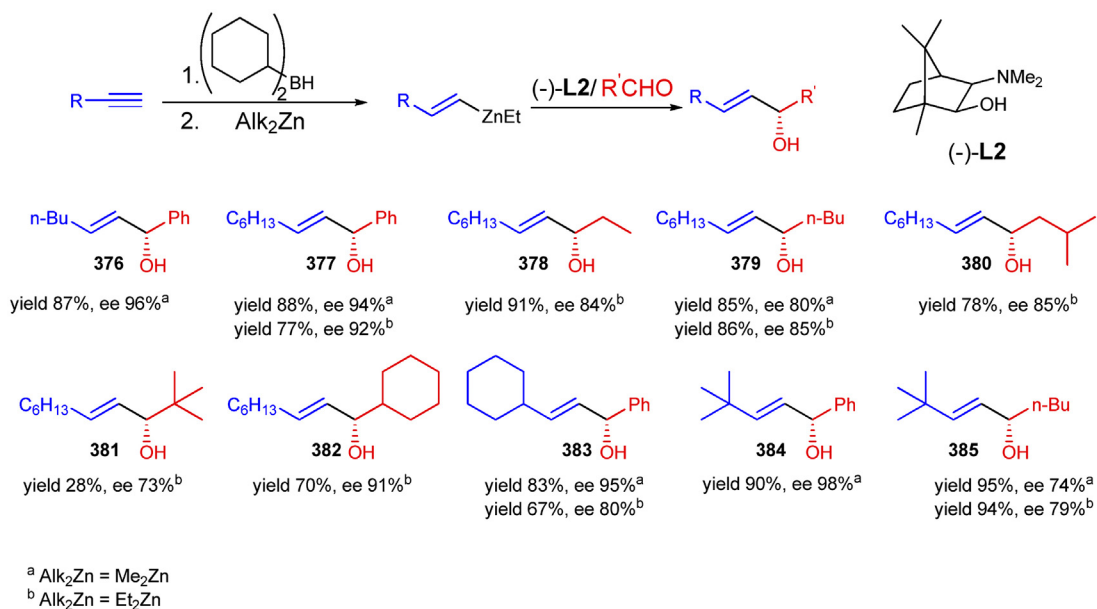
Oppolzer used divinylzinc [247] and alkenylzinc bromide [248] for an efficient enantioselective synthesis of allylic alcohols. However, the tedious preparation of such reagents, the necessity of

thorough removal of catalytically active salts of lithium and magnesium prompted him to look for another, more versatile method.

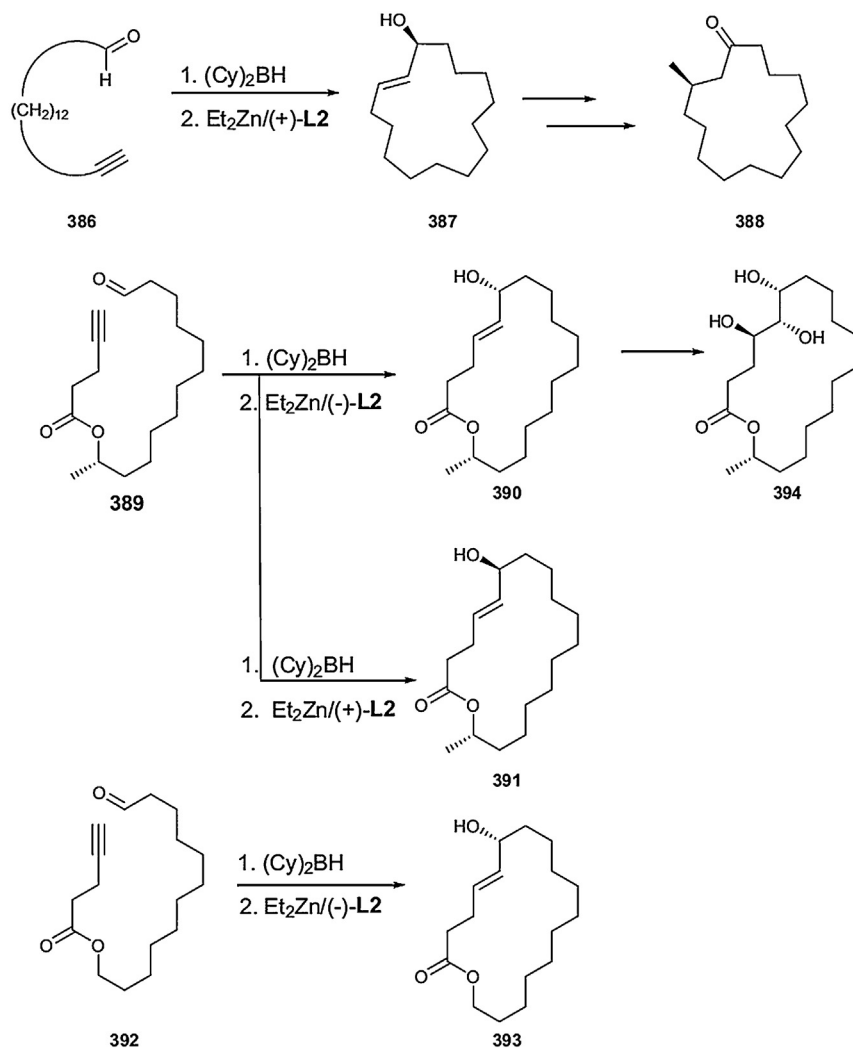
In 1992, Oppolzer and Radinov introduced a new method for an enantioselective alkenylation of aldehydes catalysed by chiral amino alcohols (Scheme 46) [249].



Scheme 46. Alkenylation of an aldehyde by Oppolzer's method.



Scheme 47. Alkenylation of aldehydes using 1 mol% of (–)-L2.

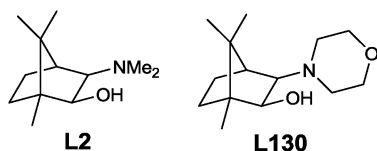


Scheme 48. Synthesis of (–)-muscone **59** and (+)-aspicilin **65**.

In this method, acetylenes are converted to alkenylboron reagents which in turn are transmetalated to mixed alkylalkenylzinc compounds. Among chiral ligands tested, (–)-DAIB (3-*exo*-(dimethylamino) isborneol) **L2** (1 mol%) gave the best results, with enantioselectivities of up to 92–98% ee. The use of more expensive Me₂Zn in most cases led to comparable or lower enantiomeric excesses, although Me₂Zn gave much higher ee's for benzaldehyde (Scheme 47, **383**). The transfer of an alkenyl moiety was complete and in neither case the products of the addition of methyl or ethyl to an aldehyde could be detected.

This new method was used for the synthesis of (*R*)-(–)-muscone **388** [250]. The key step of the synthesis is an asymmetric macrocyclization of the ω -alkynal **386**, affording 75% yield and 92% ee (Scheme 48). The same approach was applied to the synthesis of (+)-aspicilin **394** [251]. The macrocyclization of ω -alkynal **389** catalysed by (–)-**L2** gave **390** in 60% yield and dr 91:9. The catalysis by enantiomeric (+)-**L2** led to **391** in 40% yield and dr 85:15. The influence of the pre-existing stereogenic centre was then tested by the reaction of achiral **392** leading to **394** in moderate yield and 79% ee. This indicates that the key intermediate for the synthesis of (+)-aspicilin, namely compound **390**, can be obtained in the reaction of a matched pair of **389** and (–)-**L2**.

The proline derivative **L110**, which gave good results comparable to DAIB in Oppolzer's study, was applied by Soai and co-workers to the synthesis of the enantiomerically enriched (77% ee) sphingosine [252].

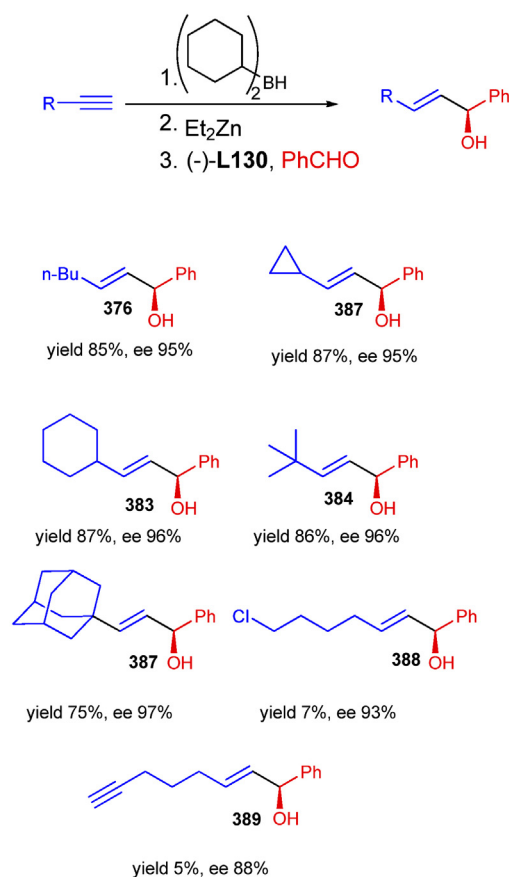


Oppolzer's method was further improved by the introduction of new, more efficient ligands. Walsh used the isborneol-derived (–)-MIB ligand [253] **L130** (2 mol%), and the products of the vinylation of benzaldehyde were obtained with an excellent enantioselectivity [254]. Various substituted acetylenes were used as substrates and enantioselectivities over 95% were obtained, regardless of the bulkiness of their substituents (Scheme 49).

(+)-MIB is also readily available from the opposite camphor enantiomer, which greatly enhances the applicability of this method. A weak positive nonlinear effect was observed. The transition state model proposed for this reaction is shown in Fig. 10.

The products of the alkenylation reaction were further converted to allylic amines using Overman's [3,3]-sigmatropic rearrangement of trichloroacetimidates, and the amines were, in turn, transformed into α -amino acid derivatives (Scheme 50). Both D- and L- α -amino acids are readily accessible just by the proper choice of the enantiomer of **L130**.

All steps proceeded with the conservation of the enantiomeric purity, except for the cyclopropyl derivative, where a slight loss of ee was observed. Moreover, the trichloroacetamide-protected



Scheme 49. Alkenylation in the presence of 2 mol% of (–)-MIB ligand **L130**.

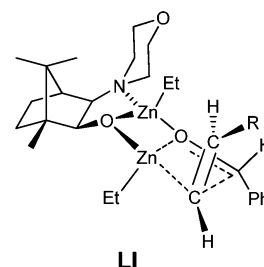
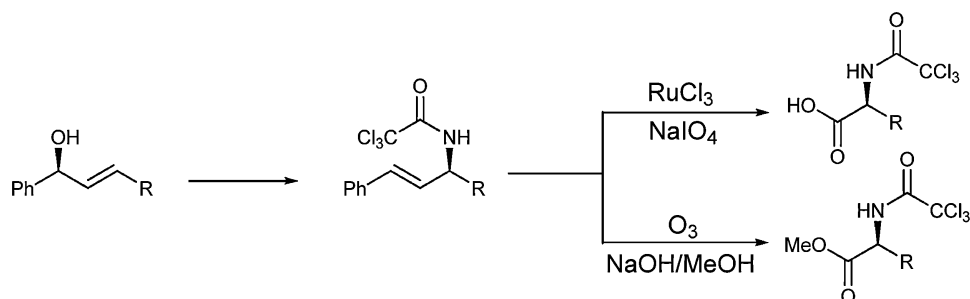
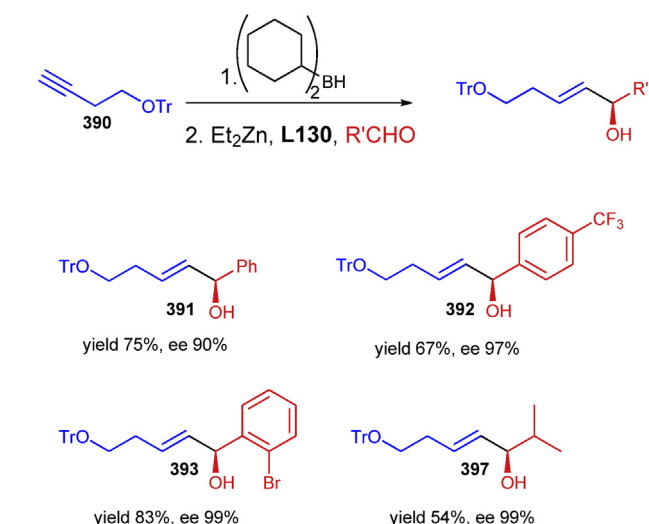


Fig. 10. Noyori's model for the transition state of MIB-catalysed reactions.

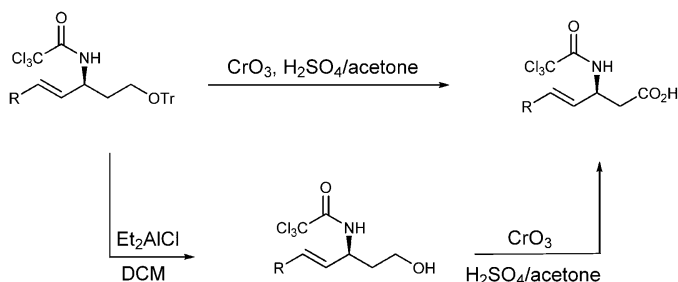
amines are crystalline and a single recrystallisation could enrich ee up to 99%. The cyclopropyl derivative required the conversion to *N*-Boc amine before a successful recrystallisation. Allylic diamine obtained from hepta-1,6-diyne was converted into *N,N*-diprotected diamino pipelicolic acid in only 3 steps.



Scheme 50. Synthesis of α -amino acid derivatives using Overman's [3,3]-sigmatropic rearrangement of trichloroacetimidates.



Scheme 51. Alkenylation using 1-trityloxybut-3-yne in the presence of 4 mol% of **L130**.



Scheme 52. Synthesis of unsaturated amino acids and amino alcohols.

The scope of the reaction was further explored and trityl ether of 1-butyn-3-ol was reacted with several aromatic and aliphatic aldehydes in the presence of 4 mol% of (–)-MIB [255]. The proper experimental procedure for this reaction was very important. The classical approach used by Oppolzer and Walsh in the above-described reports (transmetalation of vinylborane at -78°C followed by the addition of an aldehyde at 0°C) resulted in low yields. Better results were obtained in the slow addition of vinylborane to a cold mixture of **L130**, Et_2Zn and an aldehyde. Also, the use of trityl protection of alkynol was not accidental. The use of various other silyl protecting groups led to low yields of the reaction and problems with oxidation of the addition product. Under modified conditions, the reaction proceeded with good to moderate chemical yields and an excellent asymmetric induction for all aldehydes except for the aliphatic, long chain one (Scheme 51 and Supplementary materials – Scheme S24).

The addition products were efficiently transformed into unsaturated amino alcohols and/or γ -unsaturated β -amino acid derivatives (Scheme 52).

These pivotal works were further extended and the very interesting methodology for the synthesis of the otherwise difficult to obtain, enantiomerically enriched compounds was developed. Walsh used his method of alkenylation for the enantioselective

synthesis of epoxy alcohols (Scheme 53) [256]. The diastereoselectivity of the epoxidation step was in the range of 2.9–4.5:1 and enantiomeric excesses of the key alkenylation step were usually over 90% ee.

The synthesis of β -hydroxy aldehydes required a substantial modification of the basic procedure. The addition to aldehydes proceeded with a moderate enantioselectivity, while ethoxy acetylene **399** was converted to vinylborane using dicyclohexyl- or diethylborane. However, the preparation of tris(vinyl ether)borane **400** by the reaction of **399** with $\text{BH}_3\cdot\text{Me}_2\text{S}$ complex led to much higher ee's in resulting enol ethers **401–407** (Scheme 54) [257].

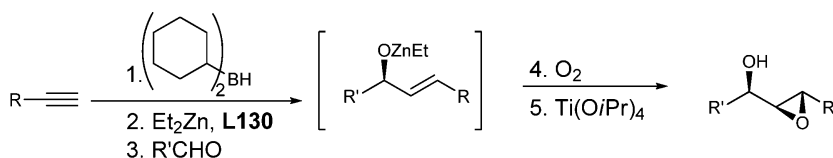
The thus obtained enol ethers were, after protection of the free hydroxyl as silyl and benzyl ethers, converted into β -hydroxy aldehydes **408** and **409**, respectively. The idea of the double asymmetric induction was then investigated. The addition of vinylborane **400** to aldehydes **408** and **409** in the presence of (+)-**L130** led to the preferred *anti*-diol **410** with a good diastereoselectivity, indicating (+)-MIB and (S)-aldehyde as a matched pair. The reaction catalysed by (–)-**L130** led to *syn*-diol **411** with lower dr.

The asymmetric synthesis of (Z)-disubstituted allylic alcohols by organometallic addition to aldehydes is a very challenging process.

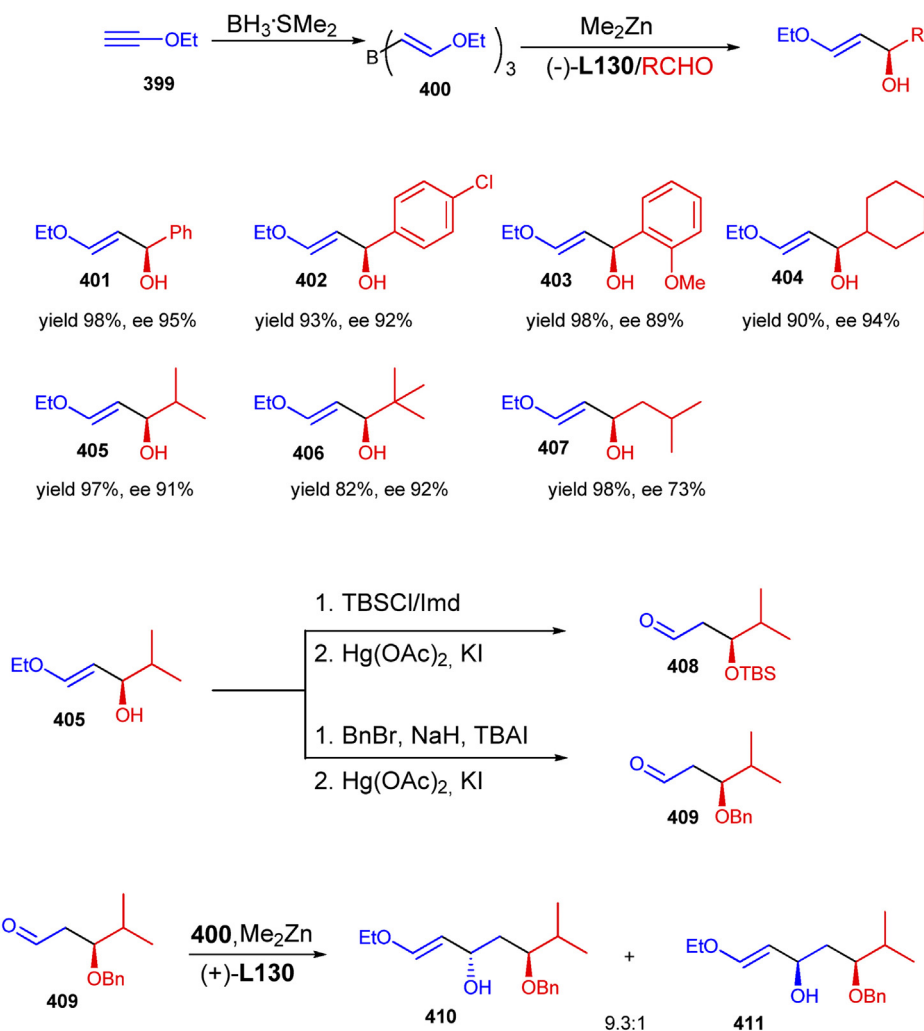
An excellent solution to this problem was proposed by Walsh. On the basis of the earlier method [258], which allowed for the synthesis of (Z)-disubstituted allylic alcohols in high yield but only in a racemic form in spite of the use of the otherwise excellent chiral ligand **L130** (Scheme 55), he developed a new approach [259]. The lack of enantioselectivity could be explained by the predominating background reaction catalysed by Lewis acidic LiBr produced in the second step of the synthesis. LiBr could be removed by filtration or centrifugation, but a more convenient method was needed, especially for a larger scale synthesis. The selected approach, already successfully applied for the synthesis of diarylmethanols [260], was the addition of *N,N,N',N'*-tetraethylethylenediamine (TEEDA). This can make a strong chelate with lithium chloride. Chelates of LiCl are much more stable than those of LiBr; therefore bromoacetylenes were replaced by chloroacetylenes (Scheme 56). Finally, the synthesis of (Z)-disubstituted allylic alcohols was accomplished in good yields and excellent enantioselectivities. Absolute configurations of selected allylic alcohols were determined by X-ray. It was assumed that all other products had the same configuration. Selected examples are presented in Scheme 56 and Supplementary materials – Scheme S25.

This approach has been extended to another very challenging problem, namely to the synthesis of (Z)-trisubstituted allylic alcohols.

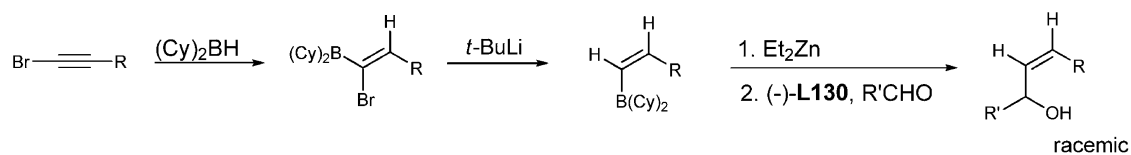
The Walsh group developed an excellent method for the synthesis of (Z)-trisubstituted allylic alcohols (Scheme 57) and applied it for the highly diastereoselective addition of α -chiral aldehydes and for the enantioselective addition to prochiral aldehydes in the presence of diamine [261]. The key step of this method is the dialkylzinc-mediated 1,2-metalate rearrangement **LII** with the formation of C–C bond, followed by the transmetalation leading to (Z)-trisubstituted vinylzinc species (Scheme 58). Despite the use of bromoacetylenes, the less expensive TMEDA could be used instead of TEEDA without an adverse effect on the enantioselectivity. R^2 could be ethyl or cyclohexyl. The method was inefficient for the synthesis of α -methyl-substituted alcohols.



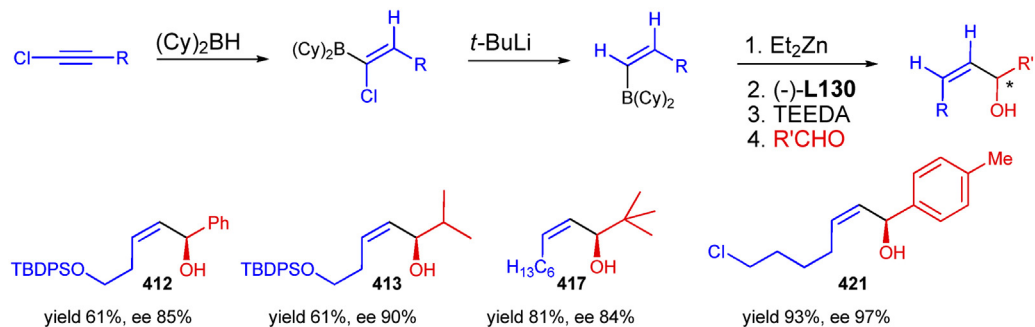
Scheme 53. Synthesis of epoxy alcohols.



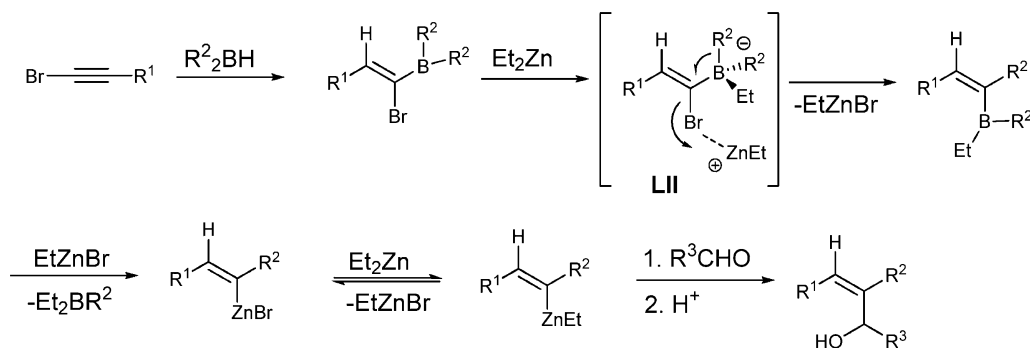
Scheme 54. Synthesis of enol ethers and application of **L130**-catalysed alkenylations to the synthesis of β -hydroxy aldehydes.



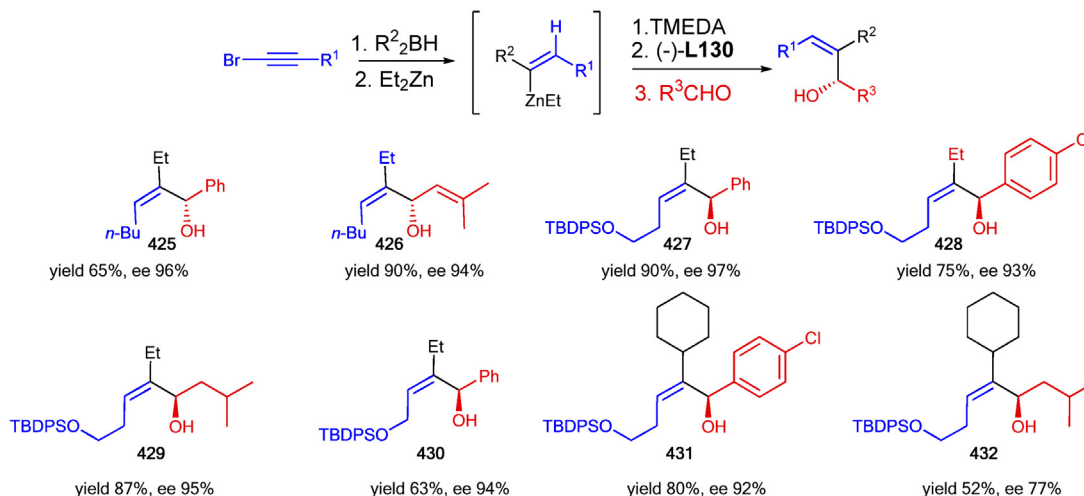
Scheme 55. Synthesis of (Z)-disubstituted allylic alcohols.



Scheme 56. Synthesis of (Z)-disubstituted allylic alcohols in the presence of 5 mol% of (–)-**L130**.



Scheme 57. The mechanism of the synthesis of (Z)-trisubstituted allylic alcohols.



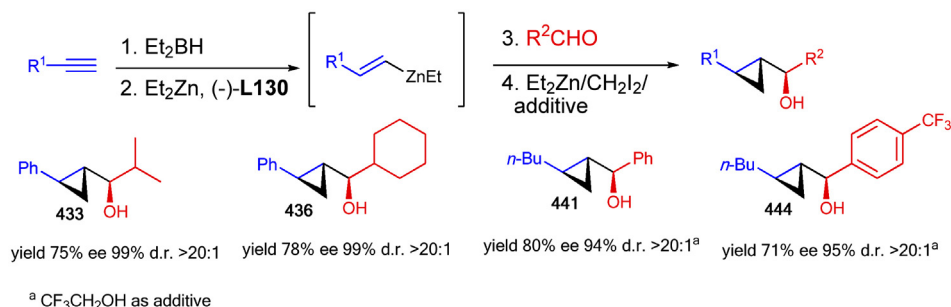
Scheme 58. Enantioselective synthesis of (Z)-trisubstituted allylic alcohols.

The methodology for the synthesis of synthetically important compounds, based on an enantioselective alkenylation of aldehydes, was further extended and the tandem addition-cyclopropanation reaction was presented (Scheme 59, additive = none) [262]. Cyclopropyl alcohols **433** and **436** were obtained with very good to excellent enantio- and diastereoselectivities from isobutyraldehyde and cyclohexanecarbaldehyde; additions to other aldehydes were less efficient. The evaporation of triethylborane formed in the transmetalation step was required to achieve good chemical yield and diastereoselectivity.

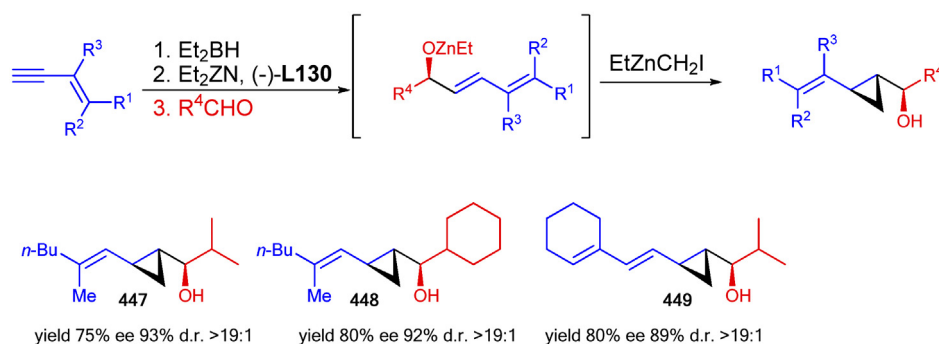
With some modifications, this method was successfully extended to arylaldehydes [263]. However, the trick with the removal of triethylborane did not work for aromatic aldehydes. A possible explanation of this different behaviour could be an interaction of zinc carbenoids with π -system of the aldehyde. The increased rate of the cyclopropanation step could be a possible

solution to overcome this problem. Additives forming more reactive zinc carbenoids were used at an increased concentration. This approach proved to be highly successful and cyclopropyl alcohols **441–446** were obtained in very good yields, with good to excellent ee and dr (Scheme 59 and Supplementary materials – Scheme S26).

The dienylation of aldehydes using Oppolzer's procedure was reported for the first time and applied for the synthesis of vinylcyclopropyl alcohols [264]. The previously established methodology was extended and enynes were used as substrates. However, the enantioselectivity was lower than in the case of simple vinylations, probably due to an increased rate of the background reaction. The problem was overcome by increasing the amount of **L130** to 10 mol% and a dropwise addition of the aldehyde. The conditions [262] for the following cyclopropanation also required modifications due to the formation of (bis)cyclopropyl alcohols, but finally the expected vinylcyclopropyl alcohols were obtained



Scheme 59. Synthesis of the cyclopropyl alcohols in the presence of 4 mol% of (-)-L130.



Scheme 60. Synthesis of vinylcyclopropyl alcohols in the presence of 10 mol% of (–)-**L130**.

with excellent diastereoselectivity >19:1 in all cases (Scheme 60 and Supplementary materials – Scheme S27).

The synthesis of *cis*-disubstituted cyclopropyl alcohols [264] was based on the earlier method for the synthesis of enantiomerically enriched (*Z*)-allylic alcohols [259]. Reported enantioselectivities were even slightly better than those presented in Scheme 56 and diastereoisomeric ratios were better than 19:1 (Scheme 61). The full account of the synthesis of cyclopropyl alcohols has been presented by Walsh [265].

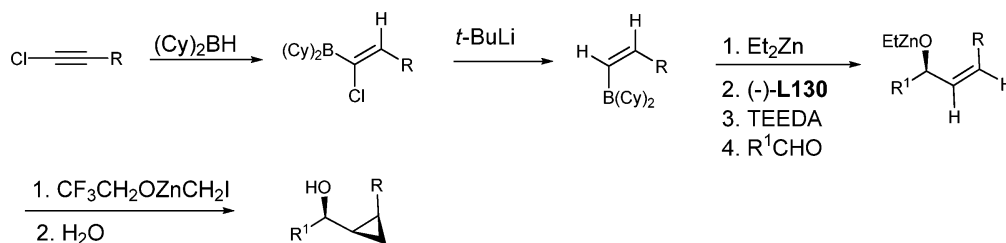
The same methodology was applied for the first one-pot catalytic asymmetric synthesis of β -hydroxy enamines [266]. The conversion of ynamides proceeded with excellent enantioselectivities. Good to very good yields were obtained only when one of the substituents on the ynamide nitrogen atom was a tosyl group. The low ee obtained for enamine **457** indicates that the presence of α -branching in an aldehyde is crucial for the high selectivity (Scheme 62 and Supplementary materials – Scheme S28).

Table 13
Alkenylation in the presence of amino thiols **L132** and **L133**.

L132

L133

Entry	Product	L* (mol%)	Yield (%)	Ee (%)
1		L132 (5)	90	>99
2		L132 (2)	86	99
3		L132 (2)	72	66
4		L132 (2)	81	70
5		L133 (5)	94	>99
6		L133 (2)	90	98
7		L133 (2)	86	93
8		L133 (1)	90	94



Scheme 61. Synthesis of *cis*-disubstituted cyclopropyl alcohols in the presence of 5 mol% of (–)-**L130**.



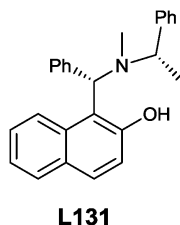
Scheme 62. Synthesis of β-hydroxy enamines.

Thus-obtained β-hydroxy enamines were subjected to the modified Simmons-Smith (see above) cyclopropanation or hydrogenation with an excellent diastereoselectivity (>20:1).

Under standard Oppolzer's conditions, (–)-2-*exomorpholinoisobornane*-10-thiol **L27** (10 mol%) was successfully used by Uang et al. for the highly enantioselective synthesis of substituted allylic alcohols [104]. The reaction of octyne-1 with various aromatic aldehydes proceeded with ee's over 95%, up to >99.5% for the 4-CF₃ substituted benzaldehyde. A substantial increase in the enantioselectivity was noticed when the transmetalation was performed using Me₂Zn instead of Et₂Zn.

Zhao and co-workers used alkenylboronic esters as substrates for the transmetalation with Et₂Zn and subsequent reactions with aldehydes catalysed by ligand **L112** (5 mol%). While chemical yields were low to moderate, the enantioselectivity was high for the majority of aromatic aldehydes used [267]. Much better results were obtained by Braga and co-workers in the presence of aziridine ligand **L111**. The addition of octyne-1, phenyl- and *tert*-butylacetylenes to benzaldehyde and some 4-substituted benzaldehydes proceeded in very good yield and enantioselectivities from 88 to 97% ee were obtained [268].

Ligand **L104b**, which also gave excellent enantioselectivities for the arylation of aldehydes, was very inefficient in alkenylation reactions and products were obtained with low to moderate ee's [269].

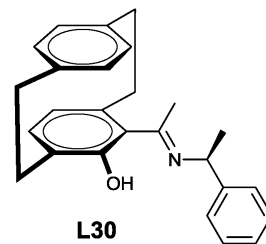


L131

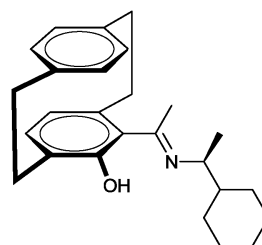
The alkenylation with hexyne-1 in the presence of Betti's base type ligand **L131** (15 mol%) proceeded with a very good to excellent enantioselectivity for several aromatic aldehydes (Supplementary materials – Scheme S29). The synthesis of ligand **L131** was accomplished in one step, which made it an excellent candidate for a large scale use [270].

A long series of amino thiols was synthesised; excellent enantiomeric excesses were obtained for two amino thiols **L132** and **L133** (1–5 mol%). The alkenylation of benzaldehyde and selected aryl aldehydes with octyne-1 proceeded with an enantioselectivity of up to 99% ee. However, these ligands were only moderately efficient for cinnamyl and cyclohexyl aldehydes (Table 13, entries 3 and 4) [271].

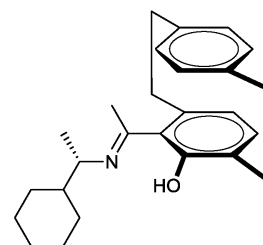
[2,2]Paracyclophane **L30** (2 mol%) was used under modified Oppolzer's conditions which allowed for the addition of the aldehyde in one portion but also required the use of 1.5 equiv. of borane and 2 equiv. of Et₂Zn. The original Oppolzer's procedure uses equimolar amounts of reagents. The alkenylation of some aromatic aldehydes and, additionally, cyclohexanecarbaldehyde and pivaloyl aldehyde, with octyne-1 proceeded with a very good to excellent enantioselectivity (Supplementary materials – Scheme S30). A much worse asymmetric induction was observed for 3,3-dimethylbut-1-yne, until Me₂Zn was used instead of Et₂Zn in the transmetalation step. The enantiomeric excess rose from 64 to 89% ee, probably due to the formation of a less sterically demanding zinc species. The method worked only for bulky acetylenes and no improvement was observed for the linear ones [272].



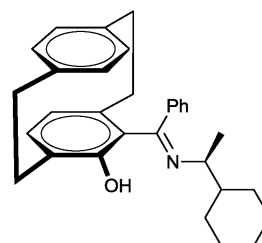
L30



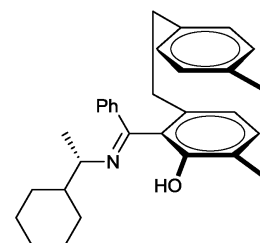
L134



L135

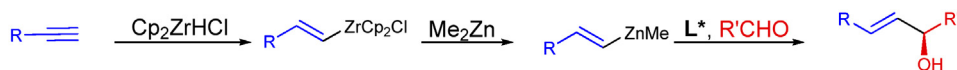


L136



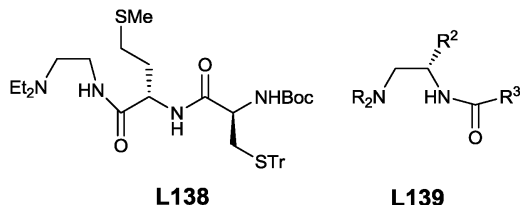
L137

The influence of the modification of the ligand's structure on the enantioselectivity was investigated [273]. Excellent results were achieved for α-branched aliphatic aldehydes with ligands **L134**–**L136**, but not with **L137**. For other aldehydes, the observed asymmetric induction was only moderate.



Scheme 63. Alkenylation of aldehydes by Wipf's method.

A combinatorial library of 59 dipeptide ligands were synthesised and these ligands were screened as catalysts for the alkenylation reaction. The best results were obtained for ligand **L138** [274].



Contrary to the above-mentioned results for paracyclophane **L30**, both octyne-1 and 3,3-dimethylbut-1-yne transmetallated with Me_2Zn in the presence of ligand **L138** gave high enantiomeric excesses in the addition to benzaldehyde and **376** and **484** were obtained with 91 and 93% ee, respectively. The alkenylation of various 4-substituted benzaldehydes, cyclohexanecarbaldehyde and 2-naphthaldehyde with 3,3-dimethylbut-1-yne also furnished allylic alcohols with high ee's – 91–95% (Supplementary materials – Scheme S31).

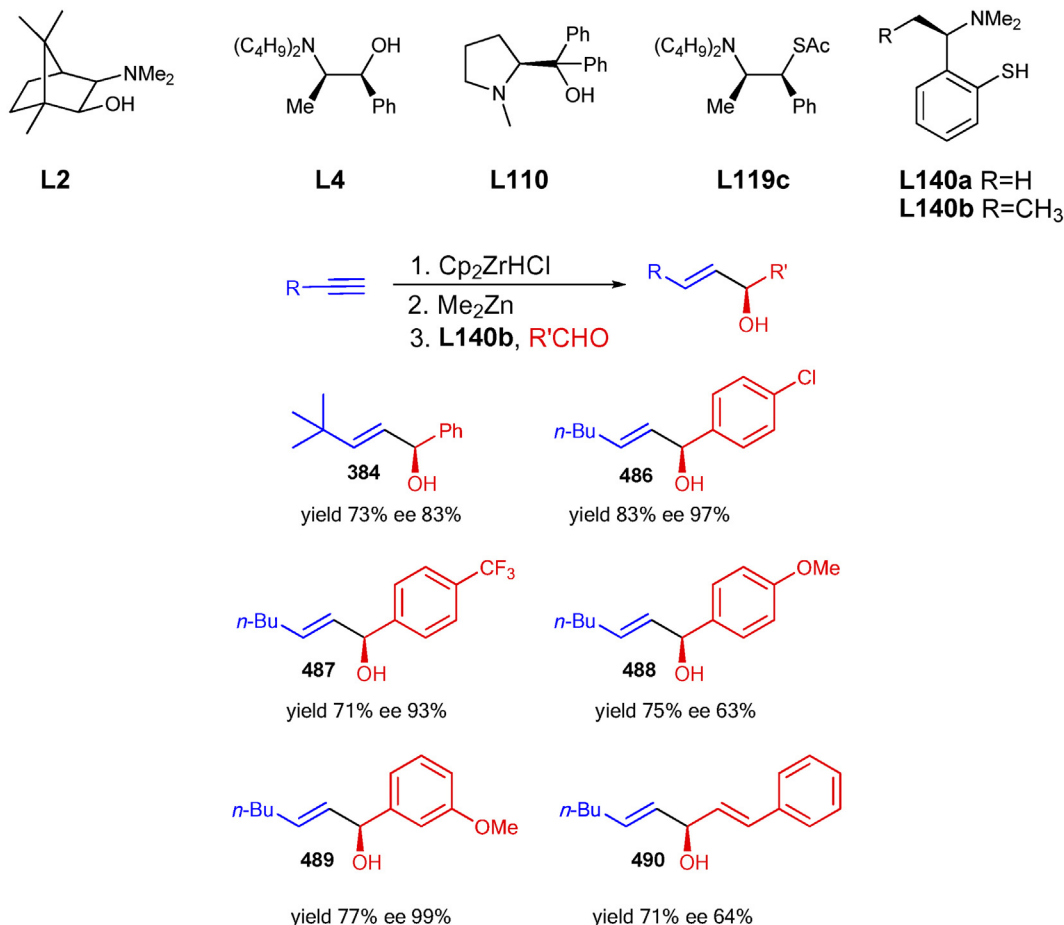
The same group reported the synthesis of a series of ligands having the general structure **L139** and its application to the alkenylation reaction. Enantioselectivities obtained in this study ranged from moderate to good, not exceeding 90% ee [275].

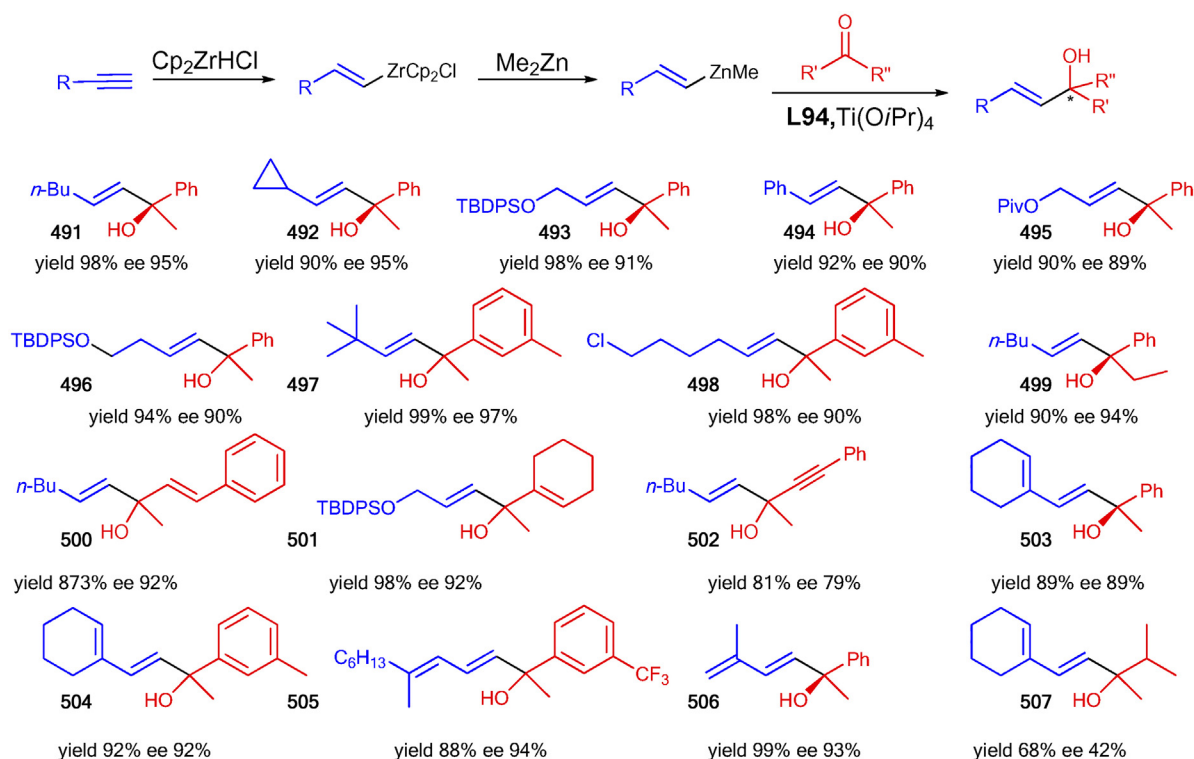
4.2. Alkenylation based on zirconium–zinc transmetalation

4.2.1. Addition to aldehydes

Another method for the enantioselective addition of alkenylzinc reagents to carbonyl groups is the zirconocene–zinc transmetalation method developed by Wipf et al. (Scheme 63).

In the initial communication, Wipf reported the reaction of hexyne-1 with benzaldehyde in the presence of 8 mol% of ligand **L110**, which afforded allylic alcohol with only 38% ee [276]. In this seminal paper, he proposed a protocol for an efficient alkenylation of aldehydes. The reaction proceeds efficiently even in the absence of any additional ligand, leading to racemic allylic alcohols, therefore the proper choice of the reaction conditions and chiral ligands is extremely important for successful enantioselective reactions [277]. Several amino alcohols (**L2**, **L4**, **L110**) and amino thiols (**L119c**, **L140**) have been checked in this reaction and, quite surprisingly, amino thiols **L140** significantly outperformed compounds containing hydroxy groups. However, amino alcohol-catalysed reactions were faster by about 50% than the reactions carried out in the presence of amino thiols, and the latter ones had reaction rates comparable to the results obtained in the absence of a chiral ligand. Best ee's, up to 99% for the alkenylation of 3- and 4-substituted benzaldehydes with hexyne-1, were obtained for amino thiol **L140b** (Scheme 64).

Scheme 64. Alkynylation using zirconocene–zinc transmetalation catalysed by 10 mol% of ligand **L140b**.



Scheme 65. Enantioselective alkenylation and dienylation of ketones in the presence of 5 mol% of **L94**.

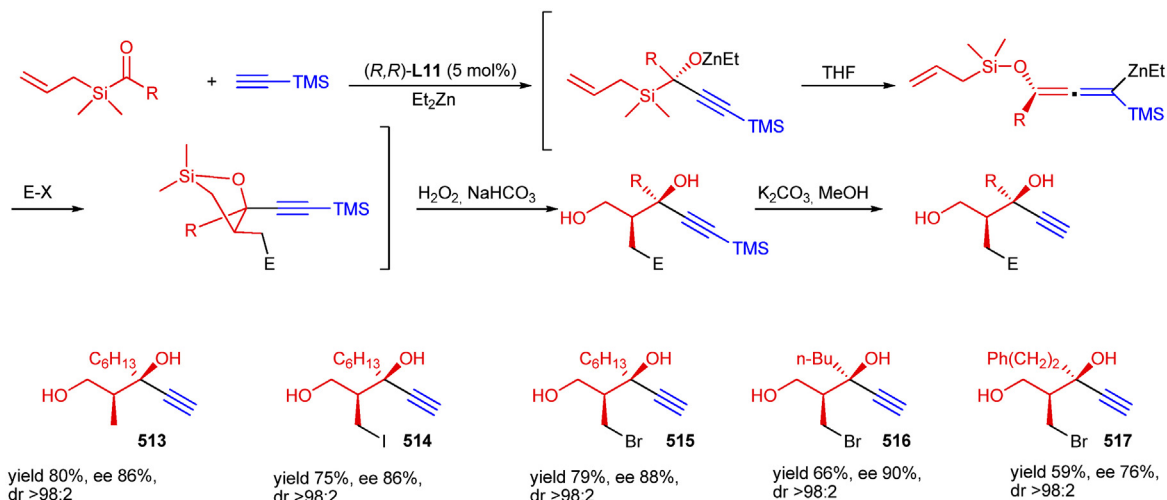
The deepened discussion of the reaction mechanism was later presented [278]. The presence of stoichiometric amounts of zirconocene and zinc species was probably responsible for the effects observed in this reaction. While ligand **L140b** exhibited a very strong positive nonlinear effect that fits perfectly Kagan's model [279], the results obtained for the amino alcohol **L110** were very unusual. When the amount of **L110** was increased from 5 to 15 mol%, the enantiomeric excess rose from 9 to 81% and then dropped to 60% upon a further increase to 20 mol%. Furthermore, a purely linear dependence of the asymmetric induction on ee of ligand from 100 to 50% ee was observed, but at 35–20% ee a reversal of the direction of the asymmetric induction was noticed and the opposite enantiomer was obtained. Ligand **L132** and its *S*-acetate were also tested and showed enantioselectivities comparable to that of **L140b**. Amino thiol ligand **L132** was somewhat more

selective than its acetate and much more selective than the respective amino alcohol, and the enantioselectivity observed for Wipf's protocol was slightly higher than that for Oppolzer's. Moreover, the higher activity of **L132** allowed for an efficient use of 5 mol% of the ligand [280]. The superiority of sulphur-containing ligands could be explained by the high affinity of thiols towards zinc and the reduced tendency to diminish the Lewis acidity of the zinc species. The latter was responsible for the higher activity of the system.

The same process was ineffective in the presence of ligand **L30**, and the enantiomeric excess did not exceed 13% [273].

4.2.2. Addition to ketones in the presence of $\text{Ti}(\text{O}i\text{Pr})_4$

The alkenylation of ketones is possible only using Wipf's method. Walsh has shown that the reaction carried out under Oppolzer's conditions in the presence of ligand **L130** (10 mol%)



Scheme 66. Synthesis of substituted propargyl diols possessing a tertiary hydroxyl group using (R,R) -**L11** catalysed alkynylation.

led to achiral unsaturated tertiary diols as the major product, while the use of a Schwartz reagent and ligand **L94** gave tertiary allylic alcohols with an excellent enantioselectivity [281]. A series of substituted acetylenes, namely hexyne-1, cyclopropylacetylene, phenylacetylene, *tert*-butylacetylene, 1-chloro-5-hexyne, as well as the protected propargylic and homopropargylic alcohols gave ee's from 87 to 95% for additions to acetophenone and its derivatives. The addition to dialkyl ketone was less selective and 79% ee was reported. The addition to α,β -unsaturated ketones was also investigated; 92–97% ee's were obtained. In the full account [282], the original procedure was modified, and better yield and enantioselectivity were achieved in the presence of only 5 mol% of ligand **L94** (Scheme 65). The authors imply a beneficial role of 2-propanol liberated during the modified process for the catalyst preparation. This is rather counter-intuitive, though well explained by the suggested sequestering of Cp_2ZrMeCl , an achiral Lewis acid competing with the chiral catalyst in the background reaction. This methodology has been successfully extended to the dienylation of ketones with a very high asymmetric induction [282].

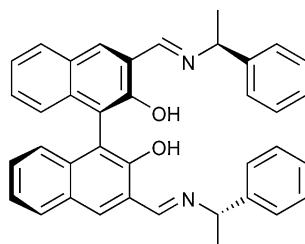
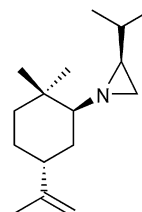
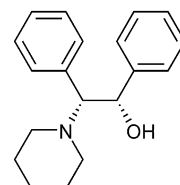
A very similar ligand, **L141**, catalyses the alkenylation of ketones with a slightly lower enantioselectivity, probably due to a greater conformational freedom of **L141** [283]. Also a non-symmetrical

The first application of phosphine-containing ligand to the dialkylzinc-mediated alkynylation reaction was presented [286]. The BINOL based ligand **L144** (5 mol%) was used for the titanium-free addition of arylacetylenes to aromatic aldehydes. Under such conditions good yields and moderate enantioselectivities were obtained. However, the presence of additives, namely a 1:1 mixture of *n*-BuLi and CaH_2 , led to the reversal of the enantioselectivity with a significant increase of enantiomeric excesses, in some cases up to 99% (Supplementary materials – Scheme S33). Initial mechanistic studies suggest the presence of dimeric or oligomeric species with multi-metal centres involving calcium, lithium and zinc alkoxides.

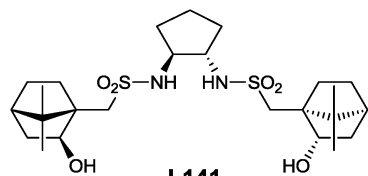
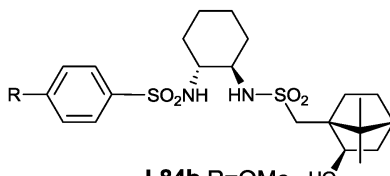
Amino alcohol **L145** was used for the dimethylzinc-mediated addition of phenylacetylene to aliphatic and aromatic aldehydes in the presence of $\text{Ti}(\text{O}i\text{Pr})_4$ with moderate to very good enantioselectivities of up to 95% ee [287]. The ^1H NMR studies suggest that the presence of several multinuclear titanium-based species.

Wanga and Pu applied a series of BINOL-derived Schiff bases for the addition of phenylacetylene to some aromatic aldehydes. Good enantioselectivities were obtained in the presence of **L146**, the best member of the series [288].

Limonene-derived aziridine **L147** catalysed the addition of phenylacetylene to several aldehydes with enantioselectivities of 91–95% ee [289].

**L146****L147****L148**

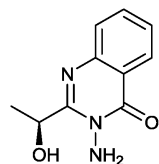
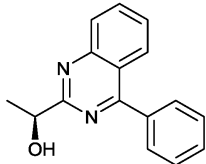
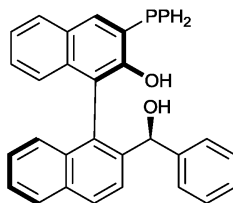
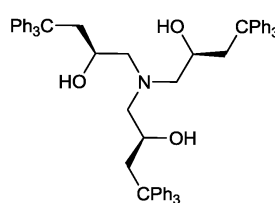
ligand, **L84b**, was used in this process and catalysed the alkenylation of simple acetophenones using phenylacetylene, hexyne-1 and TMS-acetylene as a source of the vinyl group with an enantioselectivity in the range of 74–94% [243].

**L141****L84b** R=OMe

5. Recent references

5.1. Alkynylation

Two structurally similar ligands, **L142** [284] and **L143** [285] were recently synthesised and applied for the addition of phenylacetylene to aldehydes. The presence of $\text{Ti}(\text{O}i\text{Pr})_4$ was indispensable for a high (or any, in the case of **L142**) enantioselectivity and good yield (Supplementary materials – Scheme S32).

**L142****L143****L144****L145**

The addition of TMS-acetylene to acylsilanes was conducted in the presence of ProPhenol ligand (*R,R*)-**L11** [290]. Respective adducts were obtained in very good yield and excellent enantioselectivities up to 96% ee. In addition to (*R,R*)-**L11**, several other modifications of **L11**, containing substituents other than phenyl in the prolinol moiety were tested; however, none gave a good balance of chemical yield and enantiodiscrimination. Propargylsilanols were then converted into tertiary propargylic alcohols by

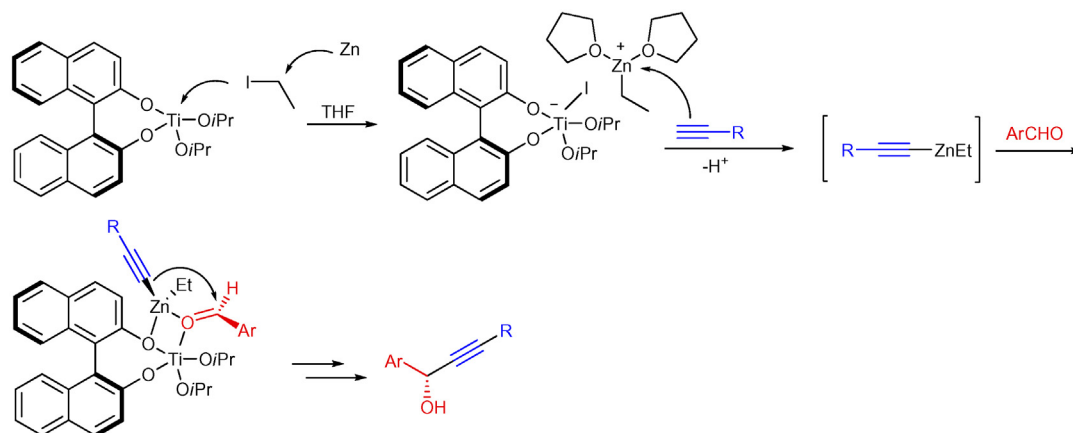


Fig. 11. Addition of alkynes to aromatic aldehydes using ethyl iodide and zinc.

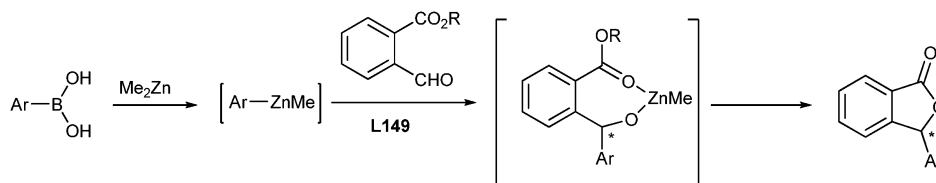


Fig. 12. Cascade reaction leading to chiral phthalides.

the one-pot allenyl-Zn-Brook rearrangement and Zn-ene-allene cyclization (selected examples are presented in Scheme 66). The second stereogenic centre was formed with an almost complete transfer of chirality.

Ligand **L148** catalysed the addition of phenylacetylene to aryl glyoxylates in good yield and a moderate enantioselectivity [291]. However, the minor enantiomer decomposed during a prolonged reaction, which resulted in a kinetic resolution of the post-reaction mixture. As a consequence, the enantiomeric excesses over 90% were accompanied by low chemical yields (34–41%).

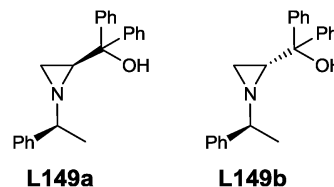
The last reference in this section does not describe diethylzinc-mediated alkynylation [292]. In fact, it describes how to avoid the use of pyrophoric Et_2Zn and perform an addition of alkynylzinc to aldehyde. Pu and co-workers found that the reaction of ethyl iodide (EtI) and metallic Zn in THF is activated with a mixture of $\text{Ti}(\text{OiPr})_4$ and BINOL, leading to ethylzinc-THF complex. The $\text{Ti}(\text{OiPr})_4$ /BINOL complex forms a stronger Lewis acid than $\text{Ti}(\text{OiPr})_4$ alone, thus allowing for such a transformation. The mixture of EtI, BINOL (*S*)-**L59** and $\text{Ti}(\text{OiPr})_4$ is stirred for at least 24 h to produce ethylalkynylzinc. The addition of an aldehyde led to propargylic alcohol with very good yield and enantioselectivity (Fig. 11).

Allowing myself for a little bit of artistic licence, this is a “diethylzinc-mediated alkynylation without diethylzinc”. Selected results of the addition to aromatic aldehydes are presented in Supplementary materials – Scheme S34. If this approach was proven more general, it would open a new area in alkynylation reactions and beyond.

5.2. Arylation

The arylation of aldehydes is now a well-developed methodology. The main goal now is an efficient synthesis of both the enantiomers of diarylcarbinols. The most popular method, the interchanging of aryl moieties, has its limitation and does not always lead to acceptable results (see above). The use of enantiomeric ligands is a possible solution; however, the second enantiomer is not always readily available. The second approach relies on the synthesis of diastereoisomeric ligands, which are

capable of acting as pseudoenantiomers. One of the best examples of such approach is synthesis and application of aziridinecarbinols **L149** [293].



Under optimised conditions these ligands gave pairs of enantiomeric diarylcarbinols with a very high and comparable enantioselectivity; the only exception was phenyl(*o*-tolyl)methanol **316** (Supplementary materials – Scheme S35). This research was extended and a cascade process leading to chiral phthalides was developed successfully (Fig. 12). However, only boronic acids that were unsubstituted or substituted with electron-donating groups furnished products with high enantioselectivities. Boronic acids with *ortho*-, *meta*-, or *para*-chloro substituents gave low to very moderate ee values.

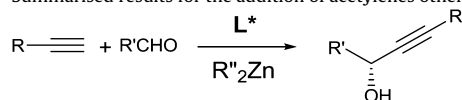
6. Conclusions and future prospects

Despite the large number of publications, there is still room for development of the three types of reactions reviewed. This is especially noticeable for the asymmetric alkynylation reaction, where most results were obtained for phenylacetylene, which is an excellent model compound, but the thus obtained adducts are of limited – if any – value as starting materials for the total synthesis. Other alkynes, like TMS-acetylene, alkyl propiolates, or long-chain alkynes have not been studied as comprehensively.

Although several catalytic systems have been developed for the enantioselective addition of alkynes to carbonyl compounds, many of them suffer from various drawbacks, such as a high catalyst loading, the necessary presence of some additives and the unavailability of both enantiomeric forms of the ligand. Therefore, the challenge

Table 14

Summarised results for the addition of alkynes other than phenylacetylene.



Entry	Ligand (mol%)	Ti(OiPr) ₄	R	R'	Yield (%)	Ee (%)	Ref.
1	L11 (10)	–	TMS	2,6-diMeOC ₆ H ₃	79	97	[78]
2	L11 (10)	–	CO ₂ Me	(E)-PhCH=CH–	92	95	[79]
3	L11 (10)	–	CO ₂ Me	C ₆ H ₁₃ CH=CH–	86	97	[79]
4	L11 (10)	–	CO ₂ Me	(CH ₃) ₂ CHCH=CH–	97	97	[79]
5	L11 (10)	–	CO ₂ Me	PhCH ₂ CH ₂ –	65	95	[79]
6	L11 (10)	–	CO ₂ Me	TIPSO(CH ₂) ₄	52	95	[79]
7	L11 (10)	–	TIPS–C≡C–	2-Furyl	95	98	[80]
8	L11 (10)	–	TIPS–C≡C–	c-C ₃ H ₇	97	97	[80]
9	L11 (10)	–	TIPS–C≡C–	(CH ₃ O) ₂ CHCH=CH–	91	95	[80]
10	L11 (10)	–	TIPS–C≡C–	(CH ₃) ₂ CHCH=CH–	89	96	[80]
11	L11 (10)	–	TIPS–C≡C–	(E)-C ₇ H ₁₅ CH=CH–	93	97	[80]
12	L11 (10)	–	TIPS–C≡C–	C ₆ H ₁₃ CH=CH–	93	97	[80]
13	L11 (10)	–	CO ₂ Me	(E)-c-C ₃ H ₅ CH=CH–	85	95	[83]
14	L11 (10)	–	CO ₂ Me	CH ₃ –	74	98	[83]
15	L16 (10)	–	TIPS–C≡C–	Ph	90	95	[96]
16	L17 (10)	–	TMS	2-MeC ₆ H ₄	77	95	[95]
17	L17 (10)	–	TMS	3-MeOC ₆ H ₄	72	96	[95]
18	L17 (10)	–	TMS	4-FC ₆ H ₄	66	96	[95]
19	L17 (10)	–	TMS	4-ClC ₆ H ₄	62	96	[95]
20	L17 (10)	–	TMS	4-MeC ₆ H ₄	76	95	[95]
21	L17 (10)	–	TMS	4-MeOC ₆ H ₄	84	97	[95]
22	L17 (10)	–	TMS	1-Naphtyl	84	95	[95]
23	L20 (10)	–	t-C ₄ H ₉	Ph	71	95	[99]
24	L20 (10)	–	TMS	Ph	85	97	[99]
25	L20 (10)	–	TMS	4-FC ₆ H ₄	85	95	[99]
26	L20 (10)	–	TMS	3-FC ₆ H ₄	75	96	[99]
27	L20 (10)	–	TMS	3-MeOC ₆ H ₄	82	96	[99]
28	L20 (10)	–	SiMe ₂ Ph	4-FC ₆ H ₄	88	99	[99]
29	L41 (22)	–	Cl(CH ₂) ₃ –	Ph	60	95	[118]
30	L41 (22)	–	Cl(CH ₂) ₃ –	4-MeOC ₆ H ₄	72	95	[118]
31	L41 (22)	–	Cl(CH ₂) ₃ –	4-MeC ₆ H ₄	74	96	[118]
32	L59 (40)	+	CO ₂ Me	1-Naphtyl	87	95	[135]
33	L59 (40)	+	CO ₂ Me	4-ClC ₆ H ₄	84	95	[135]
34	L59 (40)	+	TIPS–C≡C–	CH=CH ₂ (CH ₂) ₂ –	94	95	[140]
35	L61 (20)	+	TMS	C ₄ H ₉	65	97	[145]
36	L64 (20)	+	CO ₂ Me	C ₄ H ₉	76	95	[145]
37	L64 (20)	+	CO ₂ Me	(CH ₂) ₂ CH=CH ₂	76	95	[145]
38	L64 (20)	+	CO ₂ Me	(CH ₂) ₃ CH=CH ₂	67	95	[145]
39	L64 (20)	+	CO ₂ Me	c-C ₆ H ₁₁	84	95	[145]
40	L64 (20)	+	CO ₂ Et	c-C ₆ H ₁₁	84	99	[145]
41	L66 (20)	+	CO ₂ Me	Ph	80	95	[146]
42	L66 (20)	+	CO ₂ Me	4-MeC ₆ H ₄	90	95	[146]
43	L66 (20)	+	CO ₂ Me	2-MeOC ₆ H ₄	83	98	[146]
44	L66 (20)	+	CO ₂ Me	4-MeOC ₆ H ₄	80	95	[146]
45	L66 (20)	+	CO ₂ Me	4-ClC ₆ H ₄	89	98	[146]

to find an ideal or, being more realistic, a highly versatile catalyst is still ongoing.

Table 14 summarises all above-presented examples of enantioselective addition using alkynes other than phenylacetylene, which achieved ee values equal or higher than 95%. This last restriction seems to be very important, since it is not always possible to enhance the enantiomeric purity of compounds obtained with a lower enantiomeric selectivity.

Only 9 out of 82 ligands were able to deliver adducts of alkynes (other than phenylacetylene) to aldehydes with an enantioselectivity equal to or exceeding 95%. Nearly a third of the listed entries are for a ProPhenol ligand. The addition to ketones looks even worse. There are only eight examples of adducts obtained with >95% ee, all of them formed from the addition of phenylacetylene. These results were presented in 12 articles – this is roughly 10% of all reports

published for dialkylzinc-mediated alkyne additions to carbonyl groups.

Another drawback of methods developed so far is their high catalyst loading, in some cases as high as 40 mol%. While being not so important for the model studies, this precludes the use of such catalysts for large scale syntheses, not to mention their industrial use. One of the reasons responsible for this situation is the lack of mechanistic investigation, which would allow for an analysis of the transition states for some types of ligands, thus leading to a successful development of highly efficient ligands. Gau's and Walsh's excellent mechanistic studies on dialkyl zinc additions have been extended by some authors to alkyne additions without experimental background. In conclusion, a lot of work has to be done before this methodology can be called useful.

A substantial development has been achieved for the arylation reaction. The expensive and difficult-to-handle reagent diphenylzinc has been replaced with a combination of diethylzinc and air-stable arylboronic acids and further by arylboroxines. The latter can be used with only *ca.* 4 equiv. of Et₂Zn and boroxine itself is used in an amount of *ca.* 0.4 equiv. (Zn:B:ArCHO = 3.7:0.45:1.0); in some reports such an amount could be decreased further [216]. Additionally, the reported transmetallation time can vary from 0.5 to 10 h for very similar combinations of reagents. A further effort is also needed in order to improve the efficacy of ligands, which are generally used in an amount of at least 10%. Moreover, ligands which are highly active for the arylation with Ph₂Zn/Et₂Zn, like **114**, show a much more diminished activity in reactions with boroxines (Ref. [102] vs. Refs. [103] and [104]). Elegant theoretical and experimental studies on the mechanism of the arylation reaction, performed by Bolm, Norby, and Pericàs, added much to our knowledge of this reaction. However, the work clarifying the differences in experimental procedures is needed urgently. The arylation of ketones obviously still awaits its time. Only rare examples of additions leading to tertiary alcohols with enantiomeric purity over 90% are known.

Alkenylation reactions, which are based on the Oppolzer's protocol, are generally very effective, especially in the presence of MIB ligand. A low catalyst loading (1–4%) makes it a good candidate for large scale preparations. A well-developed methodology for the enantioselective synthesis of otherwise difficult to obtain compounds has emerged from early studies. Except for early papers by Wipf himself, the Wipf's protocol remains highly unexplored. This method awaits new ligands, capable of delivering products with an enantioselectivity of over 95% and active and efficient in amounts close to 1–2 mol%. This last statement seems to be quite general and can be applied to most reactions reported in this review.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.ccr.2015.03.025>

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